

Special Relativity from AASC

The Necessary Kernel of Physical Objecthood and Its Lorentzian Realization

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Reference Edition

Abstract

Non-degenerate determinate physical objecthood necessarily instantiates Admissibility, Standing, Reference and Irreversibility. This necessity applies anew to every richer physical construction, including its selectors, calibrations, source maps and realized laws. We develop this premetric kernel and its derived AMetric boundary through the AASC source constructions to an inhabited special-relativistic realization. The proof reproduces the intervening arguments: physical observable descent forces the characteristic constraint quotient on the regular Hamiltonian branch; stable overlap and transport construct comparison and persistence records; licensed metricity fixes their order of physical dependence. On the stated comparison branch, complete action records and unique repetition construct the affine receiver. Completed causal order and the actual rest action force strict clock separation and a finite bound; intrinsic isotropy constructs the spatial quadratic form and Lorentz cone. Completed interval comparisons fix the common rest calibration, and composable frame comparisons remove conformal scale. Exact source refinement preserves the original bearer, joint operations, domains and failures through these constructions. Attached clock and endpoint readouts yield time dilation, length contraction and relativity of simultaneity. Comparison-preserving minimal transport fixes the Levi–Civita connection; invariant attached response and closed source equations give the Ward identity. A complete invariant source prescription supplies active response protocols; controlled impulse balance gives the premetric inertia role its familiar numerical realization $E_0 = m_i c^2$, with the mass shell obtained afterward. An explicit joint fixed-background model combines a regular characteristic source, an energetic internal clock, finite balanced signal protocols and a driven Hamiltonian control retaining the complete boosted clock phase and every original marked record. A common-history refinement verifies its vacuum-Einstein and full-ray Schrödinger projections, exact metric matching and SR continuation together. Its positive laboratory response and vacuum gravitational source remain distinct queries; a proved obstruction excludes identifying them as one complete source on unchanged flat geometry. The assembled theorem establishes kernel necessity at every construction stage, the retained source lineage, and uniqueness of the familiar SR laws on an explicitly inhabited target. Each stage displays the physical branch conditions it receives; kernel necessity, physical occupation, metric licensing and numerical selection have distinct proved roles in the same derivation. Exact reception interfaces supply these SR results to the existing Wheeler–DeWitt, symmetry and Standard Model carrier constructions.

Keywords: AASC; special relativity; premetric persistence; causal propagation; clocks; inertial response; mass–energy equivalence; structural unification.

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1 Physical objecthood and the order of dependence

A physical derivation concerns determinate objects and occurrences: a state belongs to an identified carrier, a comparison concerns identified relata, a clock records its own ticks, and a measured response belongs to the body whose transport state changes. These commitments arise with the physical subject matter itself. They obtain whether a theorist names them, whether a measurement is performed, and whether anyone adopts the AASC vocabulary.

AASC identifies their minimal necessary kernel,

$$K = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}).$$

Reference fixes the bearer and its conditions of sameness. Standing concerns actual qualification for the specified relation or use. Admissibility keeps identification, qualification and combination jointly accountable to the original conditions. Irreversibility retains which occurrence instantiated which conditions against substitution by a later occurrence or repair. The kernel necessity theorem reconstructs this work from non-degenerate determinate objecthood and excludes its functional deletion [1, Prop. 2.4, Thms. 3.1, 3.6, 7.1]. The implementation can be implicit or distributed; the four roles need not be four independently installed mechanisms.

The aim is an account and demonstration of the origin of realized special-relativistic objects and laws. Origin here is the order of necessary dependence and physical construction; no temporal creation sequence is asserted. The familiar formulas are the endpoint of the construction.

Special relativity therefore begins with objects and comparisons that already incur the kernel. Its usual coordinates, worldlines, clocks, fields and momentum records are determinate physical presentations only insofar as they retain that work. AASC's contribution is to expose this dependency and carry it through the construction of the physical geometry and its laws. Mathematical equations can be studied after their formal structures have been supplied. Their use as equations of identified physical objects additionally answers to the necessary conditions of those objects.

The corpus develops this dependency before a numerical spacetime metric is supplied. Gravity as Structural Necessity reconstructs the constraint role in its regular Hamiltonian comparison class [6, Thm. 8.2]. The realized-physics sequence develops comparison, witnessed transport, field and particle persistence, inertia and irreversible continuation [7, 8, 9, 10, 11, 12, 13]. The AMetric boundary develops the kernel's restrictions on warranted discrimination, selection and transport without primitive metric or probabilistic authority [2, Sec. 1.4, Thm. 7.5]. CSP-R receives these roles in one distributed persistence bearer [14, Prop. 1.2]; licensed metricity determines when metric descriptors carry already established physical comparison [15, Thm. 6.2]. The Wheeler–DeWitt arc constructs occupied clocks and their exact readout and continuation uses [18, 19, 20, 17].

The present paper reproduces the essential upstream forcing and construction arguments in Section 3, then carries their outputs into the SR receiver. The characteristic quotient constrains which event and source records may descend; the comparison and persistence constructions supply the original records and operations; metric licensing fixes the physical use of the subsequently constructed form. Theorem 5.2 obtains the affine carrier from homogeneous comparison composition. Section 10 derives the minimal compatible transport and the attached source conservation law used later. These arguments make the source architecture part of the proof. The paper then constructs the exact refinement connecting that architecture to its Lorentzian regime and proves uniqueness of the resulting measurement and response laws. Theorem 4.8 makes the source-to-specialization step explicit; Table 1 identifies the original role, its completion and the physical records supplying its warrant. The same-event, same-clock,

same-standard and same-source requirements are the inherited conditions that make each derived relation a statement about the claimed physical system.

1.1 Necessary foundation and physical specialization

The proof has a definite hierarchy. Kernel governance is incurred by the existence of non-degenerate determinate physical content. Source-role theorems then refine what such content must support in the specified physical comparison classes. Realization conditions identify the particular propagation, frame, device and response regime under study. Calibration fixes its numerical representatives. The Anchor–Tensor–Skin decomposition makes these roles precise [4, Thm. 10.1].

A finite propagation bound, for example, is a substantive relation between determinate displacements and durations. Its content supplies a bound; its physical meaning already requires fixed endpoints, qualified comparisons and retained occurrences. The later physical conditions in this paper are consequently refinements of kernel-governed physical content. Calling them additional conditions of the SR theorem does not place their objects or their warrant outside the kernel.

Necessity applies at every level of this development. The propagation and symmetry constructions, the tests selecting a physical branch, and the further laws obtained from them are themselves determinate physical content. Each necessarily incurs the same kernel in its own right, as Proposition 2.4 makes explicit. The proof then traces the substantive dependencies among those constructions. On the completed transport and comparison target specified below, those dependencies force the Lorentzian form.

The relativistic results also have a receiving role in the corpus. Section 15.4 states how the Wheeler–DeWitt, symmetry and Standard Model carrier constructions receive the proved geometry, clock and source-response laws on their identified domains. The existing coupled construction is identified in Section 15.3; its detailed realization of a further laboratory is a receiving application, while the joint flat occupation asserted in this paper is complete.

1.2 Results and proof order

The dependency is

$$\begin{aligned} &\text{determinate physical objecthood} \implies \text{necessary kernel work,} \\ &\text{standing on the stated physical branches} \implies \text{quotient and comparison construction,} \\ &\text{homogeneous comparison composition} \implies \text{affine displacement receiver,} \\ &\quad \text{complete physical context} \implies \text{faithful event/operation receipt,} \\ &\quad \text{isotropic finite propagation} \implies \text{round causal cone,} \\ &\text{frames and common rest calibration} \implies \text{Lorentz law,} \\ &\quad \text{clock, endpoint and impulse uses} \implies \text{relativistic effects.} \end{aligned}$$

Every construction, selector, map, object and relation on the later lines independently incurs the first dependency by its own claimed determinacy. Each arrow is read with its displayed source domain and physical branch conditions. Their conclusions are used at the following stage through the original records and operation graphs, with the exact source maps retained. These are orders of presupposition and realization. A later successful metric calculation can establish consequences of an already standing physical target; it cannot confer that target’s original identity or retrospectively replace its conditions.

Theorem 2.1 states the unavoidable foundation. Corollary 2.2 applies it to every physical premise used in the derivation. Theorem 15.2 joins it to the cone, frame, measurement and inertia results. The constructed common-source continuation then situates SR, Einstein dynamics and Schrödinger dynamics as compatible realizations whose physical terms already incur the same kernel.

2 The necessary kernel and its physical realization

2.1 Objecthood incurs the kernel

The term non-degenerate determinate physical domain means a nonempty realized fragment containing at least one successful identity-bearing assessment or use, with its actual relata, physical relations and original occurrences fixed. Appendix A states the original-incidence commitments and the distinction between successful identification and successful practical use. A failed measurement or unsupported use can itself be a determinate incidence; failure of that use does not abolish its objecthood. The source necessity argument concerns actual qualification and incidence, including cases in which no observer can decide or record them [1, Prop. 2.4 and Sec. 2.3].

Theorem 2.1 (Necessary kernel of determinate physics). *Every non-degenerate determinate physical object, state, carrier, record, endpoint or incidence necessarily instantiates the functional work of K. No further assumption that the domain accepts or adopts AASC is required. Genuinely removing any one of the four functions, including its implicit or distributed implementation, loses part of the claimed determinate physical construction.*

Proof. Fix an actual non-degenerate determinate physical incidence, with its original relata and conditions. Its identification distinguishes those relata from substituted ones and permits their reidentification under a faithful presentation. This is the reference work.

The incidence also distinguishes what physically obtains under those conditions from what would obtain only of another arrangement, state or occurrence. A proposed continuation, comparison or use must answer to that distinction. The actual qualification it thereby retains is the standing work; a name or an observer’s endorsement cannot supply it.

Preserving that qualification through construction and combination requires the proposed use to remain accountable to the identified bearer and conditions. A changed support, source or operand cannot silently inherit the qualification of the original arrangement. This joint preservation is the admissibility work.

Finally, the original occurrence remains the occurrence it was under its original conditions. New data may improve knowledge of it; an intervention may produce a later occurrence. Neither operation makes a substituted occurrence the original one. Retention of this distinction is the irreversibility work. It applies to the physical incidence whether or not a record is made.

These are the objecthood commitments reconstructed in the kernel source’s Proposition 2.4 and Theorem 3.1. They have been identified from the claimed physical incidence, without beginning with a predicate asserting that the kernel applies. For functional minimality, suppose one function had genuinely been removed while the same determinate physical construction remained. Applying the preceding reconstruction to that construction recovers the supposedly removed work. Either an implicit implementation still performs it, or the original determination has been lost. The former contradicts functional deletion and the latter contradicts retention of the target, as in the source’s Theorem 3.6 [1]. $\square$

This is a constitutive necessity argument about determinate physical incidence. Its starting point includes that physical significance; it is not a set-theoretic theorem that an uninterpreted element of a set possesses a rich physical structure. The source establishes the necessary functions, while the physical domain fixes the substantive relations those functions preserve. The two claims have different logical tasks.

Irreversibility here concerns original attribution and fixation. A reversible dynamical law, an inverse Lorentz map or a time-reversed process can have its own determinate physical realization. Such a realization cannot change which occurrence an earlier claim concerned. Thermodynamic irreversibility is a further physical matter [1, Lem. 3.3].

Corollary 2.2 (Necessary kernel of physical premises and selectors). *Every determinate physical instance of a premise used in this paper already incurs K . This includes propagation bounds, comparisons, calibration records, clock devices, endpoint standards, total sources, impulse histories and projected states. Refining the physical target by such a premise cannot create a kernel-independent physical foundation.*

Proof. A physically instantiated premise concerns its actual identified bearers, relation and occurrence. Apply Theorem 2.1 to these incidences. A calibration is likewise a determinate comparison with a retained standard and conditions, so the same argument applies. Refinement changes or adds substantive physical content, while each new determinate instance again satisfies the theorem. Thus the additional content cannot remove the dependency it incurs through its own physical realization. $\square$

Corollary 2.3 (No functional exemption by physical redescription). *A physical description retaining non-degenerate determinate objects and uses cannot exempt itself from kernel work by changing its vocabulary, primitive equations or representation. A description retaining that work implicitly already instantiates it; one genuinely deleting it loses the corresponding determinate physical content.*

Proof. Theorem 2.1 quantifies over the physical incidence rather than its chosen vocabulary. Changing the representation while retaining that incidence preserves the theorem's premise. Functional minimality then gives the asserted alternatives. $\square$

This consequence applies to conventional metric and quantum descriptions as well as to explicitly AASC presentations. The claim concerns a dependency of their physical content; it neither requires their practitioners to use AASC terminology nor establishes historical priority for recognizing each individual role.

2.2 Necessary kernel at every construction stage

A source construction is itself a determinate bearer of further questions. The same holds for a selector, a calibration procedure, an operation graph, a source map, a composite system and a proposed law with its physical interpretation. Their greater structural richness does not end the necessity established in Theorem 2.1.

Proposition 2.4 (Necessity under determinate construction). *Let $\{C_i\}_{i \in I}$ be any family of constructions claimed as non-degenerate determinate physical content. Each C_i necessarily instantiates K in its own right. In particular, this applies when C_i is a richer realization constructed from earlier sources, or is itself the source, selector, calibration, map or complete receiving structure used at a later stage. Every physically realized incidence asserted between such constructions incurs the same necessity.*

Consequently every stage of a finite sequence of determinate physical constructions, and every determinate composite or completion actually obtained from it, necessarily incurs K. No extra applicability condition is introduced at a later stage. The functional implementation may change, be implicit or be distributed, while the four necessary functions remain.

Proof. Fix i . The claim that C_i is non-degenerate and determinate supplies the subject of Theorem 2.1. Apply that theorem to its own identity-bearing incidences. This conclusion does not depend on whether C_i was primitive in the description or constructed from an earlier C_j . Apply the same argument to each asserted physical relation, including the actual application of a selector or source map. For a finite construction sequence this reasoning applies at every stage. If a composite or completed target is itself claimed determinate, apply the theorem directly to that target as well. Genuine deletion of one of the functions contradicts its claimed determinacy by the same functional minimality argument. Merely changing where the work is implemented does not delete it. $\square$

This proposition is the repeated application of one universal necessity result. It requires no distinct kernel at each level and introduces no new hierarchy of physical primitives. It also makes no inference from finite-stage occupation to occupation or regularity of an unconstructed limit. A proposed completed transport target must have its own stated construction and records. Once it is non-degenerate and determinate, it incurs the kernel directly.

Governance and standing for a further use. A false description, an unsuccessful calibration and a failed selection can each be determinate occurrences. Their own kernel governance retains what happened and the applicable failure. Positive standing for their use as a description, calibration or selection of another source still requires the original source conditions. Thus necessary governance of a map does not certify that it preserves another bearer's answers. Lemmas 4.2–4.3 and Theorem 4.8 supply the exact preservation tests used here. Those tests, their applications and their richer receiving constructions in turn fall under Proposition 2.4.

Physical selectors within the construction. An additional physical condition specifies further content whose own determinate realization necessarily incurs the kernel. Its selection rule, its actual truth or failure, and the construction it qualifies all answer to their original identities and conditions. The further proof question is the precise dependence between these constructions: which selector follows when a previously specified role is completed, which source records establish it, and which numerical law then follows. The role-completion lemmas below address these implications without making kernel governance an optional premise. Their jointly occupied flat target is supplied by Theorem 15.10.

2.3 From necessary objecthood to necessary preservation

A false or unfaithful description can itself remain a determinate, kernel-governed object. Its governance does not confer positive standing as a description of another bearer. Once a physical source has incurred the kernel, its faithful descriptions must preserve the reference and qualification already belonging to that source. They cannot decide afterward which occurrence a successful formula should count as having described. Their obligation is therefore inherited from physical objecthood.

For a receiving map r and an original complete physical readout a , the obstruction is exact:

$$r(x) = r(y) \quad \text{and} \quad a(x) \neq a(y) \quad \implies \quad a \text{ cannot be a well-defined readout of } r(x).$$

If an operation is defined on x and undefined on y , the same encoded-input collision also prevents exact operation descent. Section 4 proves the necessary and sufficient fibre conditions, with original failures and joint inputs retained.

These criteria give the dependency mathematical force in the SR construction. A proposed frame comparison that changes the event, substitutes a clock record or drops an applicable failure cannot retain standing as a description of the unchanged physical question, even if its displayed coordinates satisfy a Lorentz identity. Conversely, a faithful coordinate change can alter coordinate numbers because the coordinate question includes the frame. The original physical comparison and the full transformation relation are what must be preserved.

2.4 Kernel necessity and same-query conservation

For a bearer b and a complete original query q , write $J(b, q, v)$ for independently warranted result incidence. Determinacy means

$$J(b, q, v) \wedge J(b, q, w) \implies v = w.$$

The kernel's representation-conservation theorem applies this fact to independently warranted presentations of the same bearer and query [1, Thms. 6.2, 6.9]. It does not say that all coordinate numbers remain unchanged. A coordinate-time question includes its reference frame and synchronization rule. Two frames may assign different numbers while retaining the same events and preserving the complete transformation relation.

This observation prevents an immediate error in an AASC treatment of relativity: same-event reference cannot be used to demand equal coordinate-time answers to different frame-indexed queries. The invariant task is the joint family of event identifications, frame roles, calibration relations, and lawful readouts. Its mathematical representation will contain covariant quantities as well as invariant scalar answers.

2.5 Structural gravity

The structural-gravity theorem is imported within its regular AER/AER-P envelope. Its source conditions include the regular Hamiltonian standing structure and the declared freedom of physically equivalent constraint presentations. Its conclusion provides a coisotropic standing surface, characteristic redescription directions, local first-class constraints, and a multiplier Hamiltonian description [6, Thm. 8.2 and Rem. 8.4].

More explicitly, its AER interface is a regular Hamiltonian observable-evaluation target: a symplectic carrier (S, ω) , a smooth standing surface C , a quotient submersion $\pi : C \rightarrow Q$, and physical observables pulled back from Q . Writing $D = \ker d\pi$ and $\omega_C = \omega|_{TC}$, it requires extension-independent Poisson descent, observable differentials spanning $\text{Ann } D$, and separation of every nonzero class in $TC/\ker \omega_C$. The source theorem then identifies

$$D = \ker \omega_C = (TC)^{\perp_{\omega}} \subseteq TC.$$

The additional AER-P condition requires presentation-parametrized Hamiltonian evolution to induce the same reduced continuation, up to inert reduced differences. These are the explicit regularity and observable-completeness conditions of the imported result, rather than consequences of the four kernel names alone [6, Defs. 5.1, 5.12, Axiom 7.7].

For the present paper this result fixes how the gravity role enters: physical comparison and localization are governed upstream, and lawful representations must respect the retained characteristic quotient. A frame change in the SR target is not automatically one of those

characteristic flows. It must induce a well-defined operation on the quotient observables. The structural-gravity theorem itself does not fix Lorentzian signature. The passage to the cone below has its own witnessed propagation content.

2.6 The distributed persistence bearer

CSP–R uses the partial distributed assignment

$$\Phi_D : \text{dom } \Phi_D \subseteq \text{LCD}_D \longrightarrow \text{CarProf}_D, \quad \text{CarProf}_D(U) = (B_U, S_U, I_U, R_U, P_U, Q_U, \partial U, T_U).$$

These entries retain support, standing, identity memory, restoration, perturbation burden, comparison, boundary, and transport. Only participating local domains are included. The operational source also retains actual configurations, restrictions, overlap agreements, and witnessed actions; the displayed tuple alone is not assumed to encode every possible operation [14, Defs. 1.1, 2.1].

Premetric priority here has a precise meaning. Locality, comparison, and persistence already occur, while a numerical spacetime metric has not yet been supplied. A local continuation domain is not simply renamed a spacetime point. The construction of an event representation must preserve independently identified coincidences and continuation records. CSP–R proves exact response reconstruction on its structural target and a compatible physical reconstruction on its occupied target [14, Thm. 3.5, Prop. 3.6]. Extending the original query family to SR experiments therefore requires the explicit receiving construction below.

2.7 Transport, ordering, and inertia

R4 constructs comparison content from overlap and transport witnesses. R5 supplies admitted path mediation and bounded perturbation transport. Its distinction between structural transport and later physical speed is retained: a numerical speed requires displacement, duration, units, and a propagation law [8, 9]. R11 supplies irreversible ordering and causal dependence of retained continuation; it does not turn that ordering into a universal simultaneity relation [13].

R9 defines the inertia role by restricting the mass-response profile to identity-preserving changes of transport state [12, Defs. 3.5–3.7, Thm. 7.1]. Its values are comparison content. The numerical coefficient inferred from an impulse experiment is constructed later, with the same persistence class and actual exchange records. No mass value, momentum formula, or mass shell is inserted into this premetric role.

2.8 The physical interpretation of a metric

Metricity as Licensed Description allows a metric to articulate already standing-bearing physical comparisons within its licensed domain [15, Thm. 6.2]. In this paper the quadratic form is first constructed from the propagation cone and fixed scale. It becomes a physical interval descriptor only through the event, clock, and ruler readouts that support its use. Distinct local licenses do not by themselves identify calibrations across all charts.

Null Propagation as the Causal-Limit Invariant is used after this construction. Its theorem assumes an admissible relativistic metric model and vacuum-null propagation [16, Thm. 1]. It identifies the physical invariant in that regime, while the value assigned to c remains unit dependent. The paper’s first cone is supplied by witnessed reachability and its stated calibration.

3 From kernel necessity to the physical source of relativity

The kernel becomes load-bearing through a sequence of construction obligations. A physical description must first preserve its original objects and uses; a regular Hamiltonian realization must then realize that obligation through its characteristic quotient; a persistence realization must carry actual comparison and transport; and a metric must describe those established relations. This section supplies the arguments at these interfaces. The Hamiltonian and persistence branches concern the same source only when actual joint records warrant that identification. Their mathematical carriers are not identified by notation.

3.1 The inherited quotient obligation

Let X be the actually occupied presentation set of a fixed physical source. Fix its original same-source equivalence $\sim$, and fix a family of totalized physical queries a_j , including applicability, original reference and distinguished failures. These are independently specified physical data: their values are not chosen to fit a proposed receiving description. A complete physical description at this target must retain every a_j and every originally admitted continuation.

Proposition 3.1 (Objecthood forces faithful physical receipt). *Every physically faithful description $r : X \rightarrow Y$ of this unchanged target satisfies*

$$\ker r \subseteq \bigcap_j \ker a_j. \quad (\text{U1})$$

If it identifies precisely the original redescription classes, all a_j must be constant on $\sim$. For an original partial operation f , exact reception also requires its domain to be saturated under the input identifications and the output class to be independent of the chosen input representatives.

Proof. A physical answer concerns the identified bearer and original question, by Theorem 2.1. If $r(x) = r(y)$ but $a_j(x) \neq a_j(y)$, a single receiving input would have to give two different answers to that same question. Giving only one loses an original answer; changing the question or its reference changes the target. Thus $a_j = \bar{a}_j r$ is necessary and implies (U1). Conversely that inclusion defines $\bar{a}_j(r(x)) = a_j(x)$ uniquely on the attained image. If f is defined at one representative and not another, its receiving domain cannot reflect original applicability. If it has two different receiving output classes, the receiving operation is not a function. Domain saturation and output constancy suffice by the same representative formula. The argument retains failures as answers and applies to joint input records rather than replacing their original incidence relation by a Cartesian product. $\square$

This proves an obligation incurred by the source's physical determination. It does not assert that every kernel-governed description meets the obligation: an erroneous description is itself determinate. The distinction between objecthood and faithful description is what gives the test its force [1, 2]. The regular Hamiltonian branch now gives a nontrivial geometric consequence of satisfying it.

3.2 The forced architecture of regular Hamiltonian comparison

Work locally on a finite-dimensional regular stratum. The physical entrance data are a symplectic carrier (S, ω) , an embedded standing surface $C \subset S$, and an attained quotient submersion $\pi : C \rightarrow Q$ with locally connected fibres. Put

$$D = \ker d\pi, \quad \omega_C = \iota^* \omega, \quad \mathcal{O} = \{\tilde{f} \circ \pi : \tilde{f} \text{ an admitted smooth physical observable}\}.$$

Three requirements specify complete Hamiltonian observable evaluation at this target [6, Defs. 5.1, 5.12; Axioms 5.8–5.10]:

- (H1) *Extension-independent evaluation.* For any $f, g \in \mathcal{O}$ and any smooth ambient extensions F, G , the restriction $\{F, G\}|_C$ is independent of both extensions and descends to the same quotient observable. The zero observable belongs to $\mathcal{O}$ and has zero bracket.
- (H2) *Quotient separation.* The observables separate distinct quotient classes, and $\{df_x : f \in \mathcal{O}\}$ spans $\text{Ann}(D_x) \subset T_x^*C$ at each x .
- (H3) *Complete tangent detection.* If $v \in T_xC$ lies outside $\ker \omega_{C,x}$, some $f \in \mathcal{O}$ satisfies $df_x(v) \neq 0$.

These clauses fix the Hamiltonian branch of the physical question. H1 implements independence from the ambient representative used to evaluate a physical bracket. H2 says the declared quotient retains the detectable distinctions. H3 excludes a nonzero presymplectic direction hidden from every admitted observable. The kernel requires accountability for the target; it does not introduce the symplectic carrier or assert completeness for a partially observed system.

Theorem 3.2 (Forced characteristic quotient). *Under H1–H3,*

$$(TC)^{\perp\omega} = D = \ker \omega_C \subseteq TC. \quad (\text{U2})$$

Consequently C is coisotropic, its characteristic leaves are the local redescription fibres, and there is a unique local symplectic form ω_Q satisfying $\pi^\omega_Q = \omega_C$. Its Poisson bracket is the descended physical bracket.*

Proof. Use $\iota_{X_F}\omega = dF$ and $\{F, G\} = dG(X_F)$ consistently. Fix $x \in C$ and $\xi \in (T_xC)^{\perp\omega}$. The covector $\omega_x(\xi, \cdot)$ annihilates T_xC , so regular defining coordinates give a function H with $H|_C = 0$ and $dH_x = \omega_x(\xi, \cdot)$. Hence $X_H(x) = \xi$.

First establish tangency. For every K vanishing on C , both H and K extend the zero physical observable. H1 gives $\{H, K\}|_C = 0$, so $dK_x(\xi) = 0$. These differentials span $\text{Ann}(T_xC)$; therefore $\xi \in T_xC$. Next, for every $f \in \mathcal{O}$ and any ambient extension F , H1 gives $dF_x(\xi) = 0$. Tangency now makes this $df_x(\xi) = 0$. By H2 and finite-dimensional double-annihilator duality, $\xi \in D_x$. Thus $(TC)^{\perp\omega} \subseteq D \subseteq TC$, proving coisotropy and $\ker \omega_C \subseteq D$.

For the reverse inclusion, $v \in D_x$ annihilates every df_x because f is pulled back from Q . H3 then forces $v \in \ker \omega_{C,x}$. This proves (U2). The vertical distribution of a submersion is involutive: brackets of vector fields related to zero are again related to zero. Its local leaves are the connected fibre pieces.

The closed form ω_C annihilates vertical vectors. For each vertical field Z , Cartan's identity gives $\mathcal{L}_Z\omega_C = d\iota_Z\omega_C + \iota_Zd\omega_C = 0$. It is therefore basic on a local product chart with connected fibres and descends uniquely. If a tangent vector of Q pairs to zero with every other one, any lift belongs to $\ker \omega_C = D$, so that vector is zero. Thus ω_Q is symplectic.

Finally an extension F of $f \in \mathcal{O}$ has dF zero on $D = (TC)^{\perp\omega}$; hence X_F is tangent to C . Pairing with tangent lifts shows $d\pi(X_F) = X_{\tilde{f}}$, independently of the extension. Therefore $\{F, G\}|_C = \pi^*\{\tilde{f}, \tilde{g}\}_Q$, as claimed. $\square$

Corollary 3.3 (First-class constraints and presentation multipliers). *Choose any regular local defining functions $\phi_1, \dots, \phi_k$ for C . Under Theorem 3.2,*

$$D = \text{span}\{X_{\phi_a}|_C\}_{a=1}^k, \quad \{\phi_a, \phi_b\} = \sum_c C_{ab}^c \phi_c \quad (\text{U3})$$

locally, for smooth coefficients C_{ab}^c . If a presentation family H_α has flows preserving C and induces the same reduced Hamiltonian vector field, then locally

$$H_\alpha = \hat{h} + \sum_a u_\alpha^a \phi_a + c_\alpha, \quad (\text{U4})$$

where h is one reduced Hamiltonian, $\hat{h}$ is one chosen extension of π^*h , and c_α is locally constant. With zero reduced evolution the constant-normalized Hamiltonians are pure multiplier combinations.

Proof. The $d\phi_a$ form a conormal basis. Symplectic duality and (U2) therefore give the spanning assertion. Each X_{ϕ_a} is tangent to C , so $d\phi_b(X_{\phi_a}) = 0$ on C . To exhibit the coefficients, use local coordinates (y, z) with $z_a = \phi_a$. Every smooth F vanishing at $z = 0$ satisfies

$$F(y, z) = \sum_a z_a \int_0^1 (\partial_{z_a} F)(y, tz) dt.$$

Apply this formula to each bracket to obtain (U3).

Since X_{H_α} is tangent to C , for $v \in D$ one has $d(H_\alpha|_C)(v) = \omega_C(X_{H_\alpha}, v) = 0$. Connected local fibres give $H_\alpha|_C = \pi^*h_\alpha$. The descended vector fields coincide, so nondegeneracy of ω_Q gives $d(h_\alpha - h) = 0$. Subtract the local constant c_α , choose the single extension $\hat{h}$, and apply the same displayed division formula to $H_\alpha - c_\alpha - \hat{h}$. This proves (U4). In the zero-field case $dh = 0$, allowing its constant to be absorbed as well. $\square$

The proof recovers the source's regular AER constraint architecture directly [6, Thms. 6.3, 7.5, 7.8, 8.2]; no constraint basis, characteristic foliation or multiplier law was inserted into H1–H3. An arbitrary smooth defining basis suffices, so the present proof needs no special coisotropic Darboux chart. The qualifier local is essential for quotient charts, defining functions and the Hamiltonian constants.

Lemma 3.4 (Attaching the physical records to the forced quotient). *On a local connected-fibre quotient chart of Theorem 3.2, a smooth real-valued physical record $a : C \rightarrow \mathbb{R}$ satisfies the equivalent conditions*

$$a = \bar{a} \circ \pi \quad \Longleftrightarrow \quad da(D) = 0 \quad \Longleftrightarrow \quad \{\phi_b, A\}|_C = 0 \quad (1 \leq b \leq k), \quad (\text{U5})$$

where A is any smooth ambient extension of a . Discrete reference, applicability and failure records descend precisely when they are constant on complete local fibres; a differential test is not substituted for that requirement. An original partial operation $f : C_0 \rightharpoonup C_1$ between such source sorts descends with exact applicability if and only if $\text{dom } f$ is saturated under π_0 and $\pi_1 f$ is constant on admitted π_0 -fibres.

Proof. The first equivalence is the local submersion theorem and connected-fibre constancy: in coordinates (q, y) with $\pi(q, y) = q$, the condition says $\partial_y a = 0$. Corollary 3.3 spans D by X_{ϕ_b} , so the second equivalence follows from $\{\phi_b, A\}|_C = da(X_{\phi_b}|_C)$. This value is extension-independent because X_{ϕ_b} is tangent to C . For an arbitrary set-valued record, define its value on $\pi(x)$ from its value on x ; this works exactly under fibre constancy. Finally define

$$\text{dom } \bar{f} = \pi_0(\text{dom } f), \quad \bar{f}(\pi_0(x)) = \pi_1(f(x)).$$

Saturation proves equivalence of the original and receiving domains, while output constancy proves well-definedness. Conversely an exactly descended partial map has both properties. Joint input sorts retain their complete incidence records in the same argument; genuinely relational primitives retain their original witnessed graph sort. $\square$

For the common-source realization below, these tests are applied to its actual event identifications, transport records, clock uses and response protocols. They are certificates that those records concern the forced physical quotient, rather than additional formal quantities on its ambient carrier. They give a direct testable connection between the constraint proof and the SR target.

The resulting gravity role constrains every later physical localization or observable: its source dependence must descend through π . A frame comparison is a physical redescription of events only after it is defined on these quotient objects. It need not be a characteristic motion in C . The Einstein construction receives this architecture in its separate faithful Lorentzian metric branch and then fixes its first-curvature response; its projection conditions are stated in Appendix B [21, Defs. 14.2, 20.3; Thm. 30.1]. The present SR cone construction supplies its own numerical geometry from transport.

3.3 Constructing premetric comparison from actual witnesses

The persistence branch begins with actually participating local continuation domains U . Their adjacency expresses admitted boundary-compatible continuation. Stable overlap means the same shared trace remains reconstructible in both domains. Transport is a witnessed partial operation retaining boundary trace, identity, restoration status and original standing. These are the physical entrance conditions of R4, rather than metric balls or coordinate distances [8, Defs. 3.2–3.8].

For each ordered pair (U, V) let $W(U, V)$ be the set of actual admitted comparison witnesses. Include source endpoints and all original contexts in the record $b(w)$: a context may continue or restore a witness, compare its overlap, or test its original status. Where composition is admitted, include its domain tests and its output tests after composition, with fixed sibling operands and shared-input contexts that vary repeated occurrences together. Equality of b then defines an equivalence $\equiv$ of witnesses. Write

$$\mathcal{C}_D = \coprod_{(U,V)} W(U, V)/\equiv, \quad \text{Cmp}_D(U, V) = \{[w] : w \in W(U, V)\}. \quad (\text{U6})$$

The pair is retained in the coproduct. Distinct source domains are never identified merely because a numerical display matches.

Proposition 3.5 (Witnessed comparison carrier). *The carrier (U6) retains every original comparison, boundary and transport-context answer. Original operations descend on their admitted incidence domains; they retain identities and composition wherever those laws hold at the source. Every complete receiving comparison description maps uniquely onto this attained carrier at its response quotient. The pair comparison is single-valued exactly when $\text{Cmp}_D(U, V)$ is a singleton; it is unavailable when that set is empty.*

Proof. Set the answer at $[w]$ equal to its original answer at w . Record equality makes this well-defined. For a unary operation a , equal records include equal activity tests and equal answers to every context that applies a before testing its output. Thus definedness agrees and $[a(w)] = [a(w')]$. For several inputs, replace representatives one at a time in the fixed-sibling contexts. Their domain tests preserve the actual incidence condition at every replacement, and the output class remains fixed. No operation is assigned to unoccupied or unlicensed tuples. Source identity and composition equations survive because the quotient maps preserve the involved operations.

If another attained description $r(w)$ retains all the answers, $r(w) = r(w')$ implies $b(w) = b(w')$. The map $r(w) \mapsto [w]$ is therefore well-defined and unique. At its complete response

quotient it is bijective. Finally a single value depending on (U, V) alone exists precisely when all its admitted witnesses have one class; choosing one class out of several would discard an original distinction. $\square$

This is the carrier-induction argument underlying R4’s proto-metric role, including its explicit relational case [8, Defs. 3.9–3.10; Secs. 7–10]. It constructs stable comparative content. A numerical cone requires the further addition, scaling, completion and isotropy of transport in Section 5; those conditions will act on the witnessed comparison structure just constructed. Structural bounded transport controls admitted routes and burdens before a speed unit exists [9]. It does not already assert a numerical upper speed.

3.4 A common operative persistence bearer

An occupied distributed persistence realization supplies local profiles

$$\Phi_D(U) = (B_U, S_U, I_U, R_U, P_U, Q_U, \partial U, T_U)$$

and actual configurations, restriction maps, overlap witnesses, transport and restoration graphs. The entries retain support, standing, identity memory, restoration, perturbation burden, comparison, boundary and transport. Profiles are assigned only on participating domains. Bounded continuation and comparison make these assignments carriers of actual local persistence rather than unrelated point labels [10, 11]; CSP–R retains their original operations in its operative source $\mathcal{B}_D^{\text{op}}$ [14, Prop. 1.2; Def. 2.1].

For precision, a distributed state is an actual family of local configurations together with its original restriction and overlap witnesses. Compatible formal tuples are not all assumed occupied. The source provides a nonempty set Z of these actual families and their original partial operations. Its response record $R(z)$ contains the original standing, identity, comparison, perturbation and transport tests, closed under their admitted contexts. A graph-valued primitive remains a witnessed graph sort with its endpoint tests; it is not replaced by an arbitrarily chosen functional branch.

Theorem 3.6 (Constructed persistence target and exact loss test). *The quotient $\mathcal{P}_D = Z / \ker R$ is an occupied complete response target for the operative persistence source. It preserves the original partial operations, joint incidences and failures. Any further identification of distinct classes loses an original contextual answer. A proposed smaller profile $b : Z \rightarrow B$ supports recovery of this target exactly when*

$$\ker b \subseteq \ker R. \tag{U7}$$

For any enlarged independently warranted query family $R' = (R, R_{\text{new}})$ closed under its own admitted contexts, the attained quotient $\mathcal{P}'_D = Z / \ker R'$ has a unique surjection onto $\mathcal{P}_D$ preserving all old answers and operations.

Proof. Each retained answer is constant on a quotient class by construction. Context closure gives exact operation descent by the proof of Proposition 3.5, including domains, fixed siblings and graph sorts. If $[z] \neq [z']$, the records differ at some original context, proving the loss claim. A recovery map on $\text{im } b$ must satisfy $\bar{R}(b(z)) = R(z)$, so it exists if and only if (U7) holds; the formula proves both sufficiency and uniqueness. The enlarged equivalence is finer, hence $[z]_{R'} \mapsto [z]_R$ is well-defined, onto and preserves the retained structure. Its nonemptiness follows from $Z \neq \emptyset$. If new witnesses require restricting Z , occupation must instead be certified on that restricted domain. $\square$

This proves the response construction used by CSP–R’s local realization and reconstruction arguments [14, Prop. 2.2, Lem. 2.3, Thm. 3.5, Thm. 5.4]. Recovering the finer original source class also requires its complete joint source diagrams [14, Prop. 3.6]. A present eight-entry profile is therefore not asserted sufficient for every future response. The theorem gives the exact test of that assertion and the conservative enlargement used when SR clock, endpoint and impulse questions are added.

Premetric inertial response is attached to an identified persistence class. Its transport-facing profile is the restriction of its admitted deformation-response record to identity-preserving transport-state changes [12, Defs. 3.5–3.7; Thm. 7.1]. This identifies the object whose impulse response the mass theorem computes. It introduces neither a scalar mass value nor an energy–momentum formula.

3.5 Metric use on the constructed physical source

To join the two branches, fix actual joint records J with maps $j_H : J \rightarrow C$ and $j_P : J \rightarrow Z$. They certify that the Hamiltonian and persistence data concern the claimed common source. Fix the original relation identifying equivalent joint presentations; require the induced maps to Q and $\mathcal{P}_D$ to respect it. Neither nonemptiness of C and Z separately nor a formal product $C \times Z$ supplies these physical witnesses. The refinement construction of Theorem 4.8 receives precisely such actual joint records and their independently warranted added queries.

Proposition 3.7 (Licensed metric interpretation and its obstruction). *A metric descriptor can answer an unchanged physical comparison of this source only on its original qualified domain and with its original relata, continuation and calibration retained. Its totalized physical answers must satisfy Proposition 3.1 on the joint source. In particular, an event, clock or source readout defined on C must be constant on complete π -fibres to become a readout on Q . For a smooth real-valued readout, $da(D) = 0$ is equivalent to this condition locally on connected fibres. A later successful metric calculation cannot make a failed original query into a successful query about the same original occurrence.*

Proof. An unfixed relatum does not specify which source comparison is being answered. A changed calibration or applicability condition specifies a different query. Once those data are fixed, exact reception is Proposition 3.1. Fibre constancy is necessary and sufficient for a function on the attained quotient; smooth local descent follows in submersion coordinates. Differentiating a descended function gives $da(D) = 0$, and this differential condition conversely gives constancy on connected local fibres. Original failure is itself a retained answer; changing it while claiming the same original query violates (U1). $\square$

This is the metric-licensing requirement at the actual interface used here. The full licensing theorem excludes making the descriptor a new boundary-standing authority in a fixed background [15, Lem. 6.1, Thm. 6.2]. Its Corollary 6.3 explicitly leaves metric generation to the physical construction. Here the later transport theorem constructs the quadratic form; this proposition determines when its readouts describe the established physical source. It also prevents using a metric result to choose retrospectively which source, event or clock the calculation concerned.

3.6 What enters the SR proof

The downstream argument consumes the preceding constructions in four specific ways. The event and operation receipt uses Proposition 3.1 and the complete physical profile of Theorem 3.6. In the regular Hamiltonian branch, Lemma 3.4 attaches that receipt to the forced

quotient of Theorem 3.2. Witnessed reachability is obtained from the transport operations of Proposition 3.5, before the additional homogeneous and isotropic specialization constructs its numerical cone. Clock, endpoint and impulse comparisons retain the persistence identity and original response domains, and Proposition 3.7 fixes the physical meaning of their resulting metric formulas.

Thus deleting the upstream structure has a precise consequence. Collapsing a retained distinction destroys factorization; changing an operation domain destroys continuation receipt; choosing one unwarranted comparison witness changes the target; dropping persistence reference detaches the measured response from its body; and bypassing the characteristic quotient makes an AER observable depend on an unphysical representative. The numerical SR theorems then apply to the faithfully constructed source under their displayed realization conditions. The complete implication is kernel necessity, inherited preservation, forced source architecture within each physical branch, exact reception of the witnessed joint source, and finally the Lorentzian laws. Four-dimensionality, strict completed-cone positivity, actual isotropy and occupied devices remain the specified physical content selecting that last realization.

4 Complete physical queries and faithful descent

Fix one non-degenerate determinate experimental domain D before choosing coordinates. Its source sorts include occurrence presentations, persistent occupants, witnessed paths, clock records, endpoint records, and, where needed, source and exchange records. The original query family includes coincidence and reference, participation domains, continuation composition, propagation/reachability distinctions, tick and endpoint incidence, readout calibration, and first failures. The full family is fixed independently of which presentations later succeed.

Definition 4.1 (Totalized original readout). A partial readout $a_q : X \rightarrow O_q$ is represented by

$$\hat{a}_q : X \longrightarrow O_q \sqcup \text{Fail}_q.$$

Distinct failure types are retained whenever the original task distinguishes them. The failure space is not automatically a single missing-value symbol. Complete reference and query conditions are included in the indexing of q .

Lemma 4.2 (Exact readout factorization). *Let $r : X \rightarrow Q$ be a proposed attained interface. All original totalized readouts factor uniquely through r if and only if*

$$r(x) = r(y) \implies \hat{a}_q(x) = \hat{a}_q(y) \quad \text{for every } q.$$

Proof. A factorization makes the implication immediate. Conversely define $\bar{a}_q(r(x)) = \hat{a}_q(x)$. The condition proves independence of the representative and surjectivity proves uniqueness. This is the AMetric lossless-interface criterion specialized to the full SR query family [2, Thm. 8.6]. $\square$

Lemma 4.3 (Exact partial-operation descent). *Let $f : \text{dom } f \subseteq X \rightarrow Y$ and let $r_X : X \rightarrow Q_X$, $r_Y : Y \rightarrow Q_Y$. There is a unique descended partial map $\bar{f} : Q_X \rightarrow Q_Y$ with*

$$x \in \text{dom } f \iff r_X(x) \in \text{dom } \bar{f}, \quad \bar{f}(r_X(x)) = r_Y(f(x))$$

if and only if domain membership is constant on r_X -fibres and, on that domain, $r_Y f$ is constant on the same fibres.

Proof. Necessity follows from the displayed identities. For sufficiency define $\text{dom } \bar{f} = r_X(\text{dom } f)$ and $\bar{f}(r_X(x)) = r_Y f(x)$. Domain saturation proves the equivalence, output constancy proves well-definedness, and surjectivity proves uniqueness. The proof applies to each sort and to products of input sorts, so shared arguments and joint operation domains are retained [5, Thm. 3.1]. $\square$

Proposition 4.4 (Compositional faithful receipt). *Suppose the original independently specified coincidence relation on occurrence presentations is an equivalence relation and that every required event readout and witnessed operation satisfies Lemmas 4.2–4.3 for its quotient. Then the event quotient and all declared partial continuations exist as faithful receiving objects. Finite composites preserve the original applicability, readouts, and failure records. Every other complete receiving map factors to the complete original profile on its attained image.*

Proof. Form the set of coincidence classes. The two lemmas give all readout and operation maps. If $\bar{g}\bar{f}$ is defined at $[x]$, domain reflection first gives $x \in \text{dom } f$ and then $f(x) \in \text{dom } g$. Its value is $[g(f(x))]$. The converse and equality follow the same identities. Induction gives the finite-composite statement. Retained first-failure queries factor by Lemma 4.2; a later successful composite cannot erase them.

For the final statement let $P(x) = (\hat{a}_q(x))_q$ be the complete profile, including reference coordinates and required continuation queries. If r is complete, equal r -values have equal P -values, so $r(x) \mapsto P(x)$ is a unique map $\text{im } r \rightarrow \text{im } P$. It need not be injective unless r is also minimal. This is the attained-image form of complete-interface uniqueness [3, Thm. 4.5, Cor. 4.10]. $\square$

Remark 4.5 (Identity is independently fixed). Equality of all measured outcomes alone does not establish that two physical events are numerically identical. Coincidence and source lineage must be retained as original reference content. A contextual quotient may compress presentations of one original event; it cannot erase an independently required distinction between two events. Nor does the quotient construction create a physical witness for a previously empty occurrence or device sort.

The hypotheses of Proposition 4.4 are operationally inspectable. A failed readout descent has equal encoded inputs with unequal original answers. A failed partial-operation descent has unequal applicability or unequal retained output at equal inputs. These witnesses prohibit a purported frame change from silently changing the rod, selecting a later clock tick, or discarding an inconvenient failed signal. They are also the conditions under which AER characteristic redescription, persistence comparison, and the SR receiving maps can be composed.

Proposition 4.6 (Construction by complete contexts). *Let the original tests be closed under their licensed operational contexts, including all licensed fixed-sibling single-input contexts and joint contexts for repeated occurrences. Retain exact domain status, shared-input incidence, and first failures. On each raw source sort define*

$$\mathcal{R}(x) = (\text{status}_A(x), \text{out}_A(x))_{A \in \mathcal{A}}, \quad x \sim y \iff \mathcal{R}(x) = \mathcal{R}(y).$$

Then every retained operation descends to the contextual quotient with its exact domain and output. Every complete alternative attained description maps uniquely onto this quotient. That map is an isomorphism precisely when the alternative is also minimal for the same complete original record.

Proof. The retained domain test of an operation f makes its domain constant on contextual fibres. Append any output test A to f . This is an original input context on exactly its licensed composition domain, so equal complete input records give equal domain status and equal

output-test results. Varying A proves equality of complete output records. Lemma 4.3 therefore applies. For several independent inputs, replace their representatives successively in the licensed single-input contexts with the remaining intrinsic inputs fixed. Each replacement preserves joint applicability and the complete output record. Repeated occurrences of one operand are varied together, and the original diagonal and other shared-input incidence predicates are retained. The same argument preserves the original first failure. Whole mechanisms supplied as primitives are treated as primitives rather than replaced by finite approximations.

If r is another complete description, the map $r(x) \mapsto [x]$ is well defined and surjective on its attained image. It is injective exactly when equal complete records imply equal r -values, which is minimality for this target. This constructs the complete contextual interface using the method of CSP–R, while retaining the source scope of the receiving question [14, Prop. 2.2, Lem. 2.3, Thm. 3.5]. It does not identify a coarser response quotient with a finer source class [14, Prop. 3.6]. $\square$

When every retained event context and operation domain descends through the independently fixed coincidence relation, and the retained coincidence tests separate its occurrence classes, the event sort of this construction is exactly the coincidence quotient of Proposition 4.4. Otherwise it remains a response quotient and cannot replace a finer event identity required by the original task. The source occurrence, its endpoint incidences, and any AER characteristic redescription must be received together. Quotienting unrelated marginal records does not establish a shared event.

4.1 Source refinements and the Lorentzian realization class

Section 3 constructs the characteristic source quotient on the AER branch and the premetric comparison and persistence interfaces used here. The quantitative conditions below refine those original records and operations. On that branch the event, clock and source records must first pass the characteristic descent test proved there; their numerical refinements then pass the exact construction below. Their admissibility has a positive criterion: the additional readouts and comparisons must be warranted on actual source records, descend with their original domains, and retain the old questions when the new data are forgotten. Mere kernel governance of an inscription does not establish those facts. Appendix B is part of this source identification; the following construction supplies the mathematical link.

Definition 4.7 (Witnessed source refinement). Let S be the original many-sorted source with complete query profile P_0 and its admitted partial operations. A witnessed refinement consists of:

- (i) a nonempty family R of actual joint physical records, with an independently fixed source map $\pi : R \rightarrow S$ retaining original bearer, occurrence and conditions;
- (ii) additional original questions Q_1 , independently warranted readouts, calibration records and applicability tests on R ;
- (iii) the actual partial joint operations f_R used by the target and their original source operations f_S . Whenever a lift is admitted, $f_S(\pi \mathbf{r})$ is admitted and

$$\pi(f_R(\mathbf{r})) = f_S(\pi \mathbf{r}).$$

Retain the source applicability, shared operands and original failures, together with all licensed fixed-sibling and repeated-input contexts;

- (iv) a declared selection test χ whose truth and domain are readouts of those original records, rather than definitions chosen to make a later equation true.

The complete refinement profile P_1 retains $P_0 \circ \pi$, the additional answers, χ , and their complete contexts and failures. An operation taking a selected input to an unselected output has a retained *selection exit*; that is distinct from failure or undefinedness of the original operation. The full receiver is retained with a marked selected subset and outward exit records; that subset need not be closed under the original operations. No physical witness is created by recording a proposed test.

Theorem 4.8 (Exact source-refinement construction). *For a witnessed source refinement, quotienting R by equality of P_1 constructs its minimal complete attained receiver E . The original source receiver on this attained domain is retained by the unique surjection*

$$\rho : E \longrightarrow P_0(\pi(R)), \quad \rho([r]) = P_0(\pi(r)).$$

Every original and additional readout descends; every registered partial operation descends with its exact domain, output and retained failure. On admitted operation lifts, forgetting the new data commutes with the original source operation. Selection by χ marks a well-defined subset of the full receiver, with outward successful operations retained as selection exits. This subset can be empty; an occupied selected target requires an additional record at which χ is defined and true.

Conversely, a proposed refinement cannot faithfully carry one of these original questions if its encoding identifies records with different answers, different applicability, different retained original-incidence answer or reference, or a different retained selection/failure status. Every complete alternative attained receiver maps uniquely onto E , and is isomorphic to E exactly when it is minimal for the same complete questions.

Proof. Equality of P_1 includes equality of $P_0 \circ \pi$, so ρ is well defined. Each attained old profile has a representative in R , proving surjectivity; the formula fixes the value on every class, proving uniqueness. Each retained readout is a component of the complete profile and hence constant on its equality fibres.

The context closure in Definition 4.7 is exactly that of Proposition 4.6. Apply it to the joint operations on R . It proves saturation of each registered domain and constancy of the retained output, with shared operands varied together. Lemma 4.3 gives exact operation descent. On an admitted physical lift, the original source-projection identity is part of the original joint record; its two sides descend to the same identity after applying ρ .

The selected subset is well defined because χ descends. The complete output status distinguishes an original operation failure from a successful operation whose result leaves that subset. Retaining the full receiver and marking its selected inputs retains this distinction and all unchanged original answers. If only selected outputs are displayed, the outward records must instead appear in an explicit exit-tagged result type. For the converse, an identified-input collision with unequal original answers cannot be the fibre of a well-defined answer function; a domain collision likewise contradicts exact partial descent. These are Lemmas 4.2–4.3. Finally, a complete alternative encodes enough information to determine P_1 , giving its unique attained surjection onto E ; minimality supplies injectivity. $\square$

Proposition 2.4 applies directly to the richer record carrier, its new questions and selection rules, the attained receiver, and the actual applications of the projection. Each is itself determinate physical content. The refinement theorem establishes the additional fact that these constructions preserve the earlier source on its original questions.

When a selector is determined by a completed role. Let P_R be the complete original profile of an independently specified role, and let $\hat{\chi}$ retain a proposed selector's truth, falsity

and original failure status on its actual records. The selector is determined on that role receiver exactly when

$$P_R(r) = P_R(r') \implies \hat{\chi}(r) = \hat{\chi}(r').$$

Lemma 4.2 proves both directions. If this test fails, an attained enrichment may retain the added distinction by Theorem 4.8, but it cannot assign it to the old undifferentiated role class. To prove that completion of a role forces a selector to hold, one must additionally derive the affirmative answer from the independently specified completion properties. The transport, calibration and response lemmas below supply concrete instances of this stronger implication. Every role, enrichment, selector and successful or failed application already incurs Proposition 2.4; the test concerns their exact substantive dependence.

This theorem constructs the license to use the added physical content in the same derivation. It does not prove χ true merely because its expression is meaningful. The actual witness and its warrant precede the quotient. When the old source profile already retains the new answers, they factor through it. Otherwise E is an explicit enrichment over that profile; the new answers are not silently assigned to a coarser original class.

For example, a calibrated propagation comparison asks whether a witnessed displacement and its duration satisfy a supplied bound. It refers to the same endpoint and clock records as the transport source. The refinement retains the numerical values, units and bound test; changing an endpoint or discarding a failed comparison changes its original query. A common rest-clock comparison similarly retains the same tick pair and unit. Active source covariance requires a joint record of an actual state action and its attached response witness; passive re-expression of an unchanged state cannot occupy that role. These are three applications of one exact construction, rather than three independent permissions to introduce metric assertions.

4.2 The source-to-condition dictionary

Table 1 identifies the source role refined by every numbered physical condition. Its final column gives the completed use or sufficient role-completion result, with the independently warranted records required by Definition 4.7. The theorem supplies faithful reception of that content; its existence or truth must be provided by the declared physical target or a cited construction. Thus the dictionary is a proof interface with an exact admission test, not a claim that every proposed specialization is automatically available.

Table 1: Physical conditions as completed uses and witnessed refinements of AASC source roles.

Condition	Constructed source and original role	Completion of the role and its physical warrant
5.9	Characteristic source records and constructed comparison/persistence quotient	Proposition 5.3 constructs effective, nonperiodic comparisons on its complete role; Theorem 5.2 gives affine receipt; Theorem 5.7 constructs its spatial form. These constructions retain original event identity and composition.
5.11	R5 actual mediated paths, boundaries and partial concatenation; R11 order	Completed antisymmetry, weak clock monotonicity and actual rest comparisons give strict positivity and a finite bound by Proposition 5.6. Full boundary isotropy supplies the quadratic form. Retain executed and limiting records separately.
6.3	Original comparison composition; occupied clock/readout role	Lemma 6.2 gives the common duration from ordered interval comparisons and one unit. Actual rest-tick occupancy and common-pair coverage of frame composites remain required.
7.3	Persistence of endpoints; physical-clock and transport roles	Actual repeated boundary signals, reflection and return events, and delays. These populate a device; a cone alone supplies no occupant.
7.5	Attached physical-clock quotient, readout and continuation	Uniform coasting response, same-ray additivity, regularity and rest calibration, all on the same device's certified interval.
11.1	Attached response and Ward construction of Section 10; complete finite composite closure	Invariant response and internal equations provide closed Ward balance. Normal traces, finite charges and side-flux control remain required. Keep every internal and external contribution in its original scope.
11.3	R9 identity-preserving response; attached source; active/passive distinction	Lemma 11.4 derives solution and response transport from a complete invariant prescription. Proposition 11.5 constructs the active action. Retain the stable rest occupant and its internal profile.
11.8	R9 controlled deformation; Module A exchange accounting	Actual pulses to the coasting endpoints, calibrated impulse and unchanged final internal state. Pulse existence is distinct from orbit existence.

The same test applies to the decorated common-source recipient and the free quantum comparison: their added history, Hilbert and calibration questions must be received on the declared original target. Premetric source identification and source-faithful enrichment precede the numerical laws. The following sections prove which laws those refinements permit and exclude.

5 Homogeneous displacement and witnessed propagation

We now specify the physical regime in which the event quotient receives inertial coordinates. The construction begins with additive comparisons of witnessed displacements; the spacetime quadratic form will follow from the propagation family. The dimensional and calibration content of these comparisons is explicit.

5.1 Constructing the affine receiver from displacement operations

The homogeneous event receiver can itself be constructed from its original comparison operations. The hypothesis below concerns the composition of displacements of the event reference system; an inverse comparison does not assert a physically backward signal. This separates the affine carrier from the one-sided propagation witnesses subsequently placed in it.

Definition 5.1 (Regular homogeneous comparison target). On the attained event quotient E , let the admitted homogeneous displacement comparisons form a connected, simply connected abelian real Lie group T of dimension four. Its action on the smooth four-dimensional event receiver is smooth, free and transitive. The original calibrated frame-indexed coordinate-duration difference is a nonzero continuous homomorphism $\vartheta : T \rightarrow (\mathbb{R}, +)$. Every admitted frame comparison is a homeomorphism $F : E \rightarrow E$ conjugating these displacement operations into themselves, with its inverse doing the same. All these operations and duration records are the descended operations of the witnessed source refinement. Simple connectivity and four-dimensionality are conditions of this receiving branch.

Theorem 5.2 (Affine structure from original displacement composition). *On a regular homogeneous comparison target, the Lie algebra $V = \mathfrak{t}$ acts simply transitively on E by $v \cdot e = \exp(v)e$. This makes E an affine torsor over V without selecting an origin. This coordinate-duration difference is a nonzero real-linear functional $\tau = \vartheta \circ \exp$. Each admitted frame comparison is affine with an invertible real-linear displacement action. A vector e_0 with $\tau(e_0) = 1$ gives*

$$V = \mathbb{R}e_0 \oplus \ker \tau, \quad \dim \ker \tau = 3.$$

Neither a spatial norm nor a spacetime quadratic form is needed for this construction.

Proof. Since T is abelian, its exponential map $\exp : \mathfrak{t} \rightarrow T$ is a group homomorphism. Its differential at zero is the identity, so it is a local diffeomorphism. Its image is an open subgroup and hence is all of connected T . Its kernel is discrete. Translation of a sufficiently small neighborhood on which it is injective shows that it is a covering map. As $\mathfrak{t} \simeq \mathbb{R}^4$ and T are simply connected, this covering has one sheet; therefore $\exp$ is a Lie-group isomorphism. The given free transitive action now transfers vector addition to the event receiver. The orbit map at any event has trivial infinitesimal stabilizer; dimensions agree, so it is a bijective local diffeomorphism and therefore a diffeomorphism.

The map τ is continuous and additive. Rational homogeneity followed by continuity proves real linearity. It is nonzero because ϑ is nonzero and $\exp$ is onto, so its kernel has dimension three and the displayed splitting follows. Conjugation by F is an automorphism of the group of displacement maps. Its continuity follows by evaluating at one event and using the orbit homeomorphisms. Transport through $\exp$ gives a continuous additive automorphism A_F of V , hence an invertible real-linear map. The identity $F(v \cdot e) = A_F(v) \cdot F(e)$ is precisely affinity. Different choices of reference event merely translate an affine chart. $\square$

Proposition 5.3 (Effective comparison and unique repetition construct the affine role). *Let a connected abelian Lie group G of actual comparison operations act smoothly and transitively on a connected Hausdorff smooth event receiver E of dimension d . Suppose the independently fixed complete comparison profile P has exactly the distinctions of this action:*

$$P(g) = P(h) \iff g \cdot e = h \cdot e \quad \text{for every } e \in E.$$

Put $K = \{g : g \cdot e = e \text{ for every } e \in E\}$. The attained complete comparison receiver $T = G/K$ is a connected abelian Lie group of dimension d , acting freely and transitively on E . Original continuous duration homomorphisms included in P descend to T .

If the completed comparison records additionally verify

$$t^2 = 1 \implies t = 1 \quad (t \in T),$$

then $\exp : \mathfrak{t} \rightarrow T$ is a Lie-group isomorphism. For $d = 4$, a retained nonzero duration homomorphism and the admitted frame comparisons of Definition 5.1 therefore supply its vector-torsor and affine-receipt conclusions. Freeness and simple connectivity are obtained by this construction.

Proof. Fix $o \in E$. If k fixes o , then for $e = g \cdot o$, $k \cdot e = gk \cdot o = e$ by commutativity. Hence the stabilizer G_o is exactly K . It is closed and normal. The quotient is a connected abelian Lie group; its action is transitive and has trivial stabilizers. The orbit map has constant rank. If its rank were less than d , transitivity would make every point of E a critical value, contrary to Sard's theorem. Thus the quotient orbit map is a bijective local diffeomorphism and hence a diffeomorphism. This also proves $\dim T = d$. Equality of action maps is equality modulo K , so the complete-profile hypothesis identifies this quotient with the attained role receiver. Every included duration query is constant on its fibres and descends, with its homomorphism and continuity properties.

The exponential homomorphism of connected abelian T is a local diffeomorphism. Its image is an open subgroup, hence all of T ; its kernel Γ is closed and discrete. If $\Gamma \neq 0$, choose a nonzero $\gamma \in \Gamma$ of least auxiliary norm. Such a choice exists since bounded subsets of this discrete subgroup are finite. Then $\gamma/2 \notin \Gamma$, but $\exp(\gamma/2)^2 = \exp(\gamma) = 1$, contradicting the stated comparison record. Thus $\Gamma = 0$. The exponential is an isomorphism, and vector addition supplies the torsor. The proof of Theorem 5.2 makes the descended duration linear and the admitted conjugating frame maps affine. $\square$

The first premise requires completeness of the original comparison questions. If an operation fixes every event but changes a retained clock phase, orientation or internal answer, this event-action quotient cannot discard it. The second premise is an operational form of the absence of periodic displacement ambiguity: repeating a comparison twice gives the identity only for the identity comparison. In this Lie class it also yields unique halving. A torus comparison system fails that premise. Smooth homogeneous coverage, abelian composition and the event dimension remain stated properties of the role. The quotient, repetition test, frame maps and their actual uses each incur Proposition 2.4.

The additive functional records coordinate-duration differences in the selected comparison frame. It is not assumed to be proper duration on arbitrary displacements. The construction uses the source's actual composition law. It does not reconstruct a homogeneous action from arbitrary R5 paths. Noncommuting comparisons, periodic displacement groups and curved transport can belong to other legitimate physical targets. In particular, simply connectedness excludes a torus quotient, while commutativity excludes a nonabelian homogeneous group. These are visible physical and topological restrictions. The theorem replaces a separately introduced vector addition with the linear form of the admitted comparison composition; it does not remove that branch selection.

The intrinsic route in Theorem 5.7 can now start from this V , τ and a witnessed completed transport cone. Its rest vector is chosen in that cone's interior with $\tau(e_0) = 1$, an additional condition of the propagation target. The resulting spatial form supplies the spatial-comparison entry of Physical condition 5.9. Thus the regular-comparison route and intrinsic-cone route together construct the affine, spatial and Lorentz forms in their order of dependency.

5.2 The intrinsic propagation target

The primary construction uses the affine receiver just obtained and builds its quadratic comparison from the source's reach records. The following data describe the occupied homogeneous branch of the premetric transport role. They contain no Lorentz matrix, prescribed spatial norm, proper-time formula or mass shell.

Definition 5.4 (Intrinsic completed transport datum). On the regular homogeneous comparison target, let $W \subset V$ consist of the actual retained forward transport increments. Their translated copies, finite concatenations and positive rescalings, or their independently licensed limiting realizations, have original source witnesses and retain exact endpoint and burden checks. The causal-limit query is the closed completed reach

$$\mathcal{C} = \overline{\{\sum_{j=1}^N r_j w_j : N < \infty, r_j \geq 0, w_j \in W\}}.$$

This limiting query is distinguished from exact executed-path reachability. The attained completion has nonempty interior and satisfies $\tau(v) > 0$ for every nonzero $v \in \mathcal{C}$. A rest direction $e \in \text{int } \mathcal{C}$ has $\tau(e) = 1$.

The actual linear rest-comparison subgroup H of the admitted frame group fixes e , preserves τ , the original transport records and $\mathcal{C}$, preserves spatial orientation on $\ker \tau$, and acts transitively on

$$\partial K, \quad K = \{x \in \ker \tau : e + x \in \mathcal{C}\}.$$

All admitted frame comparisons preserve and reflect the same completed-reach query. The stated operations, limiting interpretation and symmetry are independently warranted on the common source. Strict positivity is required on the completed cone, not just on every separately executed path.

The strict clock premise may be received directly or constructed by Proposition 5.6 from completed causal antisymmetry, weak clock monotonicity and the actual rest-comparison action. These are source properties stated without a numerical propagation bound. Finite sums and rescalings make the displayed completion a closed convex cone. Its physical existence and interpretation are carried by the declared limiting-transport query. The next results construct the spatial form and its finite causal boundary from these data. They also isolate the exact finite-bound and isotropy content of the selected branch.

5.3 Constructing spatial comparison from completed transport

A positive definite spatial metric is constructed from the completed source transport and its intrinsic rest-isotropy action. The following construction also identifies the precise completion condition behind a finite speed bound. It leaves the dimension and physical isotropy of the target explicit.

Lemma 5.5 (Positive completed-cone clock and finite comparison bound). *Let V be a finite-dimensional real vector space, let $\mathcal{C} \subset V$ be a nonzero closed cone, and let $\tau \in V^*$ satisfy $\tau(v) > 0$ for every $v \in \mathcal{C} \setminus \{0\}$. Then*

$$B_\tau = \{v \in \mathcal{C} : \tau(v) = 1\}$$

is nonempty and compact. For every auxiliary norm $\|\cdot\|$ there is a finite M_τ such that

$$\|v\| \leq M_\tau \tau(v) \quad (v \in \mathcal{C}).$$

Conversely, if $\tau \geq 0$ on $\mathcal{C}$ and this inequality holds for one norm and one finite M_τ , then τ is strictly positive on $\mathcal{C} \setminus \{0\}$.

Proof. The section is nonempty by positive rescaling of any nonzero cone vector. It is closed. If it were unbounded, choose $v_j \in B_\tau$ with $\|v_j\| \rightarrow \infty$. Compactness of the unit sphere supplies a subsequence $v_j/\|v_j\| \rightarrow u$ with $\|u\| = 1$. Closedness and conicity give $u \in \mathcal{C}$, whereas continuity gives $\tau(u) = \lim_j 1/\|v_j\| = 0$, a contradiction. The maximum of $\|\cdot\|$ on the compact section gives M_τ ; rescale each nonzero v by $\tau(v)$. Conversely the inequality forces $\|v\| = 0$ whenever $\tau(v) = 0$, so only the zero vector can have zero clock value. $\square$

The auxiliary norm is used only to prove compactness and does not become a physical ruler. Strict positivity on the *completed* cone is load bearing. Positivity on each actual finite path does not imply it: the closure of $\{(t, x) : t > 0\}$ contains nonzero vectors with $t = 0$. This lemma identifies strict completed-clock positivity with a finite bound. Proposition 5.6 derives that positivity from completed order and actual rest comparisons; finite path mediation alone does not supply those completion properties.

Proposition 5.6 (Completed order and rest comparisons force clock separation). *Let $\mathcal{C}$ be a closed convex cone with nonempty interior in a finite-dimensional real vector space V . Suppose the original completed causal comparison is antisymmetric, $\mathcal{C} \cap (-\mathcal{C}) = \{0\}$, and its nonzero received clock $\tau \in V^*$ is nonnegative on $\mathcal{C}$. Let an actual linear rest-comparison group H preserve $\mathcal{C}$ and τ and fix $e \in \text{int } \mathcal{C}$. Suppose its common fixed spatial subspace is zero:*

$$(\ker \tau)^H = \{x \in \ker \tau : hx = x \text{ for all } h \in H\} = \{0\}.$$

Then $\tau(v) > 0$ for every nonzero $v \in \mathcal{C}$. Consequently Lemma 5.5 gives a compact unit-clock section and a finite comparison bound in every auxiliary norm. No spatial ruler or numerical speed bound is a premise.

Proof. First $\tau(e) > 0$: if it were zero, the interior property and the nonzero linear functional would supply a nearby cone vector of negative clock value. Rescale e to have $\tau(e) = 1$. Its symmetric order interval

$$\mathcal{O} = \{w : e + w \in \mathcal{C}, e - w \in \mathcal{C}\}$$

is closed, convex and contains a neighborhood of zero. It is bounded. Otherwise choose $w_j \in \mathcal{O}$ with $\|w_j\| \rightarrow \infty$; a convergent subsequence of the normalized vectors has limit u of norm one. Closedness and conicity applied to $(e \pm w_j)/\|w_j\|$ give both $u, -u \in \mathcal{C}$, contradicting antisymmetry. Thus $\mathcal{O}$ is a compact convex neighborhood of zero.

Every $h \in H$ preserves $\mathcal{O}$. In auxiliary linear coordinates its covariance matrix

$$S = \frac{1}{\text{vol}(\mathcal{O})} \int_{\mathcal{O}} ww^\top dw$$

is positive definite. Volume preservation gives $|\det h| = 1$ and $hSh^\top = S$. Hence $b(w) = w^\top S^{-1}w$ is a positive definite H -invariant auxiliary quadratic form.

Suppose $0 \neq x \in \mathcal{C} \cap \ker \tau$. The orbit closure $L = \overline{Hx}$ is compact, lies in $\mathcal{C} \cap \ker \tau$, and has constant positive b -value. Its convex hull is compact in finite dimensions and is H -invariant. The strictly convex function b has a unique minimum y on that hull. Invariance makes y fixed by every h , so $y = 0$ by the fixed-subspace hypothesis. This is impossible: a finite convex combination $0 = \sum_i \lambda_i v_i$ of nonzero $v_i \in L$, with $\lambda_i \geq 0$ and $\sum_i \lambda_i = 1$, would give, for a positive coefficient, $-v_j = \sum_{i \neq j} (\lambda_i/\lambda_j) v_i \in \mathcal{C}$. It would put both v_j and $-v_j$ in the pointed cone. The contradiction proves strict clock positivity, and the preceding lemma gives the bound. $\square$

The completed-order premise concerns the original comparison relation: $y - x \in \mathcal{C}$ and $x - y \in \mathcal{C}$ together force the same event. It is not identified with the kernel's retention of original

occurrence. Weak clock monotonicity passes from the original forward increments to their completed cone by continuity. The actual rest action then supplies the further comparison content needed in the proof. All of these richer determinate constructions directly incur Proposition 2.4.

The premises do distinct work. The Galilean closed halfspace has weak clock monotonicity and full rest rotations, but fails completed antisymmetry. The positive orthant with its first-coordinate clock and trivial rest group is pointed, but has nonzero fixed zero-clock vectors. The cone $t \geq |x_1| + |x_2| + |x_3|$ with signed coordinate permutations has no common fixed spatial vector and finite reach, but its unit-time body is an octahedron. Thus full boundary isotropy remains the source property used by the next theorem to obtain a quadratic spatial form.

Theorem 5.7 (Spatial comparison reconstructed from rest isotropy). *Let $\mathcal{C}$ be a closed convex cone with nonempty interior in a real $(n+1)$ -dimensional vector space V , $n \geq 1$. Let τ satisfy Lemma 5.5, and choose $e \in \text{int } \mathcal{C}$ with $\tau(e) = 1$. Put*

$$X = \ker \tau, \quad K = \{x \in X : e + x \in \mathcal{C}\}.$$

Suppose the actual linear rest-comparison group H fixes e , preserves τ and $\mathcal{C}$, and acts transitively on ∂K . Then there is a unique positive definite quadratic form g_X on X whose closed unit ball is K . In the canonical decomposition $v = \tau(v)e + x$,

$$\mathcal{C} = \{te + x : t \geq 0, g_X(x, x) \leq t^2\}, \quad q(te + x) = t^2 - g_X(x, x)$$

has signature $(1, n)$ and exactly the prescribed future cone. Every element of H preserves g_X .

Proof. Lemma 5.5 makes $K = B_\tau - e$ compact. It is convex, and 0 lies in its interior because e is an interior cone vector. Choose any linear coordinates on X and a Lebesgue measure, solely for the following construction. The matrix

$$S = \frac{1}{\text{vol}(K)} \int_K xx^\top dx$$

is positive definite: for nonzero z , the integral of $(z^\top x)^2$ is positive on the nonempty interior of K . If $h \in H$, then $hK = K$, so comparison of volumes gives $|\det h| = 1$, and change of variables gives $hSh^\top = S$. Consequently $b(x, x) = x^\top S^{-1}x$ is H -invariant.

Transitivity on ∂K makes $b(x, x)$ constant there, with value $r^2 > 0$. Since K is a compact convex neighborhood of zero, each positive ray meets ∂K in exactly one point. Every such point has b -length r , so $K = \{x : b(x, x) \leq r^2\}$. Set $g_X = r^{-2}b$. This form is independent of the auxiliary coordinates and measure because a centered ellipsoid determines its quadratic unit-ball form uniquely. For an explicit normalization, the covariance of a radius- r ball in b -orthonormal coordinates is $r^2 I / (n+2)$; the definition of $b = S^{-1}$ therefore gives $r^2 = n+2$, so

$$g_X(x, x) = \frac{x^\top S^{-1}x}{n+2}.$$

Finally, every nonzero $v \in \mathcal{C}$ has $t = \tau(v) > 0$, and $v/t = e + x/t$ belongs to B_τ . Hence $g_X(x, x) \leq t^2$. The converse follows by positive conicity and the definition of K ; the $t = 0$ slice contains only zero. The displayed direct sum gives the stated signature. Invariance follows from the construction. $\square$

This construction assumes neither a Euclidean ruler nor a named rotation matrix group. It obtains the spatial comparison from the same completed reach set and its intrinsic isotropy. It

does assume that the physical comparisons act transitively on the boundary of the unit-time reach body; this is substantive isotropy. In an independently fixed spatial unit one may write $|x| = c\sqrt{g_X(x, x)}$, recovering $c^2t^2 - |x|^2$. The value of c compares those spatial and temporal units.

Corollary 5.8 (Three-dimensional isotropy supplies the rotations). *In Theorem 5.7, let $n = 3$ and suppose H preserves spatial orientation. Then its action on X is the full group $\text{SO}(X, g_X)$. Thus this intrinsic rest-comparison condition supplies the rotation input of Physical condition 6.3.*

Proof. The preceding theorem identifies the image of H with a subgroup of $\text{SO}(3)$ transitive on the unit sphere. Choose a unit vector v and $h \in H$ with $hv = -v$. A proper three-dimensional orthogonal map with eigenvalue -1 has spectrum $1, -1, -1$, so h is a half-turn about a unit axis a . For any unit axis b , transitivity supplies $k \in H$ with $ka = b$; the conjugate khk^{-1} is the half-turn about b . Thus H contains every spatial half-turn. Every spatial rotation is a product of two half-turns: for a rotation about n through angle θ , take their axes in the plane perpendicular to n with signed angle $\theta/2$ between them. Hence $H = \text{SO}(3)$. No closure, compactness or Lie-group assumption on the physical comparison group is needed. $\square$

The cone and spatial-comparison theorems work in every finite spatial dimension. The value $n = 3$ remains part of the physical realization, exactly as dimension is an anchor in the Einstein and symmetry sources. The primary intrinsic route just proved constructs the Euclidean comparison and round cone from a clock-separated, intrinsically isotropic transport target; it does not turn the existence of arbitrary determinate objects into a proof of those target attributes.

5.4 The calibrated comparison route

For a target whose spatial comparison and calibration have already been constructed, the same cone calculation has a shorter coordinate form. We record it as an alternative entry point and for direct comparison with familiar SR hypotheses. The primary source construction above does not rely on this supplied spatial norm.

Physical condition 5.9 (Homogeneous displacement receiver). The admitted flat event receiver is a torsor for a real four-dimensional topological vector group V : for each two events there is a unique displacement taking the first to the second. In one calibration,

$$V = \mathbb{R} \oplus \mathbb{R}^3, \quad v = (t, x),$$

where t is an additive duration readout and $\mathbb{R}^3$ has a positive definite spatial comparison $|x|$. A passive frame receipt F is continuous and bijective and transports displacement independently of its base event:

$$F(e + v) = F(e) + A_F(v). \tag{1}$$

Its inverse has the same property. All readouts refer to the event and continuation quotient already constructed. The frame theorem is first established for these globally defined receiving comparisons. Patchwise comparisons below are restrictions of that calibrated family.

The physical content here is substantial: four dimensions, the $1 + 3$ splitting, additive duration, positive definite spatial comparison, and base-event-independent frame comparison are supplied. Lemma 5.10 derives linearity of displacement transport from those last continuity and homogeneity conditions; it does not derive homogeneity or dimension. The finite round cone, the Lorentzian spacetime form, and the removal of conformal scale are subsequent

conclusions. Every supplied comparison is already a determinate physical relation and therefore incurs Theorem 2.1; that dependence does not identify its particular numerical or dimensional realization. Appendix B states the source interfaces.

Theorem 5.2 constructs the torsor from the regular homogeneous source comparison group. The intrinsic route in Theorem 5.7 then supplies its spatial norm from the original completed reach body. Physical condition 5.9 also permits these constructed objects to be received directly when their source certificates are already available. An arbitrary R4 comparison carrier alone does not supply that homogeneous branch. On an admitted open patch, the statements below concern restrictions of this receiving structure and its frame comparisons. Section 14 provides a sufficient condition for a global geometric realization and distinguishes it from a local flat atlas.

Lemma 5.10 (Affinity of homogeneous receipt). *Under Physical condition 5.9, A_F is an invertible real-linear map, and F is affine.*

Proof. Computing $F(e + v + w)$ directly and by successive displacement gives $A_F(v + w) = A_F(v) + A_F(w)$. Additivity gives integer and rational homogeneity. Continuity and rational approximation give $A_F(rv) = rA_F(v)$ for every $r \in \mathbb{R}$. The inverse receipt supplies an inverse linear displacement map. Choosing one event o now gives $F(o + v) = F(o) + A_F(v)$, the asserted affine form. $\square$

R5 supplies actual transport paths and their boundary-preserving partial composition. It expressly separates this result from numerical propagation speed [9, Sec. 2, Rem. 2.1, Thm. 9.1]. R11 supplies irreversible continuation ordering without a numerical proper-time law [13, Thm. 8.1]. The following conditions identify the additional physical content used to construct a finite causal limit.

Physical condition 5.11 (Calibrated propagation family). Let $W \subset V$ be the displacements of the retained elementary forward transport witnesses. The same physical target supplies the following properties.

- (i) Every nonzero $w = (t, x) \in W$ has $t > 0$, and the zero increment is admitted as an identity.
- (ii) Translated copies and the declared finite concatenations are physically admitted with their original endpoint and burden checks. Their displacement is the sum of their increments.
- (iii) Each positively rescaled increment rw , $r > 0$, is admitted, or is the limit of admitted increments in an original limiting-transport operation. The corresponding finite combinations have the same approximation property. This is a property of possible transport in the homogeneous regime, not of every matter solution.
- (iv) A constant $M < \infty$ bounds every elementary increment by $|x| \leq Mt$. Some admitted increment has nonzero spatial part.
- (v) For every $Q \in \text{SO}(3)$, rotational comparison takes (t, x) to (t, Qx) while preserving the propagation witnesses and their admission.
- (vi) The original causal-limit query records the closure of these finite concatenations and rescalings, with their licensed limiting interpretation. It is retained separately from exact finite-path reachability.

Write $\mathcal{C}$ for this completed reach set. Thus the physical receipt gives

$$\mathcal{C} = \overline{\left\{ \sum_{j=1}^N r_j w_j : N < \infty, r_j \geq 0, w_j \in W \right\}}. \quad (2)$$

Every admitted frame receipt preserves and reflects this causal-limit query on the same source.

Equation (2) follows from the supplied concatenation, rescaling, and limiting operations. Taking a convex hull of an arbitrary collection of paths without those operations would change the physical target. Likewise, clause (iv) is a uniform calibrated propagation bound. R5 establishes the witness-mediated transport to which this bound applies; it does not establish the bound. The Euclidean spatial comparison and its three-dimensionality belong to Physical condition 5.9. No spacetime metric has yet been assumed.

Theorem 5.12 (Constructed finite causal cone). *Under Physical conditions 5.9–5.11, there is a unique c with $0 < c \leq M$ such that*

$$\mathcal{C} = \{(t, x) : t \geq 0, |x| \leq ct\}. \quad (3)$$

The set $\mathcal{C}$ is closed, convex, pointed, and has nonempty interior. The quadratic form

$$q(t, x) = c^2 t^2 - |x|^2 \quad (4)$$

has signature (1, 3), and its nonzero future-null set is exactly the nonzero boundary of $\mathcal{C}$.

Proof. Finite sums and nonnegative rescaling form a convex cone, and closure preserves these properties. For each finite sum the triangle inequality gives $|x| \leq Mt$ and $t \geq 0$. These inequalities survive closure. Consequently the only member with $t = 0$ is zero, and $\mathcal{C} \cap (-\mathcal{C}) = \{0\}$.

The section $K = \{x : (1, x) \in \mathcal{C}\}$ is closed, bounded by M , convex, and invariant under every spatial rotation. It is compact. A witness with nonzero spatial part yields a nonzero element of K by rescaling. Its rotational orbit is a sphere of positive radius, whose convex hull contains a ball of that radius. Thus K has nonempty interior. Compactness gives a largest attained radius $c > 0$. Rotating a point at this radius supplies the whole radius- c sphere, and convexity supplies its ball. The maximal-radius definition excludes any further point. Hence K is exactly that ball. Each nonzero cone member has $t > 0$ and equals $t(1, x/t)$, proving (3). The diagonal matrix of (4) is $\text{diag}(c^2, -1, -1, -1)$, which gives its signature and null boundary. $\square$

The number c is the maximal limiting reach per unit duration in the chosen calibration. Its numerical value changes with units. The existence of a proper finite boundary follows from the supplied uniform transport bound and the proved cone construction. To identify vacuum light with that boundary requires actual propagation witnesses saturating it. Theorem 5.12 does not assert that every limiting boundary trajectory is occupied by a physical signal. With such a signal receipt in place, the null-propagation classification applies to the resulting metric regime [16]. Licensed metricity controls the further physical use of the form through the original event and measurement readouts [15].

6 The frame theorem and the elimination of scale

Let $F(e) = Le + a$ be an admitted affine frame receipt, as established by Theorem 5.2 or Lemma 5.10. Preservation and reflection of the original causal-limit query imply $LC = \mathcal{C}$. This

equality, rather than preservation of selected successful paths in only one direction, is the input to the rigidity argument.

Lemma 6.1 (Conformal rigidity of the cone). *If an invertible linear map L satisfies $LC = C$, then*

$$L = a\Lambda, \quad a > 0, \quad q(\Lambda v) = q(v),$$

where a and Λ are unique and Λ preserves the future cone. If L preserves orientation, $\Lambda \in \text{SO}^+(1, 3)$.

Proof. The linear homeomorphism L preserves the boundary of C . Write

$$h(t, x) := q(L(t, x)) = At^2 + 2t b \cdot x + x^\top D x,$$

with D symmetric. For every unit spatial vector n , the vector $(1, cn)$ lies on the boundary, so $A + 2cb \cdot n + c^2 n^\top D n = 0$. Subtracting the equation for $-n$ gives $b = 0$. Then $n^\top D n = -A/c^2$ for every unit n , and polarization yields $D = -(A/c^2)I$. Thus $h = (A/c^2)q$. Since L maps the cone interior to itself, $h(1, 0) > 0$, so $A > 0$. Setting $a = \sqrt{A/c^2}$ and $\Lambda = a^{-1}L$ proves the assertion. The positive scalar factor is determined by h/q ; this also proves uniqueness. $\square$

Cone preservation allows a positive dilation. We remove it using the physical duration of a rest-clock tick pair, group composition, and spatial isotropy. A moving-clock proper-time formula is not needed at this stage.

Lemma 6.2 (Common calibration from completed interval comparisons). *Let the original signed comparisons of rest-clock intervals form a uniquely divisible, totally ordered abelian group A , with translation-invariant order. Suppose it is Archimedean: for $a, b > 0$ some integer $n > 0$ satisfies $na > b$. Retain an actual positive interval u as the common unit. Then there is a unique additive order-preserving receipt $d : A \rightarrow \mathbb{R}$ with $d(u) = 1$. It is injective and is given by*

$$d(a) = \sup\{r \in \mathbb{Q} : ru \leq a\}.$$

Any two real rest-duration receipts preserving these original comparisons and this unit agree on every common interval. In particular, a rest-coordinate duration with those properties equals the retained clock receipt on a covered calibrating pair.

Proof. The Archimedean property bounds every a above and below by integer multiples of u , so its rational cut is nonempty and bounded. Divisibility supplies the rational comparisons. Rational lower bounds for a and b add to lower bounds for $a + b$; rational upper bounds add to upper bounds. Taking arbitrarily close rational bounds proves $d(a + b) = d(a) + d(b)$, and directly $d(u) = 1$. If $a > 0$, choose n with $na > u$. Then $d(a) \geq 1/n > 0$. Thus d strictly preserves order and is injective. For any other normalized additive monotone receipt, all rational lower and upper comparisons give the same bounds on its value. Rational density forces that value to be $d(a)$. Apply this uniqueness to the physical and coordinate receipts of the same original interval. $\square$

This is an exact comparison completion: concatenation, order, subdivision and a unit are specified before a real duration scale is chosen. Signed differences concern comparisons; they do not require backward execution of a clock. Divisibility and the Archimedean property are stated properties of this ideal comparison role. They are not inferred from a discrete tick stream. Each device use retains its own finite validity interval; the role does not assert that one apparatus executes infinitely many operations. The continuous coasting clock in Proposition 15.3 supplies this role on its admitted family of comparisons. Actual unit occupancy, composable frame receipt and common-pair coverage are the further records specified below. Every comparison, subdivision, unit and duration receipt itself incurs Proposition 2.4.

Physical condition 6.3 (Frame group and common rest-clock scale). The linear parts of the admitted orientation- and future-preserving affine frame comparisons form a group G containing every spatial rotation. All such comparisons, including the composites used below, are defined on the calibrating pair and preserve its original clock records. On a patch this common-pair coverage must be supplied, or the maps must be restrictions of an already calibrated global comparison regime. There is a certified pair of distinct tick events whose displacement is $(T, 0)$, $T > 0$, in an admitted chart. Whenever that same pair remains at rest in an admitted chart, its coordinate duration in the common time unit equals its retained calibrated physical duration.

This condition excludes a nonunit positive pure dilation $sI \in G$: the same tick separation would remain at rest and acquire coordinate duration sT , while its calibrated duration remains T . The equality $sT = T$ gives $s = 1$. Only rest-clock normalization is used. A clock's invariance under passive re-presentation and the physical choice of a common unit are distinct from an assumed formula for its duration while moving.

Lemma 6.4 (A scaled boost generates a pure dilation). *If G contains $L = a\Lambda$, where $a > 0$ and $\Lambda \in \text{SO}^+(1, 3)$, then $a^2I \in G$.*

Proof. For this proof use coordinates $(y, x) = (ct, x)$, so $q = y^2 - |x|^2$, and let $e_0 = (1, 0)$. Write

$$\Lambda e_0 = u_0 e_0 + rn, \quad u_0 \geq 1, \quad r \geq 0, \quad u_0^2 - r^2 = 1,$$

where n is a unit spatial vector if $r > 0$. Define B on the plane spanned by e_0, n by

$$Be_0 = u_0 e_0 + rn, \quad Bn = re_0 + u_0 n,$$

and let it be the identity on the perpendicular spatial plane. Its displayed block preserves $\text{diag}(1, -1)$, has determinant one, and preserves the future direction. When $r = 0$, take $B = I$. The isometry $R = B^{-1}\Lambda$ fixes e_0 and consequently preserves its Euclidean spatial orthogonal complement; it is a spatial rotation. Since $R \in G$, the product $LR^{-1} = aB$ lies in G .

For $r > 0$, choose a spatial half-turn J about an axis perpendicular to n . It sends n to $-n$ and fixes e_0 , so $JB J^{-1} = B^{-1}$. Group closure then gives

$$(aB) J (aB) J^{-1} = a^2 I \in G.$$

For $r = 0$, the element $aB = aI$ already lies in G , and its square gives the result. $\square$

Theorem 6.5 (Rest-calibrated Lorentz frame theorem). *Under Physical conditions 5.9, 5.11, and 6.3, every admitted orientation- and future-preserving linear frame comparison preserves q exactly. Hence $G \subseteq \text{SO}^+(1, 3)$, and every corresponding affine comparison is a restriction of a proper orthochronous Poincaré map.*

Proof. Lemma 5.10 gives affine form, and exact receipt of the causal-limit query gives cone preservation. Lemma 6.1 writes each linear part as $a\Lambda$. Lemma 6.4 puts a^2I in G . The common rest-clock scale forces $a^2 = 1$. Since $a > 0$, it follows that $a = 1$. The translation part does not change event displacements or this clock argument. $\square$

Corollary 6.6 (Frames on the constructed source receiver). *Suppose the source has the regular homogeneous comparison target of Definition 5.1 and the intrinsic transport datum of Definition 5.4. Let its composable oriented frame group and common calibrated rest-tick pair satisfy Physical condition 6.3, with the tick displacement Te , $T > 0$. Then the affine receiver, its spatial quadratic comparison and its calibrated Lorentz frame law are constructed from those same source operations and records.*

Proof. Theorem 5.2 constructs the affine carrier and the linear frame actions. Theorem 5.7 constructs g_X and the cone form $q_0(te + x) = t^2 - g_X(x, x)$. Corollary 5.8 supplies all spatial rotations as members of the actual frame group because H is its subgroup. In g_X -orthonormal coordinates, Lemma 6.1 applies to this constructed cone, with numerical speed one in the paired units. Lemma 6.4 and the same retained rest-tick pair remove the positive scale exactly as in Theorem 6.5. An independently fixed spatial unit writes the form as $c^2t^2 - |x|^2$. Every operation and tick used here is a descended original record, so the resulting comparison belongs to the same witnessed source refinement. $\square$

Topological closedness of G is not required for the scale proof; algebraic closure under composition and inverse suffices. Its use of all spatial rotations is essential to the displayed conjugation argument. Alternatively, preserving the value of q for one common nonzero timelike pair would remove the scale immediately. However, invariance of a physical duration alone does not supply that value without a metric-duration receipt in both frames. Theorem 6.5 avoids this stronger moving-pair premise.

The geometric rigidity belongs to the causal and kinematic reconstruction tradition [35, 37, 36]. Here it has been proved within the declared receiving conditions. The AASC requirements fix the physical events, allowed compositions, calibrated clock pair, and exact reachability question to which the geometric proof applies.

Proposition 6.7 (Coverage of the full connected frame group). *Suppose, in addition, the physical frame family acts transitively on the future q -unit timelike vectors. Then $G = \text{SO}^+(1, 3)$. If every translation is admitted on the whole affine event receiver, the full affine comparison group is*

$$\mathbb{R}^{1,3} \rtimes \text{SO}^+(1, 3).$$

Proof. Theorem 6.5 proves the inclusion into the Lorentz group. In the coordinates (ct, x) , let $\Lambda \in \text{SO}^+(1, 3)$ be arbitrary. Transitivity supplies $g \in G$ with $ge_0 = \Lambda e_0$. Thus $g^{-1}\Lambda$ fixes e_0 and is a spatial rotation R . Since $R \in G$, $\Lambda = gR \in G$. For the affine assertion, every group element has an admitted linear part and translation. Conversely any admitted affine map with linear part Λ can be composed with a translation to remove its offset, yielding the pure linear map. All translation–Lorentz pairs are therefore present. $\square$

Transitivity is a supply condition on actual frames. Preservation alone cannot create moving frames: the spatial rotation subgroup satisfies the preservation theorem but lacks boosts. For a particular comparison, it suffices that the needed nonzero boost pair be supplied. On a proper open region, the frame maps are Poincaré restrictions between their actual domains; the full ambient group need not preserve that region.

Corollary 6.8 (Aligned boosts). *For a supplied rotationless comparison with relative velocity v along a spatial unit vector n , let $x_{\parallel} = x \cdot n$ and $x_{\perp} = x - x_{\parallel}n$. With the passive convention that the primed spatial origin has $x_{\parallel} = vt$ in the unprimed chart,*

$$\begin{aligned} t' &= \gamma \left(t - \frac{vx_{\parallel}}{c^2} \right), & x'_{\parallel} &= \gamma(x_{\parallel} - vt), & x'_{\perp} &= x_{\perp}, \\ \gamma &= \frac{1}{\sqrt{1 - v^2/c^2}}, & |v| &< c. \end{aligned} \tag{5}$$

The inverse has v replaced by $-v$.

Proof. The primed time axis is future timelike, so $|v| < c$. In the $(ct, x_{\parallel})$ plane, the rotationless future Lorentz block constructed in Lemma 6.4 has equal positive diagonal entries u_0 and

off-diagonal magnitude r , with $u_0^2 - r^2 = 1$. Sending $x_{\parallel} = vt$ to $x'_{\parallel} = 0$ fixes the passive sign and gives $r/u_0 = v/c$, where r is now signed along n . Therefore $u_0 = \gamma$ and $r = \gamma v/c$. The transverse identity is the choice of aligned rotationless comparison. Substitution gives (5); changing v to $-v$ gives its inverse. $\square$

Corollary 6.9 (Velocity transformation and composition). *For a differentiable trajectory of unprimed velocity $u = u_{\parallel}n + u_{\perp}$, wherever the transformed time derivative is nonzero,*

$$u'_{\parallel} = \frac{u_{\parallel} - v}{1 - vu_{\parallel}/c^2}, \quad u'_{\perp} = \frac{u_{\perp}}{\gamma(1 - vu_{\parallel}/c^2)}. \quad (6)$$

For future-directed causal trajectories the denominator is positive. Two successive aligned passive boosts with parameters v and w compose to the aligned boost with parameter

$$v \oplus w = \frac{v + w}{1 + vw/c^2}. \quad (7)$$

Proof. Differentiate (5) along the trajectory and divide the spatial derivatives by $dt'/dt = \gamma(1 - vu_{\parallel}/c^2)$. If $|u| \leq c$ and $|v| < c$, this factor is strictly positive. Multiplication of the two $(ct, x_{\parallel})$ boost blocks gives diagonal factor $\gamma_v \gamma_w (1 + vw/c^2)$ and off-diagonal factor $-\gamma_v \gamma_w (v + w)/c$. Their ratio gives (7), and

$$1 - \frac{(v \oplus w)^2}{c^2} = \frac{(1 - v^2/c^2)(1 - w^2/c^2)}{(1 + vw/c^2)^2}$$

fixes the positive normalization. Noncollinear boost products can contain a spatial rotation; the group in Proposition 6.7 already includes it. $\square$

The invariant interval is now $q(\Delta t, \Delta x)$ for the same two event classes. Its physical interpretation will be established through clocks and endpoint comparisons. In particular, the algebra does not imply that two frames select the same endpoint pair in a length measurement. The distinction between fixed events and measurement selection is present already in Einstein's operational treatment [33]; the following sections retain it explicitly in the source quotient.

7 Physical clocks and proper duration

Theorem 6.5 fixes the inertial frame law using a common rest-clock calibration. A physical interpretation of duration requires more: a retained clock bearer, identified tick events, and a readout attached to those events. A scalar coordinate alone supplies none of these device records. The distinction is explicit in the AASC clock sources. The physical-clock certificate retains quotient descent, adequate readout, coherent continuation, and a validity domain [17, Def. 12.1, Thm. 12.2]. WDW-4 supplies an occupied local scalar-clock construction, while retaining the qualification that an unattached readout supports relational conditioning only [18, Rem. 43.2, Thm. W4.6].

This section reconstructs that clock method on the licensed flat target and proves the calibrated inertial-duration law. Its physical realizations belong to a declared fixed-background, test-system regime. The occupied compact-hyperbolic Einstein-scalar source of WDW-4 is not identified with a Minkowski vacuum source. In particular, a clock carrying nonzero stress cannot be inserted into an exact vacuum Einstein solution without checking the coupled source equation.

7.1 Scalar-flow duration

Let q be the quadratic form fixed by Theorems 5.12–6.5. Use the metric sign convention

$$g = -c^2 dt^2 + dx_1^2 + dx_2^2 + dx_3^2, \quad q(w) = -g(w, w).$$

Here t has time units, spatial coordinates have length units, and c converts them. The calibrated physical interpretation of this metric remains subject to the standing and comparison conditions of licensed metricity [15, Thm. 6.2].

Lemma 7.1 (Duration along a scalar-clock flow). *Let (M, g) be a licensed Lorentzian region and let $\phi \in C^2(M)$ have a past-directed timelike gradient. Set*

$$\chi = g^{-1}(d\phi, d\phi) < 0, \quad V_\phi = \frac{\nabla\phi}{\chi}.$$

Then V_ϕ is future directed, $d\phi(V_\phi) = 1$, and its integral curves are timelike. The level sets of ϕ are spacelike. Along an integral curve of V_ϕ , metric duration is

$$d\tau_g = \frac{d\phi}{c\sqrt{-\chi}}. \quad (8)$$

If $\psi = f(\phi)$ on an overlap with $f' > 0$, the two scalar descriptions give the same duration element there.

Proof. The negative denominator reverses the past direction of $\nabla\phi$, and

$$d\phi(V_\phi) = 1, \quad g(V_\phi, V_\phi) = \frac{1}{\chi} < 0.$$

The orthogonal complement of a timelike vector is positive definite, so the regular scalar levels are spacelike. Parameterizing an integral curve by ϕ gives $d\gamma = V_\phi d\phi$, hence

$$\frac{\sqrt{-g(d\gamma, d\gamma)}}{c} = \frac{d\phi}{c\sqrt{-\chi}}.$$

For $\psi = f(\phi)$, one has $\chi_\psi = (f')^2\chi_\phi$. Since $f' > 0$, the factors cancel in (8). $\square$

The flow equations used here are those constructed in WDW-4 [18, Eqs. (58)–(60), p. 25]. In its $c = 1$ convention, the anchor value $\chi = -\varepsilon^2$ gives $d\tau_g = d\phi/\varepsilon$. Equation (8) restores the speed conversion in the stated metric convention. It converts a scalar label along its specified flow; it does not turn every monotone variable into an observed clock.

Nor does the lemma identify different clock flows. On an arbitrary timelike curve γ , duration is computed from $g(\dot{\gamma}, \dot{\gamma})$, rather than from the gradient norm of an unrelated scalar. Two moving inertial networks generally have times related through both position and time, rather than through a single function $t' = f(t)$. WDW-7 retains the function-valued physical clock on its source handoff [19, Def. 7.1, Thm. W7.1]; WDW-10 compares full readings at one common source event, with their continuation and failure data [20, Eqs. (27)–(28), Thms. W10.10–W10.12]. Those source comparisons do not identify a clock-overlap Jacobian with a Lorentz dilation factor.

7.2 A constructive recipient on the flat target

Let E be the affine event space, let $\Omega \subseteq E$ be its admitted region, and choose an event o and a future vector U with $g(U, U) = -c^2$. Define

$$t_U(p) = -\frac{g(U, p - o)}{c^2}, \quad X_U(p) = p - o - t_U(p)U, \quad H_U = U^\perp. \quad (9)$$

The restriction of g to H_U is Euclidean, and $X_U(p) \in H_U$. The normalization makes U a displacement per unit clock time.

Proposition 7.2 (Flat realization of the scalar-clock method). *The scalar $\phi_U = t_U$ obeys*

$$\nabla \phi_U = -U/c^2, \quad \chi_U = -1/c^2, \quad V_{\phi_U} = U. \quad (10)$$

Its integral curves form the inertial congruence

$$\gamma_X(s) = o + X + sU, \quad X \in H_U, \quad (11)$$

restricted to their connected allowed intervals in Ω . Its levels are spacelike affine hyperplanes and its metric duration on each curve satisfies $d\tau_g = ds$.

If the faithful event receiver additionally supplies a physical readout attached to these curves, with the common rest calibration, coherent record continuation, and a stated validity domain, the scalar, flow, and readout constitute a physical clock presentation on that domain.

Proof. Differentiating (9) gives $d\phi_U(W) = -g(U, W)/c^2$, so $\nabla \phi_U = -U/c^2$. The normalization of U gives the remaining identities in (10). The flow is (11), and $\phi_U(\gamma_X(s)) = s$. Every event has the unique decomposition (9); restricting an open patch gives a local nonempty flow interval. Lemma 7.1 proves the duration and spacelike-level statements.

For the physical claim, let ρ denote the faithful source-to-event receiver. Quotient-equivalent representatives have the same received event and clock record. Thus $\phi_U \circ \rho$, the attached readout, and their applicability domains are constant on quotient fibres. Lemmas 4.2–4.3 give descent of the readout and continuation. The same factorization retains the specified first failures. The resulting tuple has a candidate, declared use, event and unit anchors, physical quotient, readout, continuation, and failure boundary, as required by the clock certificate [18, Eq. (62), Thm. W4.6]. $\square$

This construction is natural under a proved Poincare comparison $F(p) = Lp + a$: transforming U to LU and o to $F(o)$ gives

$$-\frac{g(LU, F(p) - F(o))}{c^2} = -\frac{g(U, p - o)}{c^2}.$$

The two presentations therefore refer to the same scalar reading on the same clock event. They do not actively move the clock or change its source.

The construction has two branches of input. Theorem 5.12 fixes the causal-limit geometry of admitted transport. The condition below additionally supplies a boundary-signal family, reflecting endpoints and readout events. Their conjunction constructs the pulse clock. Thus a limit of possible propagation displacements and an actually occupied repeating signal are different physical claims; neither the kernel nor the cone theorem permits the latter to be inserted without its witness.

Physical condition 7.3 (Resources for a pulse-clock realization). The flat target admits two persistent comoving reflecting endpoints at $X = 0$ and $X = \ell n$ in H_U , where $\ell > 0$ and $g(n, n) = 1$. It admits repeatable limiting signals on the straight null segments joining them, identifiable emission, reflection, and return events, and either zero reflection delay or an independently calibrated correction. The required segments and records remain inside the declared domain. These are physical device and signal witnesses; a cone boundary alone does not supply them.

Proposition 7.4 (Pulse records populate the flat clock). *Under Physical condition 7.3, a signal emitted from the first endpoint at time s_0 returns at $s_0 + 2\ell/c$. Repetition gives an admitted sequence of tick events and attached readouts*

$$p_n = o + \frac{2n\ell}{c}U, \quad r(p_n) = \frac{2n\ell}{c}, \quad (12)$$

for the nonnegative integers n realized during the operating interval, after fixing the initial phase at o . The scalar ϕ_U agrees with this readout at each tick.

Proof. An outgoing displacement $aU + \ell n$ is null precisely when $-c^2a^2 + \ell^2 = 0$. Future orientation selects $a = \ell/c$. The return displacement $aU - \ell n$ has the same positive duration. Repeatable reflections give (12). The endpoint and signal witnesses identify each return as a clock event, and the faithful receiver preserves these identities and their continuation. Finally $\phi_U(p_n) = 2n\ell/c$ by (9). $\square$

The proposition constructs recorded ticks under explicit physical resources. It does not claim a continuously populated clock from a finite sequence. Interpolation between ticks requires a further readout model or resolution convention. The exact tick-to-tick statements already supply nonempty physical clock uses.

The failure boundary retains endpoint loss or drift, missing signals, unregistered delays, failed reflections or readout attachments, and exit from the chart or common physical source. A mathematical scalar may continue after a device ceases to function; the physical clock certificate then ends. The use of a declared background or test-system clock is permitted by the source's explicit anchor rule [17, Thm. 10.2, Rem. 10.3]. Interpreting the flat model as an approximation to an occupied Einstein–matter solution requires a separate error estimate for backreaction.

7.3 Uniqueness of calibrated inertial duration

We next treat a class of continuously read ideal clocks. The idealization fixes a device type and calibrated internal state whose uniform coasting rate carries no additional environmental, aging, orientation, or acceleration-history dependence. Such dependence, when present, belongs to a richer physical response profile.

Physical condition 7.5 (Uniform ideal inertial readout). A declared class of calibrated clocks supplies a duration function $\mathcal{D}$ on its certified future timelike inertial intervals with the following properties.

1. Within the declared uniform device state, duration depends on the displacement and not on the location or coordinate name of the experiment.
2. Along one uniform inertial run, successive collinear intervals add:

$$\mathcal{D}((s + t)w) = \mathcal{D}(sw) + \mathcal{D}(tw),$$

whenever those intervals and their concatenation are admitted. The interval domain admits subdivision. The readout is positive and continuous, or monotone, in positive interval length.

3. Admitted frame comparisons preserve the scalar duration of the corresponding same clock interval, using one response law and one unit: $\mathcal{D}(Lw) = \mathcal{D}(w)$ whenever both presentations are supplied.
4. A nonzero rest interval fixes the common time unit. For each interval to which the duration theorem is applied, an admitted comparison supplies its rest presentation. The full frame-supply condition of Proposition 6.7 is sufficient for the directional part of this requirement.

Clock existence and readout coverage are required only for the certified intervals. They are not inferred for every mathematical worldline.

Theorem 7.6 (Unique calibrated inertial duration). *After Theorem 6.5, Physical condition 7.5 fixes the duration of every certified future timelike interval w as*

$$\mathcal{D}(w) = \frac{\sqrt{q(w)}}{c}. \quad (13)$$

Proof. Along a rest ray, additivity and subdivision determine the values at rational multiples of a calibrated interval. Continuity, or positivity and monotonicity, extends the linear law to the realized real intervals. Thus $\mathcal{D}((s, 0)) = ks$, and the common rest calibration fixes $k = 1$.

For $w = (\Delta t, \Delta x)$ future timelike, set $v = \Delta x/\Delta t$, so $|v| < c$. The admitted boost of Corollary 6.8 sends w to

$$\left(\frac{\sqrt{q(w)}}{c}, 0 \right).$$

The same-record covariance and the calibrated rest law therefore give (13). Every certified direction has such a rest presentation, so this value is unique. $\square$

The proper-time expression is a conclusion of the proved frame law and the stated ideal-clock properties. Additivity has been imposed only on successive intervals along one inertial ray. It is not imposed on noncollinear timelike displacements: the sum of elapsed durations along two differently directed runs generally differs from the straight endpoint duration.

The pulse clock provides a compatible discrete witness. For an interval spanning n completed cycles, $w = (2n\ell/c)U$, so $\sqrt{q(w)}/c = 2n\ell/c$, exactly its attached readout. Proposition 7.2 gives the same value through the scalar flow. The pulse records, scalar-clock method, and ideal inertial-duration law therefore agree wherever their certified uses overlap.

Corollary 7.7 (Accumulated duration under a local ideal-clock condition). *Let a clock follow a future timelike piecewise C^1 path γ . Suppose its continuation certificate additionally identifies accumulated readout with the continuous limit of calibrated local inertial increments and excludes acceleration or history corrections on that domain. Then*

$$\tau[\gamma] = \frac{1}{c} \int \sqrt{-g(\dot{\gamma}, \dot{\gamma})} d\lambda. \quad (14)$$

Proof. Theorem 7.6 fixes each local inertial increment. The additional continuation condition identifies the accumulated reading with the integral limit. Positive homogeneity of the integrand proves invariance under an orientation-preserving reparametrization of the path. $\square$

The additional condition in this corollary is the local ideal-clock hypothesis. It is unnecessary for uniform-motion time dilation and does not follow from inertial calibration alone. Real acceleration-dependent device effects remain part of the declared response and error domain.

8 Radar synchronization and rulers

A single clock supplies an ordered sequence of readings on one worldline. An inertial coordinate-time network additionally compares distant sites. These are distinct clock uses, as the physical-clock source requires [17, Sec. 9.2]. Radar constructions also distinguish a monotone clock parameter from a calibrated standard clock; arbitrary observer families need not yield reciprocal and transitive synchronization [40]. We now prove the required result for the constant-velocity congruence (11).

Choose an orthonormal basis of H_U . The spatial labels X and the function t_U are affine. Before their origins are synchronized, clocks at occupied network sites read $t_U + \delta_X$. Their common rate is fixed by Theorem 7.6; the offsets δ_X remain to be compared.

Definition 8.1 (Radar-valid domain). A radar comparison between comoving sites is admitted when its outgoing and returning straight limiting-signal segments lie in the event region, its emission, reflection, and reception records are uniquely identified, the involved devices remain in their validity domains, and any reflection delay is recorded. A sufficiently small convex flat neighborhood contains the short geometric segments used within it. Actual execution of the signals remains a physical witness.

Theorem 8.2 (Radar realization of affine inertial time). *Let $X, Y \in H_U$ be comoving occupied sites with rest distance*

$$r = \sqrt{g(Y - X, Y - X)}.$$

On a radar-valid domain with zero reflection delay, emission at X at time t_- , reflection at Y at time t , and reception at X at time t_+ satisfy

$$t - t_- = t_+ - t = r/c, \quad t = \frac{t_- + t_+}{2}. \quad (15)$$

Midpoint synchronization recovers t_U up to one common origin. It is reciprocal and transitive on the connected graph of valid comparisons, and offset transport has zero sum around every admitted closed comparison loop.

Proof. The outgoing signal displacement is $(t - t_-)U + (Y - X)$. Orthogonality, $g(U, U) = -c^2$, and nullity give $c^2(t - t_-)^2 = r^2$. Future orientation selects the positive root. The return displacement has spatial part $X - Y$, so the same argument gives its positive travel duration. This proves (15).

For unsynchronized readouts, the midpoint of the two readings at X equals $t + \delta_X$, whereas the reflection reading at Y is $t + \delta_Y$. Their difference measures $\delta_Y - \delta_X$. Reversing the comparison reverses that difference. Along successive edges the differences telescope, and around a closed loop their sum is zero. Fixing one origin therefore fixes every connected site's origin consistently. $\square$

If the endpoint waits a known duration $d \geq 0$ before returning the signal, it is at rest in the same congruence and Theorem 7.6 identifies that wait with a coordinate duration. The arrival and departure events have times

$$t_{\text{arrival}} = \frac{t_- + t_+ - d}{2}, \quad t_{\text{departure}} = \frac{t_- + t_+ + d}{2}.$$

An unregistered delay fails the specified protocol. It does not alter the frame law.

The midpoint rule is the synchronization associated with the isotropic limiting-signal presentation. An alternative coordinate convention, for example $\tilde{t} = t_U + \kappa \cdot X$ within an admissible range, changes coordinate simultaneity and one-way speeds while preserving the underlying recorded round trips. The formulas below concern the calibrated affine inertial charts with the synchronization of Theorem 8.2. Such a convention change neither reverses physical causal order nor creates a preferred foliation.

The theorem uses a constant U . A rotating network has a different signal-transport problem and may exhibit a nonzero synchronization-loop discrepancy. A curved metric can introduce position-dependent lapse and finite-path effects. Neither case satisfies the stated inertial-congruence conditions.

8.1 Persistent endpoint standards

Definition 8.3 (Certified inertial endpoint standard). An endpoint standard consists of persistent worldlines

$$A(s) = o + sU, \quad B(s) = o + L + sU, \quad L \in H_U \setminus \{0\}, \quad (16)$$

with retained endpoint identities, constant comoving separation throughout the measurement interval, attached records, and specified comparison domains. Its rest length is $L_0 = \sqrt{g(L, L)}$.

A stable coasting material rod can realize this certificate. Two persistent freely coasting endpoints also suffice as an endpoint standard. The definition requires neither an infinitely rigid body nor a protocol that accelerates all its parts simultaneously. R7's identity role and R9's inertia role support the identification of the bearers, but constant numerical rest separation remains an additional physical stability condition [12].

Proposition 8.4 (Radar calibration of rest length). *For the standard of Definition 8.3, a radar-valid round trip from A to B and back, with corrected reflection delay, has clock duration $2L_0/c$ at A . Consequently*

$$L_0 = \frac{c}{2}(\tau_{\text{return}} - \tau_{\text{emit}}) \quad (17)$$

in the zero-delay convention.

Proof. Apply Theorem 8.2 with $r = L_0$ and Theorem 7.6 to the rest clock at A . $\square$

Equation (17) provides a physical length calibration after the time unit and its speed conversion are fixed. It does not determine a unit-independent numerical value of c .

Definition 8.5 (Length selected by an inertial frame). A length reading in a synchronized inertial frame selects one event on each retained endpoint worldline at equal coordinate time and takes their spatial distance. Both selected events and the comparison must lie in the respective validity domains. Rest length selects equal times in the rest frame; a different frame generally selects another pair of events.

For a complete material rod, the affine worldtube $W(s, r) = o + sU + rL$, $0 \leq r \leq 1$, is an available geometric representation when its own material persistence is supplied. Endpoint occupancy alone does not prove that the entire continuum between the endpoints is a material body. The contraction theorem needs the endpoint standard and its measurement rule.

8.2 Domains and topology

The constructions hold on flat affine patches after restricting all clocks, endpoints, and signal segments to their admitted domains. A global result requires an injective affine realization of the event region. Flatness alone does not supply it: flat manifolds can have nontrivial linear or translational holonomy, and a developing map can fail to be injective even when its holonomy is trivial. Periodic identifications can produce several signal routes between the same sites; unique radar transport then needs a restricted domain or a retained route label.

On the whole affine space, (9) is a global product decomposition. On an arbitrary open subset, t_U remains well defined, but a congruence line can leave and reenter the region and a null segment can leave its allowed domain. The connected allowed intervals and Definition 8.1 are therefore essential. A global mathematical coordinate does not establish operational synchronization across disconnected signal domains.

9 Relativity of measured duration, length, and simultaneity

The comparison unit in each result is a retained physical bearer with explicitly identified event pairs. Same-event identity and frame-dependent coordinate answers are compatible: the complete measurement query includes its clock network or endpoint-selection rule.

9.1 Time dilation and reciprocity

Theorem 9.1 (Inertial time dilation). *Let a certified ideal clock be at rest in S' , and let a, b be two retained ticks on its valid inertial run. Suppose S is a synchronized inertial frame in which the clock has constant speed v , $|v| < c$. If $\Delta\tau$ is its calibrated readout difference, then*

$$\Delta t = \gamma_v \Delta\tau, \quad \gamma_v = (1 - v^2/c^2)^{-1/2}. \quad (18)$$

Proof. The same two ticks satisfy $\Delta x' = 0$ in S' . Their rest calibration and Theorem 7.6 give $\Delta t' = \Delta\tau$. The inverse boost in Corollary 6.8 therefore gives $\Delta t = \gamma_v \Delta t'$. Equivalently, $\Delta x = v\Delta t$ in S , so (13) gives

$$\Delta\tau = \frac{\sqrt{c^2 \Delta t^2 - |\Delta x|^2}}{c} = \Delta t \sqrt{1 - v^2/c^2}. \quad \square$$

In AASC terms, one clock bearer and one tick-event pair remain fixed. Passive frame comparison preserves its completed cycles, record order, and physical standing. The observing network assigns coordinate times to the same incidences; their difference is larger by γ_v . The effect compares the device's own elapsed readout with the duration assigned by a spatially distributed synchronized network. No irreversible standing loss or scalar-label rate is being identified with γ_v .

Reciprocity follows from $\gamma_{-v} = \gamma_v$. Each frame can compare a clock at rest in the other with its own synchronized network. These reciprocal experiments generally involve different remote event selections. They do not assign two durations to one clock's same reunion records. If two clocks reunite after one has accelerated, their comparison uses the continued device certificates and, when (14) is invoked, the additional local ideal-clock condition of Corollary 7.7.

9.2 Length contraction

Theorem 9.2 (Longitudinal length contraction). *Let a certified endpoint standard be at rest in S' with positive rest length L_0 parallel to a boost of speed v . Select its endpoint events at*

equal S -time. Its length in S and the rest-frame time difference of that selected pair are

$$L = \frac{L_0}{\gamma_v}, \quad \Delta t' = -\frac{vL_0}{c^2}. \quad (19)$$

Proof. Orient the endpoint ordering so that $\Delta x' = L_0 > 0$. Definition 8.5 imposes $\Delta t = 0$. The proved boost gives

$$\Delta x' = \gamma_v \Delta x, \quad \Delta t' = -\gamma_v v \Delta x / c^2.$$

Substitution proves (19). The endpoints are at rest in S' , so their spatial coordinate difference is L_0 even at the unequal S' -times selected by this measurement. $\square$

Corollary 9.3 (Arbitrary orientation). *Let $L'_\parallel$ and $L'_\perp$ be the standard's rest-separation components parallel and perpendicular to the boost. The equal- S -time separation satisfies*

$$L_\parallel = L'_\parallel / \gamma_v, \quad L_\perp = L'_\perp, \quad L^2 = |L'_\parallel|^2 / \gamma_v^2 + |L'_\perp|^2. \quad (20)$$

Proof. At $\Delta t = 0$, the parallel coordinate transforms as in Theorem 9.2 and the transverse coordinates are unchanged. Euclidean spatial norm in the synchronized S chart gives the final equality. $\square$

The physical endpoint bearer remains the same throughout. Each frame's length question selects a simultaneous pair, and the two selections intersect the persistent worldtube at different events. Their distances differ by the proved factor. R9's persistence-compression profile need not be interpreted as microscopic crushing [12]. The theorem does not compute stresses or equilibrium changes while an actual rod is accelerated. Such changes belong to the rod's dynamical response and can alter its eventual rest length or end its stability certificate.

9.3 Relativity of simultaneity

Theorem 9.4 (Frame dependence of simultaneity). *Let a, b be a fixed pair of distinct events with $\Delta t = 0$ in a synchronized inertial frame S . For a frame S' moving with velocity v ,*

$$\Delta t' = -\gamma_v \frac{v \cdot \Delta x}{c^2}. \quad (21)$$

If $v \cdot \Delta x \neq 0$, the events are not simultaneous in S' . Their identities and spacelike separation are preserved, and no future-causal relation is reversed.

Proof. Substitute $\Delta t = 0$ into the boost time equation of Corollary 6.8. Distinctness gives $q(\Delta) = -|\Delta x|^2 < 0$. Theorem 6.5 preserves both q and the future cone, so the pair remains spacelike and neither event becomes a future-causal successor of the other. $\square$

A nonzero boost perpendicular to the spatial separation leaves this pair simultaneous. For nonzero collinear separation and boost, the time difference is nonzero. The distinction is physical: frame-indexed ordering of spacelike events may vary while their retained records, identities, and lack of causal precedence remain fixed.

A WDW scalar-clock foliation remains an admitted relational presentation on its own source. Theorem 9.4 does not remove that use. It establishes that the equal-time surfaces of all inertial networks cannot be identified as a universal physical simultaneity foliation. In the flat target, t_U and $t_{U'}$ are different network functions on one event quotient. Each has its own certified measurement use; neither becomes a new primitive source of temporal authority.

9.4 Finite measurements and the local Lorentz limit

On a curved licensed Einstein projection, orthonormal tangent frames obey the same Lorentz algebra at a given event. Finite clock intervals, separated endpoints, and radar loops generally receive curvature effects. Theorems 9.1–9.4 apply exactly to the stated flat target and its valid finite-event comparisons. Applying them to an extended curved region requires either an exact flat subregion or a controlled approximation with specified errors. Pointwise tangent-frame symmetry alone does not justify an exact global SR measurement law.

10 Minimal metric transport and attached source conservation

The numerical metric just constructed must carry the comparison and response obligations of its source. This section proves the two interfaces used by the later continuation and inertia constructions: the compatible minimal connection, and the divergence law of an attached closed response. The metric realization is already licensed; the source equations, regularity and declared transport scope remain explicit.

10.1 Comparison-preserving transport and the attached source identity

Two upstream consequences enter the common-source construction as proved mathematical inputs. The first fixes its transport; the second supplies a sufficient construction of its source-continuation identity. The starting object is the original occupied comparison and response source with its faithful smooth Lorentzian metric realization. Its events, source labels, admissible histories, original answers and first-failure records are retained as in Theorem 4.8. Licensed metricity places this realization after the qualification of those relata; its boundary theorem does not itself generate a numerical metric [15, Thm. 6.2, Cor. 6.3].

The declared minimal transport branch. Let (M, g) be that smooth realization. A local linear comparison transport assigns smooth linear isomorphisms $P_\gamma(t, s) : T_{\gamma(s)}M \rightarrow T_{\gamma(t)}M$ along smooth curves, with identity, restriction, inverse and concatenation laws. Its local first-order property means that, for every smooth vector field Y , the derivative

$$D_v Y = \left. \frac{d}{dt} (P_\gamma(t, 0)^{-1} Y_{\gamma(t)}) \right|_{t=0}, \quad \gamma(0) = x, \quad \dot{\gamma}(0) = v, \quad (22)$$

exists, depends only on v and the germ of Y at x , is linear in v , and varies smoothly with the local data. Transport preserves the licensed comparison precisely when

$$g_{\gamma(t)}(P_\gamma(t, s)u, P_\gamma(t, s)w) = g_{\gamma(s)}(u, w). \quad (23)$$

The minimal branch further requires zero torsion: $D_X Y - D_Y X = [X, Y]$. These are explicit properties of the received transport. They are the comparison and connection envelope of the minimal Einstein projection [21, Def. 14.2, Lem. 14.6]; they are not inferred from the existence of a source label alone.

Theorem 10.1 (Forced transport in the minimal comparison branch). *The derivative (22) is a linear connection. Comparison preservation is equivalent to $Dg = 0$. With zero torsion this connection is uniquely the Levi-Civita connection of g , and the original P_γ is its parallel transport. Consequently no second choice of connection is needed in Theorem 12.6 on this branch.*

Proof. Linearity in Y follows by differentiating the linear transport. For a smooth scalar f , differentiation gives

$$D_v(fY) = v(f)Y_x + f(x)D_vY.$$

Together with the stipulated pointwise linearity in v and smoothness, this is precisely a connection. Its induced derivative along a curve makes $P_\gamma(t, 0)u$ parallel: restriction and composition reduce the derivative at any parameter value to the derivative of a constant transported vector. Conversely uniqueness for the corresponding linear ordinary differential equation identifies P_γ with its connection's parallel transport.

For arbitrary parallel U, W along γ , the tensor product rule gives

$$\frac{d}{dt}g(U, W) = (D_{\dot{\gamma}}g)(U, W).$$

Equation (23) therefore makes this derivative zero. At any point the tangent $\dot{\gamma}$ and the initial vectors U, W are arbitrary, so $Dg = 0$. The same identity proves the converse.

Metric compatibility and zero torsion give the Koszul identity

$$\begin{aligned} 2g(D_XY, Z) &= Xg(Y, Z) + Yg(Z, X) - Zg(X, Y) \\ &\quad + g([X, Y], Z) - g([Y, Z], X) + g([Z, X], Y). \end{aligned}$$

Its right side is fixed by g , so nondegeneracy gives uniqueness. Existence can be checked in every coordinate chart from

$$\Gamma^\rho_{\mu\nu} = \frac{1}{2}g^{\rho\sigma}(\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}).$$

These coefficients are symmetric in μ, ν and satisfy $Dg = 0$; the intrinsic Koszul identity makes their chart expressions agree. Thus they define the unique connection already selected by the received comparison transport. $\square$

Corollary 10.2 (Source transport supplies the continuation construction). *On the marked coherent developments of Theorem 12.6, the minimal source transport supplies exactly its P_Γ and hence its anti-development Y_Γ . An oriented, time-oriented lawful isometry transports this construction naturally. The construction retains the original history references and record factor.*

Proof. Theorem 10.1 identifies the two transports by the same parallel initial-value problem. Substitution in the anti-development integral therefore gives the same curve, tangent norms and geometric duration. Under an isometry the pushed connection is metric compatible and torsion free; uniqueness makes it the receiving connection. Parallel initial-value uniqueness then gives the naturality identity used in Theorem 12.6. The original record factor is unchanged by either construction. $\square$

This result has a definite exclusion content. Permitting nonmetricity permits transported comparison values to change. Permitting torsion adds a connection tensor not fixed by the metric. For example, on Minkowski space let a be a constant one-form and A a constant g -skew endomorphism. The connection $D_XY = \partial_XY + a(X)AY$ preserves g but has, in general, torsion $a(X)AY - a(Y)AX$. It produces a different transport. The minimal torsion-free condition rules out this additional content; it does not declare every torsion-bearing physical theory inadmissible.

The source-continuation calculation. An attached continuous metric-response witness has a unique symmetric contravariant tensor-density distribution $\mathbb{T}$ with $W(h) = \frac{1}{2}\langle \mathbb{T}, h \rangle$, because it is an element of the continuous dual of $\Gamma_c(S^2 T^* M)$ [23, Thm. 5.1]. This uniqueness retains the complete original response: a different distribution changes at least one compact metric-probe answer. It does not by itself prove conservation. The following expands the source-closure bridge of the Einstein derivation [21, Prop. 19.7] and the action-presentable construction in Module A [23, Thm. 8.5].

Proposition 10.3 (Constructed Ward identity from a complete source functional). *Suppose a particular admitted source has a differentiable response functional $\mathcal{S}[g, \Psi]$, with the original metric witness as its metric first variation. Its first variation for compactly supported perturbations is*

$$\delta \mathcal{S}[h, \delta \Psi] = \frac{1}{2} \langle \mathbb{T}, h \rangle + \sum_A \langle \mathcal{E}_A, \delta \Psi^A \rangle. \quad (24)$$

Here Ψ comprises every dynamical source field, the $\mathcal{E}_A$ are its Euler–Lagrange distributions, and any integration-by-parts boundary terms vanish for these interior probes. Suppose the complete functional is invariant under the admitted compactly supported diffeomorphisms acting on all its fields, and an actual source configuration satisfies $\mathcal{E}_A = 0$ for every A . Then

$$W(\mathcal{L}_X g) = 0 \quad \text{for every } X \in \Gamma_c(TM), \quad \text{div}_g \mathbb{T} = 0.$$

Thus this source supplies the closed Ward input of Physical Condition 11.1 wherever that condition’s separate trace, integrability and flux requirements are also satisfied.

Proof. For the local flow ϕ_t of compactly supported X , differentiate $\mathcal{S}[\phi_t^* g, \phi_t^* \Psi] = \mathcal{S}[g, \Psi]$ at zero. The pullback derivatives are the corresponding Lie derivatives, giving

$$0 = W(\mathcal{L}_X g) + \sum_A \langle \mathcal{E}_A, \mathcal{L}_X \Psi^A \rangle. \quad (25)$$

Metric compatibility from Theorem 10.1 gives $(\mathcal{L}_X g)_{\mu\nu} = \nabla_\mu X_\nu + \nabla_\nu X_\mu$. Symmetry of $\mathbb{T}$ therefore yields

$$W(\mathcal{L}_X g) = \langle \mathbb{T}^{\mu\nu}, \nabla_\mu X_\nu \rangle = -\langle \text{div}_g \mathbb{T}, X \rangle.$$

The last equality defines the one-form-density divergence by duality; it is well defined for distributions and compact tests. Equation (25), with every $\mathcal{E}_A = 0$, proves zero pairing with every such X , hence zero divergence.

To exhibit the coordinate content and sign, suppose the fields are scalars and the distributions are smooth densities. Write $\mathbb{T}^{\mu\nu} = T^{\mu\nu} dV_g$ and $\mathcal{E}_A = E_A dV_g$. Since $\mathcal{L}_X \Psi^A = X^\nu \partial_\nu \Psi^A$, integration by parts gives

$$0 = \int_M \left[-\nabla_\mu T^\mu{}_\nu + \sum_A E_A \partial_\nu \Psi^A \right] X^\nu dV_g.$$

Arbitrariness of X proves the off-shell identity $\nabla_\mu T^\mu{}_\nu = \sum_A E_A \partial_\nu \Psi^A$. For tensor or other naturally transforming fields, $\mathcal{L}_X \Psi$ also contains derivatives of X ; these are retained in (25) and integrated by parts if a coordinate formula is needed. They cannot generally be replaced by the scalar expression. They vanish in the stated on-shell calculation because their original Euler–Lagrange distributions vanish. $\square$

With a declared external or boundary sector, its first variation contributes an additional functional of X . If its registered exchange density is J with that functional equal to $\langle J, X \rangle$, the same calculation gives $W(\mathcal{L}_X g) = -\langle J, X \rangle$ and $\text{div}_g \mathbb{T} = J$ on the internal source equations.

Unvaried physical backgrounds, anomalous variations and nonvanishing boundary terms must be included here; calling their omission “closed” does not eliminate them.

For completeness, this identity can also be assembled from finitely many already certified operational sectors without one common action. If $W_a(\mathcal{L}_X g) = -\langle J_a^{\text{ext}} + \sum_e B_{ae} j_e, X \rangle$ and each internal incidence column satisfies $\sum_a B_{ae} = 0$, then direct summation gives

$$W_{\text{tot}} = \sum_a W_a, \quad \mathsf{T}_{\text{tot}} = \sum_a \mathsf{T}_a, \quad \text{div}_g \mathsf{T}_{\text{tot}} = \sum_a J_a^{\text{ext}}. \quad (26)$$

The tensor equality follows from uniqueness of the continuous-dual representer; internal cancellation follows from the displayed column sums. The sector registry must retain every contribution exactly once and use the same signed exchange on both ends of each internal incidence. These are the constructive hypotheses of the composite source theorem [23, Thm. 12.1].

The dependency has explicit geometric and source uses. Comparison preservation fixes the connection used for the marked Lorentz-path recipient. The complete source variation or operational sector calculation supplies the Ward identity; on the flat finite-charge target this becomes the divergence input to Lemma 11.2. That lemma and the independently constructed active source action of Proposition 11.5 then yield the total-charge transformation and operational mass–energy theorem. The invariance of a physical functional, existence of an on-shell occupant, finite hypersurface traces and actual impulse protocols remain substantive source conditions. Each concerns the original kernel-governed bearer; none is replaced by an unattached tensor or a new source bearing the old name.

11 Total source, operational inertia, and mass–energy equivalence

The numerical mass theorem requires dynamical response in addition to the passive frame theorem. Its conclusion is rest-energy equivalence from an attached Lorentz-covariant total source and controlled impulse balance, without assuming a mass shell. The active covariance required by the mass theorem has a sufficient construction below from a complete invariant source prescription. R9 constructs an operative deformation-response profile and its restriction to identity-preserving transport changes, but does not give a scalar mass or relativistic dispersion relation [12, Props. 6.1–6.3, Thm. 7.1]. We construct the scalar from measured impulse response. This makes the physical relation to R9 explicit without assuming that its numerical realization already equals a momentum derivative.

11.1 Attached total sources and finite charges

Module A attaches a symmetric tensor-density distribution to a complete operational metric-response witness. Its Ward theorem identifies divergence with the declared exchange current, and its composite theorem cancels internal exchanges after complete finite totalization [23, Thms. 5.1, 8.2, 12.1]. Proposition 10.3 proves the invariant-functional construction of the Ward identity, and equation (26) proves its finite operational sector construction. These are the source divergence inputs used below. A general distribution has no automatic restriction to a spacelike surface. The following analytic conditions are therefore part of the receiving target.

Physical condition 11.1 (Total-source and charge target). Use $x^0 = ct$ and $\eta = \text{diag}(1, -1, -1, -1)$ for charge components on the selected flat or fixed-background target.

The spacetime metric remains $g = -\eta$ in these coordinates, as in Section 7; the charge norm uses the opposite-sign form η . The same persistence bearer has an attached total source $T^{\mu\nu}$, including fields, binding stresses, interactions and containing structure needed for closure. For every admitted state or controlled history:

- (i) T is C^1 with the integration and flux limits used below, or has an explicit distributional normal-trace and Gauss–Green certificate supplying those identities;
- (ii) on every admissible spacelike surface, including the tilted surfaces used by boosts, the integral

$$P^\mu[S; \Sigma] = \frac{1}{c} \int_{\Sigma} T_S^{\mu\nu} d\Sigma_\nu$$

is finite, with future covector surface element $d\Sigma_\nu$;

- (iii) the limiting side flux on an exhaustion between compared surfaces vanishes, both on closed endpoint histories and on the complete pulse regions used below;
- (iv) the Ward current J satisfies $\partial_\nu T^{\mu\nu} = J^\mu$, and a pulse region Ω has a convergent exchange integral

$$I^\mu[\Omega] = \frac{1}{c} \int_{\Omega} J^\mu d^4x.$$

The physical readouts are $E = cP^0$ and $p = (P^1, P^2, P^3)$. Their units, the exchange readout, and the clock/length calibration are fixed together. No mass value or mass-shell relation is stipulated.

On a constant-time slice $d\Sigma_\nu = (1, 0, 0, 0) d^3x$. Since $d^4x = c dt d^3x$, the spatial impulse is the time integral of spatial exchange density. A body/apparatus decomposition may equivalently witness it as the opposite apparatus transfer after every radiative and internal exchange has been assigned exactly once. Module A supplies the accounting law; it does not guarantee that every upstream response belongs to this finite-charge target.

Lemma 11.2 (Pulse balance and slice independence). *Under Physical Condition 11.1, with the stipulated vanishing side flux,*

$$P^\mu[\Sigma_+] - P^\mu[\Sigma_-] = I^\mu[\Omega].$$

In particular a closed history, $J = 0$, has slice-independent total charge.

Proof. Apply Gauss–Green componentwise to $\partial_\nu T^{\mu\nu} = J^\mu$ on a compact exhaustion of the intervening region. The oriented boundary consists of the future slice, minus the past slice, and the side boundary. Divide by c and pass to the limit using the assumed charge convergence and side-flux control. If a different target retains a nonzero outward side flux F_{side}^μ , the correct balance is $\Delta P^\mu = I^\mu - F_{\text{side}}^\mu/c$; recording that flux does not make it zero. The normal-trace version is precisely the alternative certificate in Physical Condition 11.1; an arbitrary distribution is not silently restricted to a hypersurface. $\square$

11.2 Active physical boosts

Physical condition 11.3 (Stable rest branch and active covariance). The physical state or solution family has an active action $U(\Lambda)$ of the proper orthochronous Lorentz group, with attached-source naturality

$$T_{U(\Lambda)S}^{\mu\nu}(x) = \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta T_S^{\alpha\beta}(\Lambda^{-1}x).$$

There is a stable rest occupant S_0 with finite positive

$$P[S_0] = (E_0/c, 0), \quad E_0 > 0,$$

relative to one declared source baseline. The rotationless active boosts B_v , $|v| < c$, send $(c, 0)$ to $(\gamma c, \gamma v)$. Their orbit $S_v = U(B_v)S_0$ preserves the persistence identity and the rest internal state. The endpoint states are closed coasting histories: $J = 0$ on each S_v . The orbit lies within Physical Condition 11.1, including its zero-side-flux condition.

An active action changes an occupant state. The passive comparison of the same events in Theorem 6.5 does not itself construct such an action or a force protocol. The corpus's symmetry analysis distinguishes these roles [28]. A massive rest branch is explicit here; a null sector has no rest state of this kind.

11.3 Constructing the active source input from operational protocols

The active condition can be established at the level of a physical prescription. This supplies a checkable route to Physical Condition 11.3; it does not infer physical symmetry from passive coordinate covariance. In particular, Module A's source theorem represents an attached response, and its naturality theorem transports that response only when the complete operational witness transports [23, Thms. 5.1, 6.1].

Lemma 11.4 (Active response from a complete invariant prescription). *Let $I(g, z)$ be a differentiable source prescription on its original configuration, boundary and probe domains. Stationarity means $D_z I(g, z)[v] = 0$ for every admitted source variation $v \in \mathcal{V}_z$; the complete attached metric response is $W_z(h) = D_g I(g, z)[h]$ on the declared compact probes. Let natural transport F_Λ include all source, boundary and apparatus data, carry the variation and probe domains bijectively, and satisfy*

$$I((f_\Lambda)_*g, F_\Lambda z) = I(g, z)$$

for every admitted Lorentz isometry and metric perturbation used here. For prescriptions expressed through finite action differences or compact variations, require this identity on those differences or variations instead. Then on the fixed flat recipient g_0 :

- (i) F_Λ maps the stationary solution set bijectively to itself;
- (ii) the complete metric-response graph transports with its original applicability and output:

$$(h, w) \in \text{MetResp}(z) \iff ((f_\Lambda)_*h, w) \in \text{MetResp}(F_\Lambda z).$$

These are the solution and response premises of Proposition 11.5.

Proof. Since $(f_\Lambda)_*g_0 = g_0$, differentiation in a source variation gives

$$D_z I(g_0, F_\Lambda z)[DF_\Lambda(v)] = D_z I(g_0, z)[v].$$

The transported variations exhaust the target variation domain. Stationarity therefore transports, and the inverse transformation proves the converse. Differentiation in $g_0 + \lambda h$, with the corresponding source representatives fixed, gives

$$W_{F_\Lambda z}((f_\Lambda)_*h) = W_z(h).$$

Bijective probe transport and completeness of the attached response graph give the claimed equivalence and original definedness. The argument uses finite derivatives and does not equate divergent total actions. $\square$

The independent input is the full source prescription and its actual symmetry, including its boundary and variation domains. Stationary solutions and metric-response transport are conclusions. The coasting clock action in (31), with its scalar internal variables and transported endpoints, supplies this prescription. Controls are transported as part of their complete protocol. A specified invariant coasting prescription does not establish access to a controlled endpoint; Lemma 15.8 provides that separate construction. The local Ward identity, finite traces and side-flux conditions retain their stated hypotheses. The source functional, its domain, the natural action and response graph each necessarily incur Proposition 2.4.

Proposition 11.5 (Operational construction of the active source action). *Write $f_\Lambda(x) = \Lambda x$ for the spacetime isometry. Let $\mathcal{C}$ be a configuration carrier on the selected Minkowski recipient with its natural pushforward F_Λ by $\Lambda \in \text{SO}^+(1, 3)$. Suppose the physical equation and boundary prescription has an admitted solution set $\mathcal{S} \subseteq \mathcal{C}$ preserved by every F_Λ , with persistence labels transported. For each $S \in \mathcal{S}$, let its complete original metric-response relation be the graph of an attached continuous linear witness W_S on $\Gamma_c(S^2 T^* M)$. Suppose the actual metric-response protocols transport bijectively, with their outputs and definedness, so that*

$$(h, w) \in \text{MetResp}(S) \iff ((f_\Lambda)_* h, w) \in \text{MetResp}(F_\Lambda S).$$

Then $U(\Lambda) = F_\Lambda|_{\mathcal{S}}$ is an active action on the admitted solution family. The unique attached tensor-density source satisfies

$$\mathbb{T}_{U(\Lambda)S} = (f_\Lambda)_* \mathbb{T}_S.$$

In the selected flat coordinates this is

$$T_{U(\Lambda)S}^{\mu\nu}(x) = \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta T_S^{\alpha\beta}(\Lambda^{-1}x).$$

For the closed finite-charge states satisfying Physical Condition 11.1, the constructed action consequently obeys $P[U(\Lambda)S] = \Lambda P[S]$.

Proof. Natural pushforward satisfies $F_I = \text{id}$ and $F_{\Lambda_1 \Lambda_2} = F_{\Lambda_1} F_{\Lambda_2}$ on the configuration carrier. Invariance of $\mathcal{S}$, including the inverse transformations, restricts these identities to an action there. This invariance may be checked directly by an equivariant equation residual together with preservation of its boundary and source data; it is not a consequence of merely rewriting an equation in new coordinates.

Graph completeness of the response and protocol transport give $W_{U(\Lambda)S}((f_\Lambda)_* h) = W_S(h)$. For every compact test section h' ,

$$\begin{aligned} \frac{1}{2} \langle \mathbb{T}_{U(\Lambda)S}, h' \rangle &= W_{U(\Lambda)S}(h') = W_S(f_\Lambda^* h') \\ &= \frac{1}{2} \langle \mathbb{T}_S, f_\Lambda^* h' \rangle = \frac{1}{2} \langle (f_\Lambda)_* \mathbb{T}_S, h' \rangle. \end{aligned}$$

Equality on every test section proves equality of the distributions. The two contravariant indices supply the two Lorentz matrices; the density Jacobian is one since $\det \Lambda = 1$. For a closed finite-charge state, changing variables $x = \Lambda y$ in its surface integral gives

$$P[U(\Lambda)S; \Sigma] = \Lambda P[S; \Lambda^{-1}\Sigma].$$

The covector surface element contracts the second Lorentz matrix. Lemma 11.2 makes the closed source's charge independent of the admitted slice, with the stated vanishing side flux. This proves the charge conclusion without a separate transformation postulate. $\square$

The protocol condition is substantive. It preserves the response while transporting the entire experiment, not while holding an untransformed preferred apparatus fixed. The proposition derives the tensor and charge laws from a complete physical-response prescription; it neither manufactures the prescription nor turns an active change of state into passive presentation skin.

Lemma 11.6 (Covariant total charge). *Under Physical Conditions 11.1–11.3,*

$$P[U(\Lambda)S] = \Lambda P[S]$$

for closed states on the orbit. Consequently

$$E(v) = \gamma(v)E_0, \quad p(v) = \gamma(v)\frac{E_0}{c^2}v.$$

Proof. In the surface integral make the change of variables $x = \Lambda y$. The covector surface element contracts the second Lorentz matrix, leaving

$$P^\mu[U(\Lambda)S; \Sigma] = \Lambda^\mu{}_\alpha P^\alpha[S; \Lambda^{-1}\Sigma].$$

Lemma 11.2 replaces the inverse-image tilted slice by any other admissible slice. Apply the resulting four-vector law to $(E_0/c, 0)$ and B_v . The control on tilted surfaces is essential; a same-time integral alone does not prove the transformation law. $\square$

This complete-system argument is closely related to the classical Laue theorem and the modern treatment of active transformations of global charges [41, 42]. The mass shell has not been assumed: invariance of $\eta(P, P)$ is now a consequence.

11.4 A received R9 response and independently measured inertia

Fix the quantitative controlled-response target before calculating a mass. Its records contain one retained R9 persistence class Π , the complete original deformation-response signature, calibrated initial/final velocities, the attached-source and external-exchange record, continuation definedness, first failure, identity retention, and the final internal-profile check. Lawful equivalence preserves these records and calibrations in both directions.

The R9 comparison carrier need not have retained every newly proposed numerical use. When a proposed readout is absent from that carrier, the proper receiving object is an explicit enriched response object over it. A coarse comparison class must not be assigned two incompatible numerical answers.

Lemma 11.7 (Operational response descent). *Let Q_{op} be the quotient of admitted controlled records by equality of their complete original response answers and definedness. Then*

$$[\delta] \longmapsto (v_-, v_+, I^\mu, \text{identity and internal-profile status})$$

is well defined and retains exactly the registered queries. It additionally factors uniquely through the original R9 comparison image if that comparison's equality fibres preserve these readouts and their definedness.

Proof. Equality on Q_{op} fibres gives well-definedness and preservation of every registered answer and failure. For the additional factorization choose a record representing an R9 image value. The stated fibre condition makes its assigned readout independent of that choice, and surjectivity onto the image gives uniqueness. If the condition fails, the proposed factor would assign two different answers to one argument. This is the exact criterion of Lemma 4.2 applied to the response target. $\square$

The domain consists of actual identity-retaining transport deformations, so its readout realizes the operative inertia role. The entire possibly multidimensional R9 profile is not identified with a single number.

Physical condition 11.8 (Controlled impulse receipt). For all v in a neighborhood of zero there is an admitted pulse δ_v beginning at S_0 and ending at S_v . The source/exchange record belongs to the preceding response target, and its calibrated exchange integral is the momentum transfer delivered by the controller. A complete apparatus/body exchange decomposition may supply this receipt. Final internal excitation, escaped radiation, and changed binding vanish or are included so that the endpoint is indeed S_v . If the numerical readout is claimed on the original R9 carrier itself, the fibre condition in Lemma 11.7 holds.

This is an accessibility and calibration condition. The existence of a symmetry orbit does not imply that physical pulses connect its states without changing their internal profile.

Define the measured rest inertial response, when the limit exists, by

$$M_{ij} = \left. \frac{\partial I_i[\delta_v]}{\partial v_j} \right|_{v=0}.$$

Equivalently, its directional readout is $Mn = \lim_{\varepsilon \rightarrow 0} I[\delta_{\varepsilon n}]/\varepsilon$. This uses impulse and velocity measurements; it presupposes neither rest-energy equivalence nor a momentum formula. Small finite pulses avoid an instantaneous-acceleration premise.

Theorem 11.9 (Same-bearer operational rest-energy equivalence). *Under Physical Conditions 11.1, 11.3, and 11.8, the rest inertial response exists, is isotropic, is independent of the admitted pulse with a fixed endpoint pair, and satisfies*

$$M = \frac{E_0}{c^2} I_3.$$

Its scalar value m_i therefore obeys $E_0 = m_i c^2$. Along the stable orbit,

$$E = \gamma m_i c^2, \quad p = \gamma m_i v, \quad E^2 - c^2 |p|^2 = m_i^2 c^4.$$

Proof. Pulse balance and $p[S_0] = 0$ give

$$I[\delta_v] = p[S_v] - p[S_0] = \frac{E_0}{c^2} \gamma(v) v.$$

The right side depends only on the endpoints, proving pulse independence. It is smooth near zero, and $\partial_j(\gamma v_i)|_0 = \delta_{ij}$. Thus $M_{ij} = (E_0/c^2) \delta_{ij}$. Lemma 11.7 descends the measured response on the same retained persistence class. Substitution in Lemma 11.6 gives E and p ; their Lorentz norm gives the final equation. No equality of R9 inertia and a momentum derivative was assumed: the balance law proved it for the received operational response. $\square$

Corollary 11.10 (Finite-speed response and work). *For a smooth sequence of certified pulses staying on one rest-profile branch,*

$$\frac{\partial p_i}{\partial v_j} = m_i \left(\gamma \delta_{ij} + \frac{\gamma^3 v_i v_j}{c^2} \right), \quad dE = v \cdot dp.$$

Proof. Differentiate $p = \gamma m_i v$ and $d\gamma = \gamma^3(v \cdot dv)/c^2$. Contracting the momentum derivative with v gives $m_i \gamma^3 v \cdot dv = dE$. $\square$

Finite-speed response is direction dependent. The rest scalar m_i remains invariant; γm_i need not be reinterpreted as another invariant mass.

The physical origin of the numerical inertia law is explicit. One identity-bearing source carries one total charge. Active transport change transforms that charge, and the controller

supplies its difference. Operational inertia records the small-velocity impulse required to preserve the bearer while changing its transport state. Lorentz covariance fixes that response coefficient to E_0/c^2 . R9's compression-response language does not supply an additional microscopic energy reservoir.

11.5 Baselines, composites, and changes of rest energy

Lemma 11.11 (Invariant finite baseline). *A finite four-charge baseline invariant under all proper orthochronous Lorentz transformations is zero.*

Proof. Spatial rotations force $b = (b^0, 0)$. Invariance under any nonzero boost then gives $b^0 = 0$. $\square$

Thus an arbitrary shift $E \mapsto E + C$ with p unchanged breaks active four-vector covariance unless $C = 0$. A covariant reference source may instead be subtracted before integration when the resulting source, exchange, integrals and reference transformation are certified. A homogeneous vacuum tensor has divergent whole-space charge on Minkowski space; Lemma 11.11 does not regularize it. Gravitational vacuum-source and cosmological-constant bookkeeping must remain consistent.

Corollary 11.12 (Composite rest-energy change). *Every stable completed composite satisfying Theorem 11.9 obeys $E_0 = m_i c^2$ for its total energy, including binding, fields, stresses and containment. Two separately certified stationary configurations A, B with common baseline and calibration satisfy*

$$m_i(B) - m_i(A) = \frac{E_0(B) - E_0(A)}{c^2}.$$

Proof. Module A's finite composite theorem gives the total tensor and cancellation of internal exchanges. Apply Theorem 11.9 to each configuration and subtract. $\square$

An actual emission process must account for body, products, apparatus and external transfers. Recoil may prevent both configurations from resting in one chart; their invariant rest energies are evaluated in their respective rest frames. A closed system can keep the same invariant mass while a subsystem transfers energy to radiation. Summing isolated constituent masses while discarding binding or stress does not compute the complete mass.

Remark 11.13 (Exact strength of the mass theorem). The theorem does not assert that every R9 class has a finite source, stable rest branch, physical Lorentz orbit or accessible lossless pulses. It proves the response law for every admitted same-bearer realization of those conditions. Numerical attachment to an earlier coarser response carrier still requires Lemma 11.7's fibre condition. Module B already proves common source-charge and inertial Hamiltonian matching in its operator-presentable stable scalar envelope, including a positive-mass Wigner representation [24, Def. 5.1, Thms. 7.3, 26.1]. That is a substantive existing matching result; its mass-shell premise is not used in the independent proof above.

11.6 An explicit ideal source and controlled response

Proposition 11.14 (An occupied ideal-body orbit and controlled pulse). *Consider the following ideal-body source model, a dimensionally calibrated version of Module A's worldline construction [23, Prop. 13.4, Cor. 13.5]. An occupant is a smooth future timelike worldline*

$z : \mathbb{R} \rightarrow M$, carrying one fixed persistence label, parameterized on all of $\mathbb{R}$ by proper length s for η , with

$$n = \frac{dz}{ds}, \quad \eta(n, n) = 1.$$

Its physical response law has a positive constant ϵ_* of energy dimension and the attached witness

$$W_z(h) = \frac{\epsilon_*}{2} \int n^\mu n^\nu h_{\mu\nu}(z(s)) ds.$$

Its admitted comparison slices are the spacelike hyperplanes, including the constant-time pulse cuts. The model admits smooth controlled timelike curves with the acceleration exchange specified below. Then its straight rest line has energy $E_0 = \epsilon_*$, its Lorentz orbit satisfies Physical Condition 11.3, and the model has a smooth finite-duration pulse from rest to every $|v| < c$. Its impulses give the operational response $M = (E_0/c^2)I_3$.

Proof. Since $dz^0/ds \geq 1$, compact test support has compact parameter preimage; boundedness of the smooth n there makes W_z a continuous order-zero distributional witness. Here h varies the manuscript's physical metric $g = -\eta$. The source-normalized proper-time response is

$$R_{g,z}(\lambda h) = -\epsilon_* \int \left[\sqrt{-(g + \lambda h)_{z(s)}(n, n)} - 1 \right] ds.$$

Since $-g(n, n) = 1$, its derivative at zero is the displayed positive witness. The minus sign is the usual proper-time action sign; the fixed normalization between action and source response absorbs the factor c . Thus the charge-sign convention does not reverse the attached source. Here sufficiently small λ preserves timelikeness on the compact intersection with the test support. The reference subtraction makes the response finite even on a complete line. The model's original metric protocol is this proper-length response, so the derivative graph supplies attachment rather than an independently chosen source tensor. Module A's representation is explicitly

$$T_z^{\mu\nu}(x) = \epsilon_* \int n^\mu n^\nu \delta^{(4)}(x - z(s)) ds, \quad J_z^\mu(x) = \epsilon_* \int \frac{dn^\mu}{ds} \delta^{(4)}(x - z(s)) ds.$$

Integration by parts against compact vector tests gives $\partial_\nu T_z^{\mu\nu} = J_z^\mu$. Thus straight lines are closed, and accelerated curves have the displayed declared exchange. Proper-length response is preserved by Lorentz pushforward of both line and test section, and the class of future timelike curves is invariant. Proposition 11.5 therefore constructs its active source action.

Every spacelike hyperplane with future unit normal N meets such a complete line once: $\eta(N, n) \geq 1$ makes its defining linear function strictly increasing and unbounded in both directions as $s \rightarrow \pm\infty$. Its normal-trace integral is obtained directly by the transversal delta-function pairing: the factor from the normal contraction cancels the derivative of the hyperplane defining function along the line. Hence

$$P^\mu[\Sigma] = \frac{\epsilon_*}{c} n^\mu|_{z \cap \Sigma}.$$

This proves existence of the required surface charges for this distributional model rather than assuming restriction of an arbitrary distribution. On any finite endpoint region the line stays in a bounded spatial tube, so sufficiently distant side boundaries have zero flux. The rest line has $n = (1, 0)$ and $P = (\epsilon_*/c, 0)$. Its boost has $n_v = (\gamma, \gamma v/c)$, giving

$$E(v) = \gamma \epsilon_*, \quad p(v) = \gamma \frac{\epsilon_*}{c^2} v.$$

No mass-shell equation or prior identification of ϵ_*/c^2 with measured inertia is used.

For an explicit pulse choose $T > 0$ and a smooth $f : \mathbb{R} \rightarrow [0, 1]$ with $f(t) = 0$ for $t \leq -T$, $f(t) = 1$ for $t \geq T$, and $f(-t) = 1 - f(t)$. Define

$$x_v(t) = v \int_{-T}^t f(u) du, \quad z_v(t) = (ct, x_v(t)).$$

Its speed is $|f(t)v| < c$, and it has a smooth proper-length reparameterization. Symmetry gives $\int_{-T}^T f(u) du = T$, so the curve agrees with the rest line before $-T$ and with $x = vt$ after T . The latter is exactly the boosted rest line as an unparameterized coasting history. The exchange is supported in the finite acceleration interval. Integrating J_z yields

$$I^\mu = \frac{\epsilon_*}{c} (n_v^\mu - n_0^\mu), \quad I_i = \frac{\epsilon_*}{c^2} \gamma(v) v_i.$$

Differentiation at $v = 0$ proves $M_{ij} = (\epsilon_*/c^2) \delta_{ij}$. The unchanged label and absence of internal degrees of freedom preserve the model's rest profile. $\square$

This example establishes mathematical occupation of the declared ideal response target and explicitly checks its tensor, charge, orbit and pulse records. Its proper-length response law and admitted external control are physical model content; their empirical realization is a separate question. The example does not by itself supply an observed apparatus or the independent actual-record warrants of Definition 4.7. A concrete actuator, extended-body binding, radiation reaction, or an Einstein backreaction certificate requires further data. In particular, the declared external exchange is not by itself a constructed closed laboratory apparatus. The general total-source theorem above covers more elaborate sources when their corresponding records are supplied and proved.

12 Common-source realization and its relativistic kinematic recipient

12.1 The completed common realization

The corpus supplies a completed certificate-relative gravity–quantum realization. The Modules A–D capstone [27], version `v1.6.0-moduleABCD-integrated-fep1.0`, imports attached source construction, common mass-charge matching, independently formed upstream continuation, and exact certificate-fibre closure [23, 24, 25, 26]. Appendix B makes the exact hypotheses and conclusions accessible here, including an explicit occupied packet. Its occupation is established for the declared target and eligibility classes.

The capstone's Theorem 79.1 supplies a certified local clock, a smooth locally Lipschitz RPI seed formed before solved downstream paths, and a nonempty fixed-clock Moore-history category with nonidentity morphisms. On the exact eligible subtype its metric image intertwines with the Einstein–Module-A Cauchy development; on its exact saturated full-ray quantum-eligible subtype the descended continuation is Wigner–Stone–Schrödinger classified. Record and compatibility processes descend from the same history, and declared exchanges form a coherent sixteen-vertex continuation cube. Complete closed and declared-open packets inhabit its mandatory coordinates [27, Thms. 16.1, 79.1].

Module D closes the exact frozen target's source-locked tensor completion, complete joint probe, maximal lawful skin kernel, eight complete role fibres, live residue, compatible occupation and one lawful assignment orbit [27, Thms. 17.1, 80.1]. This is target-relative realization, rather than a claim of uniqueness across changed source systems, potentials, states, initial data, seeds or matching packages. The capstone excludes a universal interacting quantum-gravity law and generic curved-spacetime global mass from its stated endpoint [27, Thm. 82.1].

The Einstein and Schrödinger receiving scopes remain explicit. The Einstein theorem fixes the minimal first-response equation form after a faithful four-dimensional Lorentzian projection and closed source receipt [21, Def. 14.2, Rem. 14.3, Thm. 30.1]. The Schrödinger theorem uses the standing quantum quotient, a faithful scalar closed nonrelativistic $L^2(\mathbb{R}^d)$ projection, unitary implementation and domain conditions, same-scope spatial exhaustion, and admitted action/inertial calibration [22, Def. 16.1, Thms. 25.10, 27.3, 36.1]. Every source and receiving object necessarily instantiates the kernel. The additional common-source result supplies one history with compatible scoped projections. Its realization data have substantive dynamical content: the Einstein branch needs seed-to-Cauchy compatibility, and the quantum branch needs full-ray transition preservation, regular unitary implementation and spatial generator eligibility. In the explicit closed packet, the smooth source flow advances clock and record while its quantum projection carries the specified spectral evolution. Appendix B.5 exhibits these maps. Forming the seed before the projection therefore does not mean that the seed alone determines the receiving dynamics.

This section has a precise role in the SR argument. Sections 5–6 construct Lorentz kinematics from displacement and propagation data without assuming a spacetime metric. The next theorem instead constructs a recipient of the already Lorentzian metric projection of the common source. It establishes inherited local geometry. The later Theorem 12.6 adds an explicit continuation construction. Neither step is a second derivation of signature or the source of the earlier cone theorem.

12.2 An occupied tangent recipient

Fix an admitted capstone object in its regular local target. Let (M, g) be its smooth oriented, time-oriented Lorentzian metric image, written in the $(-+++)$ convention of Section 7 and let e be a certified standing-positive event in its event/record domain. Define the tangent recipient

$$\mathcal{T}(M, g, e) = (T_e M, g_e).$$

The event already belongs to the occupied metric image. A premetric continuation neighborhood has not simply been renamed a spacetime point.

Theorem 12.1 (Tangent Lorentz recipient). *The construction $\mathcal{T}$ is functorial under lawful pointed metric redescrptions. Its oriented future-oriented orthonormal frames form an $\mathrm{SO}^+(1, 3)$ torsor. Changes between such frames are exactly proper orthochronous Lorentz transformations.*

Proof. The declared signature gives orthonormal bases by the inertia theorem for real symmetric forms. For two such bases the change matrix satisfies $L^\top \eta L = \eta$; the shared spatial and temporal orientations select $\mathrm{SO}^+(1, 3)$. Conversely each matrix in this group sends one oriented future frame to another, and the action is free and transitive. For a lawful pointed metric isomorphism f with $f(e) = e'$ and $f^* g' = g$,

$$g'_e(\mathrm{d}f_e u, \mathrm{d}f_e v) = g_e(u, v).$$

Derivatives preserve identities and composition, proving functoriality and independence of the metric representative up to isometry. $\square$

The bare tangent object retains tangent metric contractions and causal type. It does not by itself retain event identity, a physical clock, or that clock's applicability. For physical germ uses take instead the decorated recipient

$$\mathcal{T}_e^{\mathrm{rec}} = (\rho_e, T_e M, g_e, \mathcal{R}_e),$$

where ρ_e retains the original event reference and $\mathcal{R}_e$ contains the attained complete answers, domain predicates and calibration records for the declared clock/rod germ queries. These are obtained from the existing faithful metric/event/record receiver. Lawful pointed redescrptions act by their derivative and by the already descended record maps. Keeping these factors makes the corresponding readouts and failures constant on receiving fibres by construction; forgetting them would not.

This recipient need not be injective on complete states: curvature, extended-body structure and topology have deliberately been excluded from its restricted query family. If the experiment asks those questions, the recipient is insufficient. No new clock is created by recording an empty or failed clock domain.

A certified timelike clock curve $z(\lambda)$ with an admitted metric-duration readout has

$$\frac{d\tau}{d\lambda} = \frac{\sqrt{-g(z', z')}}{c}.$$

This is an exact differential relation; a monotone clock label still needs a physical calibration receipt. The capstone clock and the Wheeler–DeWitt clock method provide occupied clock structures in their respective declared domains. A selected SR recipient must preserve the actual event and calibration uses on their intersection.

Theorem 12.1 is therefore an occupied local relativistic consequence of the existing common source. It does not require inserting clock matter into a new exact Minkowski solution. It also does not replace separated events in curved spacetime by tangent displacements without curvature corrections.

12.3 Identifying the constructed and dynamical metric recipients

The cone construction and the metric-history recipient must agree on more than a source name. Their identification uses the original causal-limit query and clock calibration. This is the exact matching step between the transport-first SR construction and the already occupied Lorentzian dynamical branch.

Lemma 12.2 (Metric matching from the same cone and calibration). *Let q be the constructed form of signature $(1, 3)$ on an identified displacement or tangent space V . Let g_* be the pullback of the dynamical metric through the actual source identification, and put $h(v) = -g_*(v, v)$. Suppose its future cone equals that of q , and an independently retained nonzero timelike calibration vector w satisfies*

$$h(w) = q(w) > 0.$$

Then $h = q$ on V , hence $g_ = -q$ as bilinear forms. The statement applies pointwise on a coherently identified tangent domain. For two constant forms on a common flat affine displacement carrier, one such shared calibration pair suffices throughout that carrier.*

Proof. Choose linear coordinates in which $q(t, x) = t^2 - |x|^2$ and write $h(t, x) = At^2 + 2tb \cdot x + x^\top Dx$, with D symmetric. The common null boundary gives $h(1, n) = h(1, -n) = 0$ for every unit spatial n . Subtracting shows $b = 0$, and then $n^\top Dn = -A$ on the unit sphere. Polarization gives $D = -AI$, so $h = Aq$. The shared future interior gives $A > 0$. Evaluation on w forces $A = 1$. Polarization again identifies the bilinear forms. Pointwise application proves the tangent statement. For constant forms the same scalar A holds on the entire affine carrier, so the single retained pair fixes it everywhere. $\square$

Lemma 12.3 (Finite completed comparison certificate for metric matching). *On the already constructed four-dimensional comparison space, choose paired calibrated coordinates with*

$q(t, x) = t^2 - |x|^2$ and basis e_0, e_1, e_2, e_3 . Suppose the original signal or causal-limit role retains these nine null-direction records:

$$k_i^\pm = e_0 \pm e_i \quad (1 \leq i \leq 3), \quad \ell_{ij} = e_0 + (e_i + e_j)/\sqrt{2} \quad (1 \leq i < j \leq 3),$$

and one nonzero timelike clock standard w . If a symmetric quadratic receipt h retains their original answers,

$$h(k_i^\pm) = 0, \quad h(\ell_{ij}) = 0, \quad h(w) = q(w) > 0,$$

then $h = q$ on the entire comparison space. Conversely equality retains all ten answers. Thus these finite original comparisons separate the quadratic receiving class and supply the matching condition of Lemma 12.2.

Proof. Write $h(t, x) = At^2 + 2t b \cdot x + x^\top D x$, $D = D^\top$. The two axial equations give $A + 2b_i + D_{ii} = A - 2b_i + D_{ii} = 0$, hence $b_i = 0$ and $D_{ii} = -A$. The diagonal-direction equation is then $0 = A + (D_{ii} + D_{jj})/2 + D_{ij} = D_{ij}$. Consequently $h = Aq$. The nonzero calibration gives $A = 1$; polarization identifies the bilinear forms. The converse is immediate. $\square$

The inputs to this finite certificate are independently identified boundary comparisons and a retained clock interval, with an actual source identification. Their faithful quadratic receipt forces the metric equality. If the original clock answer is T , the last equation comes from the two certified rest interpretations $h(w) = c^2 T^2 = q(w)$. The certificate presupposes neither equality of the entire cones nor a moving-clock law. Positive rescaling of a null test preserves its answer, allowing its interval to fit the certified domain. On a constant flat recipient the tests may be covered by different registered runs on that same carrier. A causal-limit record retains that type; executed signals require the actual witnesses. The protocol family of Proposition 15.7 supplies the null directions and the rest standard. Every test, identification and metric receipt itself incurs Proposition 2.4.

The hypotheses are physical matching records: equality of the received causal-limit question and agreement of one nonzero timelike standard in common units. Kernel governance alone supplies neither record. Once they hold, every local metric norm and geometric-duration integral agrees; with the same minimal transport branch, Theorem 10.1 also identifies the connections. On an injective flat developing region this identifies the two finite SR constructions. Equality at one tangent space alone does not identify a whole curved region or its finite radar network.

The regular finite-dimensional Hamiltonian sector used in Section 3 likewise has its own scope. Joining its records to a field-theoretic or Hilbert/Banach source requires an actual common record carrier with projections onto both. The finite-dimensional symplectic proof is not being extended to an arbitrary infinite-dimensional phase space. The common-history conclusion below is conditional on that certified joint domain and the metric matching just proved; the capstone's occupation theorem is used at its original scope.

12.4 Exact flat regions and the controlled local limit

Theorem 12.4 (Flat specialization). *Let an occupied Einstein projection contain a connected region U with $\text{Riem}[g]|_U = 0$ and an injective affine developing map identifying g with the Minkowski form on its image. Then oriented inertial chart transitions on that region are restrictions of Poincaré transformations. The certified clock, ruler and simultaneity comparisons of the preceding sections apply exactly when their physical records and calibrations restrict to U .*

Proof. Zero curvature makes the Levi-Civita connection locally flat. The stated developing map supplies the otherwise nonautomatic single affine identification. Two affine orthonormal identifications differ by an affine isometry, whose linear part is Lorentz and whose constant part is a translation. The retained physical certificates identify the events and standards to which these coordinate transformations apply. $\square$

Flatness alone is insufficient for a global injective chart. Moreover, the Einstein equation imposes

$$\text{Riem}[g] = 0 \implies T = \frac{\Lambda}{\kappa} g$$

when $\kappa \neq 0$. For $\Lambda = 0$ this gives $T = 0$. A localized gravitating clock or body cannot be inserted into an exact vacuum Minkowski branch without changing that equation. Fixed-background SR is a declared test/background projection; self-gravitating finite experiments require a controlled approximation.

Proposition 12.5 (Controlled local clock geometry). *In smooth normal coordinates about e , fix a compact rescaled coordinate domain and write $g_\varepsilon(y) = g(\varepsilon y)$. Using $\eta = \text{diag}(1, -1, -1, -1)$, one has $g_\varepsilon = -\eta + O(\varepsilon^2)$ uniformly. For a fixed C^1 path $y(s)$ on a compact parameter interval $s \in [s_-, s_+]$, $-\infty < s_- < s_+ < \infty$, with uniform timelike margin*

$$\eta(y', y') \geq \delta |y'|^2, \quad \delta > 0,$$

the metric and Minkowski duration integrands differ by $O(\varepsilon^2)|y'|$. Their rescaled integrated durations differ by $O(\varepsilon^2)$; physical durations on the scaled path have an additional overall factor ε .

Proof. Normal coordinates give $g(0) = -\eta$ and $\partial g(0) = 0$. Taylor's theorem with bounded second derivatives on the compact neighborhood gives the uniform $O(\varepsilon^2)$ estimate. For sufficiently small ε the perturbed path remains timelike, and

$$\left| \sqrt{-g_\varepsilon(y', y')} - \sqrt{\eta(y', y')} \right| = \frac{|(g_\varepsilon + \eta)(y', y')|}{\sqrt{-g_\varepsilon(y', y')} + \sqrt{\eta(y', y')}}.$$

The numerator is $O(\varepsilon^2)|y'|^2$ and the margin bounds the denominator below by a positive multiple of $|y'|$. At $y' = 0$ the claimed bound is immediate. Integrate and use $x = \varepsilon y$ to obtain the physical scaling. $\square$

Application to an actual apparatus also requires a controlled family of physical paths and readouts. The result is not an exact-flatness assertion, and a uniform estimate needs the stated margin from null directions.

12.5 A constructed continuation action

The tangent construction extends to histories by an explicit operation. Fix a smooth oriented and time-oriented Lorentzian development (M, g) of signature $(-+++)$ and the common conversion scale $c > 0$. A marked history

$$\Gamma = (a, \gamma, \mathcal{R}_\Gamma), \quad a \geq 0,$$

has a piecewise C^1 future causal curve $\gamma : [0, a] \rightarrow M$, its original endpoint references, and its complete declared event, domain, calibration, and first-failure records. The parameter is the retained source-clock parameter; it need not be proper time. Positive-duration curves are admitted only where the original source and record receiver admit them. The zero-duration

history at every admitted object is an identity. The admitted family is closed under restrictions and every concatenation whose complete endpoint objects match. Those objects include the source context and applicability data needed to make matching equivalent to the original composition license; a weaker family with additional pairwise restrictions would instead require a partial category. Records concatenate in their original order, with the shared endpoint represented once; matching includes agreement of its complete endpoint record.

Each such history must lie in one coherent metric development. Successive Cauchy slabs are allowed when their existing lawful identification places their common marked history in that development. A sequence of unrelated tangent spaces, without such a development or a specified compatible connection, is not this input. These are concrete source eligibility conditions, not a supplied action on a new SR history category.

Put $V_x = T_x M$, with its Lorentz form g_x and inherited orientations. Here ρ_x retains the original marked source-object, lineage, and event reference, so different source objects are not made composable by a coordinate coincidence. An object of the receiving category consists of

$$(\rho_x, V_x, g_x, \mathcal{R}_x).$$

A receiving arrow from the object at x to that at y is a tuple

$$(a, Y, P, \mathcal{R}), \quad Y : [0, a] \rightarrow V_x, \quad Y(0) = 0, \quad P : V_x \longrightarrow V_y,$$

where Y is piecewise C^1 and future causal, P is an oriented future-oriented linear isometry, and $\mathcal{R}$ retains the declared ordered records with the specified endpoint data. For a second arrow $(b, Z, Q, \mathcal{S})$ from y to z , define

$$(b, Z, Q, \mathcal{S}) \circ (a, Y, P, \mathcal{R}) = (a + b, Y \star_P Z, QP, \mathcal{R} \star \mathcal{S}), \quad (27)$$

where

$$(Y \star_P Z)(t) = \begin{cases} Y(t), & 0 \leq t \leq a, \\ Y(a) + P^{-1}Z(t - a), & a \leq t \leq a + b. \end{cases}$$

The formulas agree at $t = a$ because $Z(0) = 0$. The identity is $(0, 0, \text{id}, \mathcal{R}_x)$. Only the arrows constructed from admitted source histories are taken as physically occupied. The ambient geometric category supplies no additional physical execution or device. The endpoint references retain the complete original source object, so a new geometric coincidence cannot create a source composition.

Theorem 12.6 (Constructed Lorentz-path continuation recipient). *For every marked history Γ , let $P_\Gamma(t) : T_{\gamma(0)}M \rightarrow T_{\gamma(t)}M$ be Levi-Civita parallel transport along its prefix. Define*

$$Y_\Gamma(t) = \int_0^t P_\Gamma(s)^{-1} \dot{\gamma}(s) \, ds, \quad P_\Gamma = P_\Gamma(a). \quad (28)$$

Then

$$R_{\text{SR}}(\Gamma) = (a, Y_\Gamma, P_\Gamma, \mathcal{R}_\Gamma)$$

defines a functor from the admitted marked Moore-history category to the receiving category above. It is natural under lawful oriented, time-oriented isometries of marked histories. For almost every t it preserves causal type and the tangent norm:

$$g_{\gamma(0)}(\dot{Y}_\Gamma, \dot{Y}_\Gamma) = g_{\gamma(t)}(\dot{\gamma}, \dot{\gamma}). \quad (29)$$

Consequently its geometric duration agrees exactly with that of the original marked curve:

$$\frac{1}{c} \int_0^a \sqrt{-g_{\gamma(0)}(\dot{Y}_\Gamma, \dot{Y}_\Gamma)} \, dt = \frac{1}{c} \int_0^a \sqrt{-g_{\gamma(t)}(\dot{\gamma}, \dot{\gamma})} \, dt. \quad (30)$$

This nonnegative duration is additive under admitted concatenation. Every retained event reference, record answer, applicability predicate, calibration and first-failure datum is preserved.

Proof. On the minimal comparison branch, Theorem 10.1 identifies the source transport with the Levi–Civita connection used here. The Levi–Civita equation along each smooth piece is a linear ordinary differential equation; uniqueness joins its solutions across the finitely many corners. It supplies prefix transport with $P_\Gamma(0) = \text{id}$. Metric compatibility implies

$$\frac{d}{dt}g(W_1, W_2) = 0$$

for parallel vector fields W_1, W_2 . Thus every $P_\Gamma(t)$ is a linear isometry. Starting at the identity and varying continuously along the path preserves orientation and the future component of the timelike cone. Equation (28) is therefore defined on the entire admitted compact parameter interval. Differentiating on its smooth pieces gives $\dot{Y}_\Gamma = P_\Gamma(t)^{-1}\dot{\gamma}$. This proves future causality, (29), and (30), including zero integrand on null pieces. The zero-duration identity has zero duration and identity transport.

For composable histories Γ and Δ , uniqueness of parallel transport gives

$$P_{\Gamma \star \Delta}(a + s) = P_\Delta(s)P_\Gamma(a).$$

Substitution in (28) yields

$$Y_{\Gamma \star \Delta}(a + s) = Y_\Gamma(a) + P_\Gamma(a)^{-1}Y_\Delta(s).$$

The value on the first interval remains Y_Γ . This is exactly (27), and the endpoint transport is $P_\Delta(b)P_\Gamma(a)$. Duration additivity follows either by splitting the source integral or by the isometry of P_Γ^{-1} in the receiving integral.

The receiving composition is associative. For three successive arrows, both bracketings produce the third curve piece

$$Y(a) + P^{-1}Z(b) + P^{-1}Q^{-1}W(s),$$

and endpoint transport SQP ; the first two curve pieces also coincide. Concatenating ordered records with an equal shared endpoint represented once is associative. Zero-duration arrows give both identity laws.

Let $f : (M, g) \rightarrow (M', g')$ be a lawful history isometry, with $\gamma' = f \circ \gamma$ and the corresponding original records. The Levi–Civita connection is natural under isometries: the pushforward connection is metric compatible and torsion free, hence equals the Levi–Civita connection of g' . Consequently

$$P_{\Gamma'}(t) \, df_{\gamma(0)} = df_{\gamma(t)} P_\Gamma(t).$$

It follows that

$$Y_{\Gamma'}(t) = df_{\gamma(0)} Y_\Gamma(t), \quad P_{\Gamma'}(a) df_{\gamma(0)} = df_{\gamma(a)} P_\Gamma(a).$$

These are the required naturality identities. The original lawful record maps supply the corresponding record equality. Taking the inherited record factor unchanged proves exact preservation of its declared answers, applicability, calibration, and first failure. No unoccupied or failed source interval is added. $\square$

The construction has a genuine flat comparison output: Y_Γ is a curve in the Lorentz vector space $T_{\gamma(0)}M$, so any oriented future orthonormal basis expresses it as an ordinary Minkowski-space curve. Changing that initial basis acts by a proper orthochronous Lorentz transformation. The output is computed from the received metric, its uniquely compatible torsion-free connection, the marked history, and the original records; its continuation action is not another postulate.

The equality of geometric durations is not an assertion that every physical device reads this integral. Agreement with an attained clock readout uses that device's existing calibration and local ideal-clock certificate, as in Corollary 7.7. The construction retains a failed or uncalibrated readout as such.

Nor does anti-development preserve arbitrary finite-pair spacetime intervals, curvature, radar networks, or the relative placement of different worldlines. Two histories with the same endpoints can have different developed endpoints or different endpoint transport. The transport is canonical along the specified path and may have curvature-dependent holonomy; it is not a canonical boost between arbitrary curved tangent fibres. Exact SR comparisons of extended experiments still require the common flat specialization of Theorem 12.4.

Corollary 12.7 (Constructed common-source continuation). *Let an occupied capstone history target have a certified nonempty marked subtype in the joint domain where its already certified Einstein Cauchy and Schrödinger recipient functors are both defined, and whose metric/event/record receiver supplies the coherent developments above. Retain an original positive-duration marked history when nonidentity continuation is claimed. The Einstein, Schrödinger, and constructed Lorentz-path recipients then have one upstream history source, and every certified metric-side continuation identity remains true after the constructed receipt.*

Proof. The existing source projection supplies the marked history and its original records. Theorem 12.6 constructs its complete receiving arrow and proves its identity, composition, domain, record, and redescription properties. Composing that proved functor with the existing marked source projection yields the Lorentz-path recipient. Applying a functor to either side of a certified equality preserves that equality. The occupied source and its marked histories are inherited from the specified source target; no additional action on continuation arrows is assumed. $\square$

This corollary constructs the SR action on continuation arrows and establishes its recipient on the stated geometric history class. The remaining source requirement is the existence of that class: smooth coherent spacetime development, an actual marked causal history, and its attached original records. An arbitrary capstone clock coordinate does not by itself supply a material worldline in the metric image. An occupied scalar-clock flow, or another certified source-to-worldline receiver, discharges this marking requirement where its domains intersect the capstone's metric and record domains.

Corollary 12.8 (Agreement with ordinary flat-history kinematics). *Suppose the marked curve lies in a region with an injective affine developing isometry $D : U \rightarrow \mathbb{R}^{1,3}$ as in Theorem 12.4. Then*

$$dD_{\gamma(0)}Y_{\Gamma}(t) = D(\gamma(t)) - D(\gamma(0)), \quad P_{\Gamma}(t) = (dD_{\gamma(t)})^{-1}dD_{\gamma(0)}.$$

Thus the constructed recipient agrees with the actual curve's Minkowski coordinate displacements throughout that region, and its concatenation is ordinary displacement concatenation in a common affine frame.

Proof. Naturality of Levi-Civita transport under D identifies source transport with constant-vector transport in the flat image. This gives the second identity. Substitute it in (28) and use the fundamental theorem of calculus to obtain the first. The common affine trivialization then changes the twisted composition (27) into ordinary addition of displacement curves. $\square$

The formulas are defined on representative developments and are natural under their lawful isometries. Passing to lawful-isometry classes gives the corresponding recipient up to that natural isometry. No derivative map is asserted to become a literal identity merely by changing its representative.

13 Controlled free compatibility with Schrödinger dynamics

The classical charge theorem and a quantum spectral realization have different types. To compare evolutions, admit the same positive mass m , action scale $\hbar$, and quantum translation realization on $L^2(\mathbb{R}^3, d^3p)$, with momentum multiplication operators and the free positive-energy branch. Classical SR alone supplies neither this Hilbert carrier nor a quantum Hamiltonian. Module B constructs a concrete stronger instance using its attached stable scalar Wigner sector and common-core contraction [24, Thms. 16.2–18.1].

Define the multiplication operators with their maximal self-adjoint domains by

$$\begin{aligned} h_c(p) &= \sqrt{m^2 c^4 + c^2 |p|^2} - mc^2, & H_c &= M_{h_c}, \\ h_{\text{NR}}(p) &= \frac{|p|^2}{2m}, & H_{\text{NR}} &= M_{h_{\text{NR}}}. \end{aligned}$$

Subtracting mc^2 is a homogeneous phase convention in one fixed mass sector. It does not permit independently shifting the physical source baselines of compared sectors.

Lemma 13.1 (Exact free dispersion error). *For every p and $m, c > 0$,*

$$0 \leq h_{\text{NR}}(p) - h_c(p) = \frac{|p|^4}{2m^3 c^2 \left(\sqrt{1 + |p|^2/(m^2 c^2)} + 1 \right)^2} \leq \frac{|p|^4}{8m^3 c^2}.$$

Proof. Set $a = |p|^2/(m^2 c^2)$ and $s = \sqrt{1 + a}$. Rationalization gives $h_c = mc^2 a/(s + 1)$, whence

$$h_{\text{NR}} - h_c = \frac{mc^2 a(s - 1)}{2(s + 1)} = \frac{mc^2 a^2}{2(s + 1)^2}.$$

Since $s \geq 1$, the estimate follows. $\square$

Theorem 13.2 (Quantitative free evolution comparison). *Let Q_P be the spectral cutoff to $|p| \leq P$, and set $U_c(t) = \exp(-itH_c/\hbar)$ and $U_{\text{NR}}(t) = \exp(-itH_{\text{NR}}/\hbar)$. Then*

$$\|(U_c(t) - U_{\text{NR}}(t))Q_P\| \leq \min \left\{ 2, \frac{|t|P^4}{8\hbar m^3 c^2} \right\}.$$

For every $\psi \in L^2$, $U_c(t)\psi \rightarrow U_{\text{NR}}(t)\psi$ strongly as $c \rightarrow \infty$, uniformly on compact time intervals.

Proof. The unitaries multiply by phases. For real a, b , $|e^{ia} - e^{ib}| \leq \min\{2, |a - b|\}$. Apply Lemma 13.1 on the momentum ball to obtain the operator-norm cutoff bound. For arbitrary ψ , choose P so that $\|(1 - Q_P)\psi\|$ is small. The tail difference is bounded by twice that norm; the cutoff term tends uniformly to zero for t in a compact interval. Density proves the asserted strong convergence. No unrestricted momentum-space operator-norm convergence is claimed. $\square$

Fourier realization $p = -i\hbar\nabla$ gives

$$H_{\text{NR}} = -\frac{\hbar^2}{2m}\Delta$$

on the standard self-adjoint domain. This is the kinetic normal form in the AASC Schrödinger theorem [22, Thms. 25.10, 27.3, 36.1]. Module B additionally supplies the explicit graph and commutator contraction and its Bargmann central mass; Theorem 13.2 gives a finite-momentum error estimate.

Matching this mass to Theorem 11.9's operational response requires the two recipients to carry the same attached intrinsic rest-charge line and calibration. A localized quantum packet can have rest-frame mean energy above mc^2 because of momentum spread; the norm of its mean four-charge is not automatically the Wigner particle mass. Module B retains that distinction. Declared potentials require separate domain, operator-convergence and relativistic-coupling control. The free comparison does not establish a covariant interacting quantum theory merely by appending a potential.

14 Local realization, globality, and alternative targets

The frame theorem concerns a supplied flat receiving regime. A Lorentzian metric at one event gives a tangent Lorentz group, and an exact flat atlas gives local affine Poincaré comparisons. Neither fact alone proves that an entire physical source has one injective Minkowski chart. Source occupancy and the global topology must be controlled separately. This is also required by fixation-local metric licensing [15, Thm. 9.3].

Theorem 14.1 (A sufficient global Minkowski realization). *Let M be a connected, simply connected, time-oriented, four-dimensional smooth manifold with a flat Lorentzian metric and geodesically complete Levi-Civita connection. Then M is globally isometric to Minkowski space. If the physical occurrence receiver is faithful on M , this isometry supplies a global affine event receiver.*

Proof. Flatness and simple connectivity make parallel transport path independent. An orthonormal frame at one point therefore extends to a global parallel frame $E_0, \dots, E_3$. The connection is torsion free, so $[E_a, E_b] = 0$. Each integral curve of E_a is an affinely parametrized geodesic because $\nabla_{E_a} E_a = 0$. Geodesic completeness makes every E_a complete. Their commuting flows define an action of the additive group $\mathbb{R}^4$ on M .

The four vector fields span each tangent space, so every orbit is open. Since M is connected, there is one orbit. The stabilizer Γ of a point is discrete, because the orbit map has invertible derivative there. Thus M is the homogeneous space $\mathbb{R}^4/\Gamma$, and the orbit map is its covering projection. Since M is simply connected, $\Gamma = \{0\}$. The resulting global flow coordinates have constant metric components equal to the inner products of the parallel frame. They give the required Minkowski isometry. Faithful physical receipt then preserves the original event distinctions under this global chart. $\square$

The conditions of Theorem 14.1 are sufficient, not necessary for using SR on a region. An incomplete open patch can already have an injective affine developing map and support every stated local comparison. Conversely, an exact globally flat Einstein branch must satisfy its actual source equation. A clock with nonzero stress cannot be added to a vacuum solution while retaining an exact vacuum equation. A test-device or negligible-backreaction regime must be declared as such; its operational idealization does not establish an exact occupied global Einstein solution.

Example 14.2 (Trivial holonomy without injective development). On $M = \mathbb{R}^4$ with coordinates (t, z, u, v) , let

$$g = -c^2 dt^2 + dz^2 + e^{2u}(du^2 + dv^2).$$

The map

$$\text{dev}(t, z, u, v) = (ct, z, e^u \cos v, e^u \sin v)$$

is a local isometry into Minkowski space with the timelike two-plane defined by its last two coordinates being zero removed. Hence g is flat. The manifold is simply connected and

has trivial holonomy, but dev identifies v and $v + 2\pi$. It is not injective. The geometry is geodesically incomplete toward the removed plane. This example shows that flatness and trivial holonomy alone do not establish a global injective event chart. Completeness supplies the missing condition in Theorem 14.1.

14.1 Which assumptions exclude the alternatives

The following diagnostic models identify the work done by the physical receiving conditions. They test the listed transport and frame implications. They are not offered as independently verified models of every theorem in the AASC corpus.

Example 14.3 (Witnessed paths without a uniform finite speed). Take events $\mathbb{R} \times \mathbb{R}^3$ and all finite piecewise-linear paths whose time coordinate strictly increases on each nontrivial edge. Retain an explicit witness for each edge and concatenation, stable occupant identities, and the original endpoint, comparison, restoration, and failure records. Every admitted transfer is path mediated; no primitive instantaneous global jump is used. The transformations

$$(t, x) \mapsto (t, Rx - vt + a), \quad R \in \text{SO}(3),$$

compose, invert, preserve the same-event reference, and carry finite path witnesses to finite path witnesses. They preserve irreversible forward record ordering and supply no distinguished inertial velocity in their frame law.

For any fixed positive duration, however, arbitrary finite spatial displacement is admitted. No uniform bound M exists. The closure of the reachable displacement set is $\{(t, x) : t \geq 0\}$, which is not pointed. Thus witnessed finite paths, exact reference, and reciprocal frame composition do not imply a finite causal cone. The model fails clause (iv) of Physical condition 5.11. This is compatible with R5's explicit separation of path-mediated transport from a numerical speed theorem [9, Sec. 2].

A nonzero Galilean boost cannot preserve the round finite cone of Theorem 5.12: it shifts the bounded spatial section by a nonzero velocity. Hence it cannot be an alternative frame receipt on that same propagation target. The exclusion follows from the physically supplied finite-boundary content, not from abstract closure alone.

Example 14.4 (A finite anisotropic cone). Allow elementary spatial velocities $\pm e_1, \pm e_2, \pm e_3$, together with positive rescaling and admitted finite concatenation. The limiting reach cone is

$$\mathcal{C}_1 = \{(t, x) : t \geq |x_1| + |x_2| + |x_3|\}.$$

It is closed, convex, pointed, has nonempty interior, and satisfies a uniform Euclidean propagation bound. It fails full spatial rotational isotropy. Its boundary is not the zero set of a nonzero Lorentzian quadratic form. Indeed, a quadratic polynomial vanishing on the interior of a planar facet must have that facet's linear equation as a factor. The cone has more than two distinct facet hyperplanes, so such a quadratic would vanish identically. Finite propagation alone therefore does not select the round quadratic cone; the isotropy clause has independent content.

Example 14.5 (Composition without affine calibration). Let $\psi(t, x_1, x_2, x_3) = (t, x_1 + x_1^3, x_2, x_3)$. It is a smooth bijection with smooth inverse. Conjugating a Poincaré action by ψ gives the reciprocal composable family $\psi^{-1}F\psi$, which preserves the correspondingly transported event and causal records. These coordinate maps are generally nonlinear. Thus composition and exact event reference alone do not establish affine formulas in a chosen display. The witnessed additive displacement and homogeneous receipt of Physical condition 5.9 are what make Lemma 5.10 applicable.

Example 14.6 (Cone preservation without a physical scale). For $s > 0$, the map $(t, x) \mapsto s(t, x)$ preserves the entire round causal cone and all its order relations. With fixed time units it changes a nonzero rest-clock duration by the factor s . Consequently a causal cone alone determines only a conformal interval scale. Lemma 6.4 and Theorem 6.5 show how group composition, rotations, and the retained rest-clock calibration remove this freedom without presupposing a moving-clock formula.

These examples also specify the sense of necessity in the main result. Within the fully declared homogeneous, isotropic, finitely bounded, rest-calibrated physical target, the admitted frame law is forced. Every other non-degenerate determinate physical target also necessarily incurs the kernel, even when it fails one of those receiving conditions. They do not refute the conditional frame theorem, and the theorem does not classify them by silently changing their original transport or clock queries. Its exact comparison class is preserved throughout [2, 3, 5].

Corollary 14.7 (Exhaustion of laws on the unchanged receiving target). *Fix an occupied witnessed source refinement with the frame, clock, endpoint and source-response conditions used in the corresponding results above. Every admitted orientation- and future-preserving homogeneous inertial frame receipt is a Poincaré restriction. Its certified ideal duration, simultaneous-endpoint length and frame-indexed simultaneity readouts obey the proved formulas. Every completed stable source with the stated pulse receipt has $M = (E_0/c^2)I_3$.*

Consequently a candidate law differing on one of those unchanged original queries cannot also satisfy the same complete refinement and physical conditions. It either changes a retained source, query, calibration, operation domain or specialization, or fails the claimed standing as a law on that target.

Proof. Theorem 6.5 exhausts the permitted frame maps on this class; Theorem 7.6 proves uniqueness of duration in the stated ideal-clock class. The endpoint and simultaneity constructions in Theorems 9.2 and 9.4 then determine their readouts for the unchanged event-selection queries. Theorem 11.9 determines the response from the attached total source and endpoint impulse balance. Two differing answers cannot both satisfy these conclusions at fixed inputs. If an alternative identifies different original inputs, the refinement obstruction in Theorem 4.8 applies instead. Thus the exhaustion follows from the general rigidity and response proofs, not from an assertion that the illustrative examples enumerate every possible physical theory. $\square$

The result fixes these laws within the declared comparison class. It leaves distinct microscopic realizations with the same complete admitted answers, different physical targets, and richer regimes outside that class open. Equality with the entire Poincaré group still requires the frame supply of Proposition 6.7.

15 The assembled special-relativity theorem

The main result composes the physical source constructions with their Lorentzian realization. The proof has a fixed subject throughout: the originally identified source with its comparison, continuation, device and response records. The following definition specifies how the branch conditions enter that construction.

Definition 15.1 (Coherent occupied AASC construction datum). Fix a non-degenerate determinate physical domain D and its complete original questions. A datum for the primary construction consists of the following successive, jointly witnessed stages.

- (i) A certified finite-dimensional regular Hamiltonian source sector or presentation satisfying the hypotheses of Theorem 3.2, together with its realized persistence, stable comparison and transport witnesses. The actual event, clock, endpoint and response records satisfy the source-attachment criterion of Lemma 3.4. The comparison and persistence constructions of Section 3 apply to these records. A larger field-theoretic or Hilbert/Banach source is joined to this sector by actual joint record and projection maps; it is not asserted to be finite-dimensional, and its existence alone does not supply those maps.
- (ii) On the resulting event and comparison source, the regular homogeneous operations of Definition 5.1, the intrinsic transport datum of Definition 5.4, and the covered common rest calibration of Physical condition 6.3. These conditions concern the outputs and operations of the preceding source stage.
- (iii) On that same source, the occupied clock and endpoint targets used in Sections 7–9, and the attached complete response, finite charges, stable active source and controlled-pulse records used in Section 11. An invariant complete response prescription may supply its active action by Proposition 11.5; an invariant attached functional or complete sector ledger may supply its Ward identity by Proposition 10.3 or equation (26).
- (iv) When the common dynamical conclusion is claimed, a certified nonempty marked source-preserving refinement over a joint Einstein–Schrödinger eligible history domain, with its actual source projection, source-to-curve and record maps. Where its metric recipient is identified with the transport-constructed SR recipient, their original cone and calibration records satisfy Lemma 12.2, pointwise or on the declared common flat carrier. On the minimal comparison branch, Theorem 10.1 supplies its connection. Flatness or the stated controlled local approximation is certified on the domain of extended SR measurements.

At every stage, additional witnesses are attached to the previously constructed source by Definition 4.7: they retain the old questions, original applicability and failures, and the exact identities between lifted operations and their source projections. Their domains have actual joint occupants. Separate nonempty marginal examples do not establish this joint condition. Successful exits from a selected stage remain recorded as exits. The calibrated alternative of Section 5.4 can replace stage (ii) when its comparison data have already been faithfully received.

This datum contains the physical branch conditions and their attachment. It does not contain a Lorentz transformation formula, a moving-clock duration formula, a mass shell, or $E_0 = m_i c^2$ as an assumed answer. The datum is physically instantiated only where its original witnesses are independently warranted. The local ideal-body example verifies the response-model component; the capstone verifies its own declared history target. Theorem 15.10 below additionally constructs a joint occupant for the declared fixed-background branch, including clocks, finite signals, phase-preserving controls and an explicit projection to the common vacuum/quantum packet. Its Einstein source is the retained vacuum baseline; it is not identified with the positive laboratory response. Section 15.3 proves the obstruction to that stronger flat, common-total-source identification.

Theorem 15.2 (The necessary kernel of physical objecthood and its Lorentzian realization). *For every non-degenerate determinate physical domain D , K is necessary. Every coherent occupied AASC construction datum on D produces the following source-preserving chain, with the indicated calibrated special-relativistic laws uniquely determined on its declared target:*

- (a) *Every physical object, state, carrier, record, endpoint and determinate incidence in D , including each richer construction, selector, calibration, source map, realized law and physical instance of the specialization conditions, necessarily instantiates K in its own*

right by Proposition 2.4. No further applicability or adoption hypothesis is required. Theorem 3.2 and Lemma 3.4 construct the characteristic source and its physical record descent. Proposition 3.5 and Theorem 3.6 construct its comparison and persistence interface. Definition 4.7 and Theorem 4.8 then receive each next stage as a minimal complete attained interface retaining the original source questions, with an exact forgetful map back to their attained source receiver.

- (b) Where their displayed antecedents hold, Proposition 5.3 supplies the regular comparison role from complete action records and unique repetition, Proposition 5.6 supplies strict clock positivity from completed order and rest comparisons, and Lemma 6.2 supplies common rest calibration from completed interval comparisons. These are sufficient construction routes to the receiving conditions; a datum may also supply those conditions directly. Theorem 5.2 constructs affine displacement from the original comparison composition. The intrinsic transport construction, Theorem 5.7, constructs its spatial quadratic form and Lorentz cone, and Corollary 5.8 supplies its actual three-dimensional rotations. Corollary 6.6 gives the rest-calibrated Lorentz law. On the already calibrated alternative, Theorems 5.12 and 6.5 give the same conclusions. Every admitted oriented inertial frame receipt is a Poincaré restriction. With the additional frame and translation supply in Proposition 6.7, the full proper orthochronous Poincaré group is obtained. Proposition 3.7 applies to the physical interval use of this constructed form.
- (c) On the occupied ideal-clock and endpoint-standard domains of Sections 7–9, the same retained event, clock and rod source gives the unique ideal duration, time dilation, length contraction and relativity of simultaneity. Radar synchronization is obtained on its executed two-way signal domain.
- (d) On the attached total-source, stable active rest orbit and controlled-pulse target of Physical conditions 11.1, 11.3 and 11.8, operational inertia satisfies $M = (E_0/c^2)I_3$. Its scalar value obeys $E_0 = m_i c^2$, with the derived energy, momentum and invariant mass shell. Proposition 10.3 and equation (26) supply the source divergence law on their declared constructions; Lemma 11.4 supplies those protocol inputs on the complete invariant variational branch, and Proposition 11.5 constructs the tensor action from complete physical protocols; Proposition 11.14 gives a nonempty ideal response-model instance, without substituting that model for independent empirical warrants.
- (e) On a certified nonempty marked source-preserving refinement over a joint eligible domain where the occupied capstone's Einstein and Schrödinger recipient functors are both defined, Theorem 12.6 constructs the Lorentz-path continuation action from the metric, actual marked curve and retained records. On the minimal comparison branch, Theorem 10.1 and Corollary 10.2 identify its transport with the original source transport. It preserves causal tangent geometry, geometric duration and source composition. Lemma 12.2 identifies its metric with the transport-constructed form on their certified overlap; Lemma 12.3 supplies an exact finite original-query certificate for that matching; the continuation agrees with ordinary SR displacements on the common flat specialization. The metric, quantum and constructed relativistic recipients retain one upstream history source. On the additional common free quantum translation target, their relativistic/nonrelativistic evolutions satisfy Theorem 13.2.
- (f) Every alternative law satisfying the same complete source refinement, physical conditions, calibrations and original query domains has the same indicated frame, duration, endpoint and operational-response formulas. A different answer on an unchanged original query requires a changed specialization or failure of the claimed faithful law. Minimal complete receivers of the same original target are isomorphic on their attained images.

The physical refinements are those identified in Table 1 and Appendix B, with their independent witnesses and exact scopes retained. No later metric or energetic success changes an original failure or supplies an unoccupied physical use.

Proof. Theorem 2.1 establishes the universal dependency. Proposition 2.4 applies it directly at every construction stage, including the richer selectors, source maps and law receipts. Corollaries 2.2 and 2.3 include every physical witness and its faithful redescription. The characteristic forcing proof identifies the source’s invisible directions; Lemma 3.4 descends the actual joint records and operations through that forced quotient. The comparison and persistence propositions then form the record structures used by the geometric and response stages. These conclusions construct the source of the SR queries rather than merely classifying an unrelated Hamiltonian example.

For each subsequent stage let R_i be its actual joint records and π_i their declared projection to the preceding source. Theorem 4.8 gives the attained receiver, exact operation descent and the induced map retaining the previous complete profile. Composing these maps retains every earlier original answer. For admitted lifted operations the identities $\pi_i f_i = f_{i-1} \pi_i$ compose by substitution; domain and failure tests compose because they are retained answers too. Shared inputs are transported once along their joint incidences. Thus all constructions below operate on one coherent source lineage. Their joint occupants come from Definition 15.1; the quotient theorem does not create them.

The role-completion propositions provide sufficient constructions of the receiving conditions where their displayed antecedents hold. The assembled theorem receives the certified conditions and does not require those particular source presentations from every datum. Complete homogeneous action profiles and unique repetition give the effective vector comparison group; completed antisymmetry, the weakly monotone clock and the actual rest action give strict clock separation; normalized ordered interval comparisons give common rest duration. These are Propositions 5.3, 5.6 and Lemma 6.2. Their source properties retain the stated warrants and are verified in the joint occupation below. On this source, Theorem 5.2 constructs the affine displacement action. Theorem 5.7 constructs the spatial form and cone, and Corollary 6.6 removes scale. The calibrated route gives the same result through Lemma 5.10 and Theorems 5.12–6.5. Proposition 6.7 supplies the full group when its additional frame supply is present. The metric is applied to original standing-positive event comparisons, as Proposition 3.7 requires.

Theorems 7.6, 8.2, 9.1, 9.2 and 9.4 determine the clock and endpoint laws on their exact physical records. Proposition 10.3 or the finite sector calculation supplies the closed/exchange source identity on its own domain, while the charge and flux conditions supply its surface use. Lemma 11.4 derives solution-set and response-graph transport for the complete invariant prescription; Proposition 11.5 constructs the source action from those operational inputs; Theorem 11.9 then determines the response by charge covariance and actual impulse balance. The worked ideal source verifies nonemptiness of that model class without asserting its empirical realization.

The explicit joint projection retains the regular sector’s scope within the larger common source. Lemma 12.2 identifies the two metric forms from their common cone and calibration records. Theorem 10.1 then fixes the transport of the minimal metric realization. Theorem 12.6 constructs its continuation arrows, their composition and geometric-duration equality. Corollary 12.7 composes this proved action with the existing marked source projection. Corollary 12.8 identifies its output with affine SR displacements, and Theorem 13.2 proves the specified quantum overlap. Corollary 14.7 supplies the uniqueness/exclusion clause from the general rigidity and response results. Every step retains its source references and failure data through Theorem 4.8; hence the joined statement follows. $\square$

15.1 One occupied source with its clocks and finite signal protocols

The following construction populates the source, clock and signal conditions on one explicit ideal model. It is an occupation proof for the displayed physical branch. Its action is a substantive model choice, made after the general reconstruction theorem; it is not a new deduction of that action from kernel necessity. We use $c = 1$ in the construction and restore units at the end. Put $g_0 = -dt^2 + |dx|^2$ on the selected model carrier.

Choose $\mu_* > 0$. A massive clock has an unwrapped internal reading r , its conjugate energy $J > -\mu_*$, and total rest energy

$$\mu(J) = \mu_* + J > 0.$$

On an oriented timelike worldline its action is

$$I[\gamma, r, J] = \int (J \, dr - \mu(J) \, d\tau), \quad d\tau = \sqrt{-g_0(d\gamma, d\gamma)}. \quad (31)$$

The reading is an internal dynamical variable. Its contribution to the body's rest source is retained in μ .

Proposition 15.3 (Occupied continuous clock and its source). *On every finite coasting interval, (31) has solutions with constant J , a straight timelike worldline, and $dr = d\tau$. In any inertial time presentation the corresponding Hamiltonian and flow are*

$$H(p, J) = \sqrt{|p|^2 + \mu(J)^2}, \quad (32)$$

$$\dot{x} = p/H, \quad \dot{p} = 0, \quad \dot{r} = \mu/H, \quad \dot{J} = 0. \quad (33)$$

The clock is populated continuously; its rest reading has rate one, and its internal energy is included in the same body's response source.

Proof. Variation in r gives $dJ = 0$, and variation in J gives $dr = \mu'(J) d\tau = d\tau$. The remaining variation is the constant-positive-rest-energy worldline equation. In coordinate time,

$$L = J\dot{r} - \mu(J)\sqrt{1 - |\dot{x}|^2}, \quad p = \frac{\mu\dot{x}}{\sqrt{1 - |\dot{x}|^2}}.$$

The Legendre transform in $\dot{x}$ gives (32); $J\dot{r}$ is already canonical. Hamilton's equations give (33). Since H and μ are constant, its complete flow is

$$\Phi_s(x, p, r, J) = (x + sp/H, p, r + s\mu/H, J).$$

Every $J > -\mu_*$ and initial (x, p, r) gives a solution. The same action supplies the worldline metric response with weight μ ; replacing that weight by μ_* would discard the internal clock energy. $\square$

Proposition 15.4 (Regular source and exact attachment through a controlled pulse). *Let*

$$P = \mathbb{R}_x^3 \times \mathbb{R}_p^3 \times \mathbb{R}_r \times (-\mu_*, \infty)_J, \quad \omega_P = \sum_i dx_i \wedge dp_i + dr \wedge dJ.$$

An original protocol context a specifies a smooth compactly supported force $F_a : \mathbb{R} \rightarrow \mathbb{R}^3$. Put

$$H_0(p, J) = \sqrt{|p|^2 + \mu(J)^2}, \quad H_a(s, z) = H_0(p, J) - F_a(s) \cdot x.$$

Let $\Phi_{t,s}^a$ be its evolution from time s to time t . On $S = T^\mathbb{R}_s \times P$, with $\omega = ds \wedge dp_s + \omega_P$, put*

$$C_a = \{p_s + H_a(s, z) = 0\}, \quad \pi_a(s, z) = \Phi_{0,s}^a z.$$

This is a nonempty regular Hamiltonian source satisfying H1–H3 of Section 3.2. Its characteristic fibres are connected real lines, and its reduced symplectic carrier is (P, ω_P) .

For every original marked time σ , define the actual records by

$$z_\sigma = \Phi_{\sigma,s}^a z, \quad e_\sigma = (\sigma, x(z_\sigma)), \quad r_\sigma = r(z_\sigma), \quad J_\sigma = J(z_\sigma). \quad (34)$$

They are constant on complete characteristic fibres, including during and after the controlled interval. Body identifiers, marked event incidences and original clock-use domains descend with them. The zero force context recovers the complete coasting formulas $x_\sigma = x + (\sigma - s)p/H_0$ and $r_\sigma = r + (\sigma - s)\mu/H_0$.

Proof. Hamilton's equations are

$$\dot{p} = F_a(s), \quad \dot{x} = p/H_0, \quad \dot{r} = \mu/H_0, \quad \dot{J} = 0.$$

Their flow is global: $p(t) = p(s) + \int_s^t F_a(u) du$, J stays in its open interval, and x, r are integrals of smooth functions with $|\dot{x}| < 1$ and $0 < \dot{r} \leq 1$. Thus there is no finite-time escape. The constraint $\phi_a = p_s + H_a$ is regular since $\partial\phi_a/\partial p_s = 1$. On C_a , identified with $\mathbb{R} \times P$, its characteristic direction is $\partial_s + X_{H_a(s)}$ and

$$\omega_{C_a} = \omega_P - ds \wedge d_z H_a = \pi_a^* \omega_P.$$

For the last equality use the parametrization $\Psi(s, z_0) = (s, \Phi_{s,0}^a z_0)$. At fixed s the evolution is symplectic. Its mixed time–state terms in $\Psi^* \omega_P$ are $ds \wedge \Psi^* d_z H_a$, and cancel those of the displayed second term. Hence $\Psi^* \omega_{C_a} = \omega_P$ on the initial-data factor, proving the identity. Its kernel is exactly the stated line. The fibres are $(s, \Phi_{s,0}^a z_0)$, $s \in \mathbb{R}$, so π_a is an attained submersion with connected fibres and nonzero characteristic direction.

Take as admitted reduced observables all smooth functions on P . Their pullbacks are constant along the characteristic direction. Any two ambient extensions of one such observable differ locally by $h\phi_a$. Its bracket with a physical extension vanishes on C_a , because $\{\phi_a, F\}|_{C_a} = 0$; the same argument applies in the other slot. The resulting bracket is the pullback of the reduced bracket, proving H1. Smooth functions separate points and their differentials span each cotangent space of P , proving H2. Nondegeneracy of ω_P and the pullback identity prove H3. Thus Theorem 3.2 applies by direct calculation.

The evolution cocycle gives, for every h, σ ,

$$\Phi_{\sigma,s+h}^a \Phi_{s+h,s}^a z = \Phi_{\sigma,s}^a z.$$

It proves invariance of all marked records under characteristic redescription. Their numerical, identity and applicability records therefore satisfy Lemma 3.4. Event coincidence is equality of e_σ . A clock-use record additionally retains its body identifier and r_σ ; clocks at one event are not thereby identified. The force context is an original part of the query. A coasting extrapolation in the zero-force context is never substituted for the actual future record in a driven context. In particular no future query is deleted to make the pulse attach. $\square$

The prescribed control has a covariant action description. In its adapted inertial presentation put $A_a = (F_a(t) \cdot x) dt$ and add $\int_\gamma A_a$ to (31). This one-form is a declared external input, held fixed in metric variation. It gives precisely H_a above, leaves the body's metric response unchanged, and supplies its recorded external current. In covariant form the equation is $\mu dU_\mu/d\tau = (dA_a)_{\mu\nu} U^\nu$; antisymmetry makes the force orthogonal to U . The body's physical energy obtained from its stress is H_0 . The driven canonical generator H_a additionally contains the external potential; the two agree on the zero-control endpoint branches. Transforming the

complete protocol transforms A_a as a one-form along with the body. Its expression need not remain a spatially uniform force in an arbitrary fixed presentation. This covariance does not identify the controlled body with a closed actuator system.

For fixed protocol parameters the additional endpoint and signal histories below are explicit functions of the same initial source records. The finite inertial clock experiment is scheduled where $F_a = 0$. A second clock is placed at a chosen constant rest separation with synchronized reading; its initial data and coasting history are fixed. The original joint source contains these solution records and their actual shared vertices. This supplies an actual map from the regular sector to the enlarged protocol history. Parameter changes are distinct declared protocol contexts. The regular Hamiltonian assertion applies to the whole smooth body history, including its controlled pulse; ideal ray scattering vertices are retained as distributional history events.

Lemma 15.5 (Balanced finite-energy reflection). *Let U be the unit future velocity of a coasting clock endpoint and n a unit spacelike vector orthogonal to U . Two rays of equal rest energy $\epsilon > 0$, incident simultaneously with future momenta*

$$k_+ = \epsilon(U + n), \quad k_- = \epsilon(U - n),$$

can each be reflected into the opposite spatial direction with their ray identities retained. The complete vertex has zero four-momentum defect, and the endpoint may continue with exactly the same U and J .

Proof. Both rays are future null and their incoming sum is $2\epsilon U$. The two reflected momenta have that same sum. Thus the sum of the impulse transfers to the endpoint is zero, and no change of its rest energy is required. The model's local scattering prescription admits precisely this balanced elastic event; it preserves the individual ray identity while reversing its spatial momentum. More generally two opposite rays may be redirected into any opposite pair of the same energy with the same balance. This explicit scattering prescription is part of the ideal model. $\square$

Proposition 15.6 (Joint pulse clock, radar and finite response ledger). *For every $\ell > 0$, $\epsilon > 0$ and finite positive integer N , there is an occupied joint protocol with two finite-energy comoving clock endpoints at rest separation ℓ , a positive-energy main signal making N round trips, and finitely many compensating rays. Both endpoint clocks retain the same internal state throughout the certified interval. The main-signal return events occur at clock readings*

$$2\ell, 4\ell, \dots, 2N\ell$$

after its initial recorded departure. The same construction gives every finite collection of radar exchanges used on this inertial network.

The complete finite body-ray stress ledger is distributionally conserved, has finite charges, and has zero side flux on a sufficiently large spatial cylinder over every bounded time slab. The pulse and radar readouts agree with the continuous clock of Proposition 15.3.

Proof. In the common rest presentation take the endpoints at 0 and ℓn . The main ray moves at unit speed, so its successive target reflection events are fixed recursively by these two worldlines. At every such event prepare an equal-energy ray from the opposite exterior side, arriving at that same event. Its initial point on an earlier time slice is obtained by tracing its straight null line backward. Only one compensating ray per reflection is required. By Lemma 15.5, the two rays reflect while the endpoint keeps its original state. The compensating ray escapes

outward. Rays crossing away from designated endpoints do not interact in this model. All preparations and shared reflection incidences are therefore explicit finite source records.

The main ray traverses distance ℓ on each leg, giving the displayed return readings since the rest clock has rate one. The initial departure can itself be a recorded balanced reflection, so it need not be an emission that changes the endpoint's rest energy. After the certified final vertex, unmatched rays transmit under the declared scattering rule. Their original continuation is retained even when a further clock return ceases to be certified. A finite radar schedule uses the same paired preparation at its reflection sites. Its incoming and outgoing travel durations equal the rest separation, so the continuous reading satisfies the midpoint rule.

For completeness, each ray has the reparametrization-invariant action

$$I_{\text{ray}}[z, e] = \frac{1}{2} \int e^{-1} g_0(\dot{z}, \dot{z}) \, d\lambda, \quad e > 0.$$

Variation in e imposes nullity and variation in z gives constant $e^{-1}\dot{z}$ on a free segment. In the affine gauge $e = 1$ put $\dot{z} = k$, with future-positive energy. Metric variation at fixed z, e gives the attached witness $W_{\text{ray}}(h) = \frac{1}{2} \int k^\mu k^\nu h_{\mu\nu}(z) \, d\lambda$. Thus its distributional tensor is obtained from the same ray action. The massive clock's canonical term $J \, dr$ is metric-independent; its metric derivative is the massive worldline witness with weight μ . At a shared scattering event the endpoint variations of the actions cancel by total four-momentum balance. The reflection rule selects the stated balanced solution among the permitted vertex data; the free action alone does not select that constitutive rule.

The complete response ledger is the sum of these attached witnesses, with every massive body and null ray piece retained. On each free segment its divergence vanishes. Distributional integration by parts reduces the divergence at a vertex to the sum of outgoing minus incoming four-momenta, which vanishes by the balanced rule. The ledger is finite, so its charges are finite. Over any bounded time slab its finitely many finite-speed trajectories lie inside a sufficiently large spatial cylinder; no trajectory crosses the side of that cylinder. This proves the stated side-flux and charge conditions on their distributional domains. No infinite-mass reflector or zero-energy material signal has been used. $\square$

Proposition 15.7 (Occupied comparison, cone and calibration records). *The same family of clock-ray protocols supplies regular homogeneous comparison, the intrinsic completed transport datum, common rest-clock calibration, and the ideal inertial readouts used in the SR construction. The completed transport cone and the reconstructed quadratic form are*

$$\mathcal{C} = \{(t, x) : t \geq |x|\}, \quad q(t, x) = t^2 - |x|^2.$$

The comparison metric and the metric in (31) have the same cone and common rest calibration.

Proof. Whole-source translations supply the free transitive action of the connected simply connected abelian group $\mathbb{R}^4$ on events. Time difference is its nonzero additive duration record. All future null increments are occupied by straight rays. Their finite concatenations are mediated by recorded paired redirection vertices, and arbitrary positive durations and translated copies are admitted by the same formulas. Every finite nonnegative combination lies in the displayed closed cone by the triangle inequality. Conversely, if $x \neq 0$ and $t \geq |x|$, set

$$n = x/|x|, \quad a = (t + |x|)/2, \quad b = (t - |x|)/2.$$

Then $(t, x) = a(1, n) + b(1, -n)$. The rest ray is obtained with $a = b$, and zero is the identity. Thus the completed-reach query is exactly the displayed cone, with time strictly positive away from zero. Its unit-time section is the closed spatial unit ball. Rest rotations of the entire

protocol source preserve all its laws and act transitively on that ball's boundary. Theorem 5.7 therefore reconstructs the stated form.

Lemma 11.4 applies to the complete coasting worldline prescription and its transported endpoint variations. The covariant worldline action and the balanced vertex law admit all proper orthochronous Lorentz re-presentations of these same records. Their compositions and inverses retain the original calibrating clock pair. Their effective translation group has no nonidentity element whose double is the identity and its complete displacement profile is separated by its event action, verifying Proposition 5.3. The displayed cone is pointed; its clock is nonnegative, and full rest rotations have no common fixed nonzero spatial vector. Proposition 5.6 therefore applies to this same completed transport. Signed coasting-clock interval comparisons form the Archimedean divisible ordered group $\mathbb{R}$ with the retained unit, so Lemma 6.2 gives their common normalized receipt. On a rest clock, (33) gives $\dot{r} = 1$, so any nonzero interval gives the common rest calibration. Translation invariance, same-ray additivity, positivity, continuity, rest-presentation coverage and scalar preservation of the internal reading verify all clauses of Physical condition 7.5 on these coasting intervals. The endpoint clocks and finite pulse records are the actual witnesses just constructed. The metric matching follows either directly from the two forms or from Lemma 12.2 and the common clock pair. $\square$

Lemma 15.8 (A smooth pulse retaining the complete final clock phase). *Fix $T > 0$. For all sufficiently small final velocities $v \in \mathbb{R}^3$ there is a smooth subluminal velocity protocol, constant outside $[-T, T]$, whose massive clock history has*

$$x(t) = 0, \quad r(t) = t \quad (t \leq -T), \quad x(t) = vt, \quad r(t) = t\sqrt{1 - |v|^2} \quad (t \geq T).$$

The source keeps the same J and μ throughout. Thus the final coasting history agrees with the actively boosted rest history in its position and its retained physical clock phase. The compact pulse can be taken smooth in the vector v near zero.

Proof. Choose a smooth step $f_0 : [-T, T] \rightarrow [0, 1]$ with $f_0(-t) = 1 - f_0(t)$, equal to zero near $-T$ and to one near T . Choose an even nonnegative smooth function ζ compactly supported in $(-T, T)$ and positive somewhere that $f_0 \neq 1/2$, and put

$$h = (f_0 - 1/2)\zeta, \quad f_A = f_0 + Ah.$$

Then h is odd, $f_A(-t) = 1 - f_A(t)$, and $\int_{-T}^T f_A dt = T$ for every A . Moreover

$$\int f_0 h dt = \int (f_0 - 1/2)^2 \zeta dt > 0.$$

Since f_0 takes intermediate values on a nonempty open interval, $\int f_0^2 < T$. The function $A \mapsto \int f_A^2$ is strictly increasing for $A \geq 0$ and tends to infinity. Hence there is a unique $A_0 > 0$ with $\int f_{A_0}^2 = T$, and its derivative there is positive.

For b near zero define, using its smooth extension at $b = 0$,

$$F(b, A) = \frac{\int_{-T}^T \sqrt{1 - b f_A(t)^2} dt - T(1 + \sqrt{1 - b})}{b}, \quad F(0, A) = \frac{1}{2} \left(T - \int f_A^2 dt \right).$$

The extension is smooth by the Taylor expansion of the square root on a common compact parameter neighborhood. At $(0, A_0)$, $F = 0$ and $\partial_A F = -\int f_{A_0} h \neq 0$. The implicit-function theorem therefore supplies smooth $A(b)$ near zero with $F(b, A(b)) = 0$.

Set $b = |v|^2$ and

$$u_v(t) = v f_{A(|v|^2)}(t), \quad x(t) = \int_{-T}^t u_v(s) ds, \quad r(t) = -T + \int_{-T}^t \sqrt{1 - |u_v(s)|^2} ds$$

on the pulse interval. For sufficiently small v , these velocities are uniformly subluminal. The first integral identity gives $x(T) = Tv$ and $F = 0$ gives $r(T) = T\sqrt{1 - |v|^2}$. The endpoint collars make $u_v = 0$ near $-T$ and $u_v = v$ near T , so the stated exterior formulas join smoothly with all derivatives. Dependence on $|v|^2$ makes the family smooth in the vector v , including zero. Fix the selected internal state J_0 and write $\mu_0 = \mu(J_0)$. In Proposition 15.4 choose the explicit input

$$F_v(t) = \frac{d}{dt} \left(\frac{\mu_0 u_v(t)}{\sqrt{1 - |u_v(t)|^2}} \right).$$

It is smooth, compactly supported, and smooth in v . With initial $p = 0$, $J = J_0$, $x = 0$ and $r = -T$ at $-T$, the driven Hamiltonian solution is exactly the displayed history: $p/H_0 = u_v$ and $\mu_0/H_0 = \sqrt{1 - |u_v|^2}$. Thus its actual marked records attach through the full pulse by the evolution cocycle. The corresponding worldline force is $f = \mu_0 dU/d\tau$, orthogonal to U . Its divergence/current is the retained external exchange of the same body response. All intermediate clock readings remain original answers. $\square$

The velocity is allowed to overshoot or reverse briefly within the compact pulse. This freedom matters: an ordinary smooth step confined between zero and v generally leaves a nonzero clock-phase offset. The present construction removes that offset by an explicit protocol while retaining all its intermediate readings. It occupies a neighborhood of the rest response sufficient for the operational inertia derivative; it does not assert reachability of arbitrary large final velocities by this local implicit-function construction.

Normal traces for the ideal ray ledger. The finite-charge assertion includes tilted affine spacelike sections. For a null segment use an affine parameter with $\dot{z} = k$ and write

$$T_{\text{ray}}^{\mu\nu} = \int k^\mu k^\nu \delta^{(4)}(x - z(\lambda)) d\lambda.$$

For a massive segment the corresponding expression is $\mu \int U^\mu U^\nu \delta^{(4)}(x - z(\tau)) d\tau$. Let an affine spacelike hyperplane have future-positive unit normal covector n . At a transverse crossing, $n(k) > 0$; evaluation of the delta trace contributes $1/n(k)$, which cancels the normal contraction, leaving the crossing charge k^ν . The massive contribution is similarly μU^ν . Every complete future-directed ray or body trajectory in this finite protocol has finitely many vertices and straight nonzero causal asymptotic tails. Consequently $n(z)$ is strictly increasing from $-\infty$ to $+\infty$, and every such hyperplane has exactly one crossing. If a section contains a vertex, the coincident incoming and outgoing representatives give the same total trace by vertex balance. Hence the sum is finite on every affine spacelike section. Between any two such sections, each participating trajectory segment is bounded; a sufficiently large cylinder contains their finite union and has zero side flux on those segments. The compensating rays are complete incoming histories specified by their data on an initial slice. They are not created at that slice.

Theorem 15.9 (Occupied joint source, device and response sector). *There exists a nonempty declared fixed-background ideal source family in which the following data are verified together:*

- (i) *the regular characteristic source, its retained body identity, marked events, continuous clock readout and their exact attachment maps;*
- (ii) *homogeneous displacement, executed null transport, completed-cone positivity and rest isotropy, common rest calibration, persistent endpoint standards, and finite pulse-clock and radar uses agreeing with the continuous clock;*
- (iii) *a positive complete coasting body response with the clock energy included, finite closed-network charges and exact vertex balance, and a neighborhood of compact controlled*

histories preserving the same J and matching the final active orbit including the original clock phase.

The controlled histories in (iii) use the explicit ideal input class of smooth external one-form controls with compact support in the reference time. All quoted uses retain their original protocol parameters, shared incidences, domains, failures and selection exits. The Einstein–Schrödinger recipient and its source matching are additional constructions, not conclusions of this fixed-background occupation statement.

Proof. Choose $J_0 > -\mu_*$ and initial clock data. Proposition 15.4 provides the nonempty regular source and its record maps. For any finite protocol choices, Proposition 15.6 constructs the second endpoint, all ray histories and every shared vertex from those same initial records. Proposition 15.7 verifies their comparison and calibration properties. The worldline and ray response ledger and the preceding trace calculation verify its complete charge uses. Lemma 15.8 provides a smooth occupied control family on that same internal source; its explicit force F_v realizes that history in the same regular source, and its final position and clock identities prove the active-orbit matching. The unchanged J keeps the same total rest response $\mu(J_0)$. Every additional operation is defined on these actual solution records with its original incidence condition. Its totalized answers and domains therefore form the witnessed refinement of Definition 4.7; that theorem supplies its minimal complete receiver without creating any of these occupants. $\square$

The two source questions in the protocol packet. The body charge used by the controlled-inertia theorem is the charge of one retained clock body, including all of its internal clock energy: $E_{0,\text{body}} = \mu(J_0)$. The complete laboratory ledger additionally contains the second endpoint and every incoming and outgoing ray. Its charge is their sum and need not have zero spatial momentum. These are explicit distinct projections of the same retained joint protocol. Balanced reflection gives zero net current into the selected body, while a controlled acceleration supplies its recorded nonzero external current. The laboratory charge is never substituted for the body’s rest charge.

Restoring units gives

$$H = \sqrt{c^2|p|^2 + \mu^2}, \quad \dot{x} = c^2 p/H, \quad \dot{r} = \mu/H, \quad \Delta r = \frac{\sqrt{c^2 \Delta t^2 - |\Delta x|^2}}{c}.$$

The finite pulse period is $2\ell/c$. The retained total rest source is $E_0 = \mu(J_0)$, including the clock’s energy. Its operational inertial coefficient is obtained by applying the manuscript’s controlled-response theorem to this same frozen internal source, rather than by treating the Hamiltonian’s dispersion as a separate proof of that theorem.

15.2 An occupied common-history refinement and exact metric matching

The previous construction gives actual solution records of one finite protocol. Choose its pulse-clock experiment before the controlled acceleration interval. The second endpoint remains coasting and supplies the reference clock θ throughout; the response body’s own reading is retained separately during its acceleration. Include all proper orthochronous Lorentz transforms and translations of the coasting protocols, the small-velocity pulse family and its transformed copies, and their finite validity intervals. Write $\mathcal{A}$ for these explicit protocol records, including the initial reduced clock data. It is a union of finite-dimensional smooth parameter charts, indexed by the finite signal schedule. Scattering events remain labelled records, so smoothness of a field at a hard vertex is not asserted. Choose $J_0 = \mu_* > 0$ for each endpoint. The all-rest

configuration, two endpoints at positive separation, one completed cycle and zero control velocity give an explicit member a_0 .

Let (Q_*, R_t) be the full-ray oscillator carrier and unitary ray group in the occupied Module C packet [25, Defs. 24.1, 24.3, 24.5, Thm. 24.7]. Its Hamiltonian, domain and full-ray extension are those specified there. These quantum data are retained physical model content; the construction below does not derive a spectrum from clock translation.

Theorem 15.10 (Joint occupation on the declared fixed-background branch). *There exists a nonempty witnessed joint refinement carrying the regular source, clocks, endpoint standards, finite signal records, attached positive body response, stable active orbit and controlled impulses of Definition 15.1(i)–(iii). Over the Module C joint vacuum-Einstein and full-ray Schrödinger eligible packet, it has an explicit marked history projection, exact cone/calibration matching and the Lorentz-path continuation of clause (iv), with the metric source scoped to its vacuum baseline. In particular, these data have joint occupants, rather than only separately occupied marginal domains.*

The same refinement retains the positive laboratory response as a distinct source query and explicitly records its omission from the vacuum Einstein equation. It does not identify that response with the baseline’s zero stress, assert Module B’s operator-source matching, or assert a new Module D frozen-target certificate.

Proof. Work in coordinates $X = (ct, x)$ on the model carrier and write $g_0 = -(dX^0)^2 + |dx|^2$. Take the original nonempty Module C clock interval I , chosen large enough to contain the finite protocol under consideration, and put

$$\mathcal{U} = I \times \mathcal{A} \times Q_* \times \mathbb{R}, \quad u = (\theta, a, q, \rho), \quad V = \partial_\theta + \partial_\rho.$$

The last factor is an original monotone record coordinate, distinct from every device reading. On each parameter chart this vector field is smooth, clock-normalized and nonzero. Its local flow is explicitly

$$\alpha_s u = (\theta + s, a, q, \rho + s), \quad \pi u = (\theta, q, \rho).$$

The underlying flow has exactly the original Module C interval domain: the full clock segment must remain in I . For each a , let $I_a \subseteq I$ be a connected interval on which the selected finite protocol uses are valid. These tests are independent of q . They mark a selected locus in the retained full carrier; they do not change the old flow domain. A successful underlying continuation leaving I_a has a recorded selection exit, and failed added device queries retain their failure answers. Only histories remaining in the selected locus assert continued validity of that device use. Since $\pi \alpha_s = \alpha_s^* \pi$, the projection is an exact refinement over the original Module C packet. The protocol parameter and its response/clock queries enrich the old source; they are not claimed to have been present in its original frozen parameter diagram.

Regular sector and records. Proposition 15.4 supplies the nonempty regular Hamiltonian sector and the actual reduction map. For each a its explicit initial-data map and two-time evolution give the actual body, event and clock records on the entire controlled history. The force in Lemma 15.8 realizes the phase-matched trajectory in this regular source; the finite signal construction attaches its earlier coasting device experiment. The evolution cocycle preserves the original readings before, during and after the pulse. The maps from $\mathcal{U}$ evaluate those fixed protocol histories at θ and on their registered intervals. The forgetful map retains their reduced initial source and all earlier queries. Gauge changes leave the marked formulas unchanged; restrictions, frame comparisons and protocol continuation commute with these evaluations. A shared reflection vertex is one event with its separate incident ray and clock-use records. Thus the source attachment and exact operation identities are verified on the

explicit carrier. The complete query profile retains the protocol, original endpoint references, applicability and exits, so the refinement theorem applies without an identified-input answer collision.

Common dynamical projections. Let m_0 be the certified-zero-source Minkowski Cauchy datum of the original packet. Define

$$\Pi_g(u) = m_0, \quad \Pi_H(u) = R_\theta q.$$

The vacuum constraints hold directly, its independently specified vacuum Cauchy vector field vanishes at m_0 , and

$$d\Pi_g(V) = 0 = F_{\text{vac}}(m_0), \quad \Pi_H(\alpha_s u) = R_s \Pi_H(u).$$

At every fixed θ , R_θ is a bijection on the full ray carrier. Full-ray saturation, transition preservation, the strongly continuous group extension and the stated spatial generator eligibility are consequently exactly those of the original occupied packet. No Hilbert generator is being treated as a smooth vector field on every full-ray state: the smooth upstream field advances θ and ρ , and the quantum quotient is its specified clock-dependent map. Both projections are defined on the same explicit states. Therefore $\mathcal{U}$ itself is a jointly eligible source, not merely a set mapped to two unrelated examples. For each fixed protocol a , the graph of its deterministic curve and record assignments is a marked subtype over this eligible source, isomorphic to the corresponding unmarked states. Taking the stated union of protocol charts gives the marked refinement. Thus the construction meets the marked-subtype requirement as well as its explicit refinement formulation; recipient eligibility and the original source maps are proved in either presentation. Old record and compatibility outputs are pulled back through π with their original query scopes; the new signal and exchange queries are additional records.

Marking and continuation. Let $z_a(\theta)$ denote the actual response-body history, using the coasting reference clock as parameter. For every admitted segment define

$$\Gamma_{u,s}(v) = z_a(\theta + v), \quad 0 \leq v \leq s.$$

The selected body curve is smooth and future timelike throughout the balanced-reflection and controlled family; balanced vertices have zero body impulse, while ray corners remain separate records; its full source endpoint includes a, q, ρ and the original validity records. Thus geometric coincidence does not create extra source composition. Restriction and concatenation obey

$$\Gamma_{u,s+t} = \Gamma_{u,s} \star \Gamma_{\alpha_s u, t}$$

whenever the complete endpoints and original domains permit the composition. Flat parallel transport is the identity, so its anti-development is explicitly

$$Y_{u,s}(v) = z_a(\theta + v) - z_a(\theta).$$

This verifies the recipient's identity and composition laws. Its geometric duration is the actual clock increment of the same body:

$$\frac{1}{c} \int_0^s \sqrt{-g_0(\dot{z}_a, \dot{z}_a)} \, dv = r_a(\theta + s) - r_a(\theta).$$

For the accelerated body it need not equal s , the reference clock's elapsed time. This difference is retained as an answer, not removed by relabelling the two clocks. At a_0 every sufficiently small $s > 0$ gives a nonidentity marked history.

Matching and response. Proposition 15.7 constructs the reach cone and calibrated form from this protocol family's comparison records. In the common units they give $q(X) =$

$(X^0)^2 - |x|^2 = -g_0(X, X)$. The same retained coasting clock pair supplies a nonzero timelike calibration. The protocol family covers the nine null-direction comparisons of Lemma 12.3, with their original signal records on this constant flat carrier. Together with that rest pair, they provide its finite matching certificate. Thus the hypotheses of Lemma 12.2 hold on the identical event carrier, throughout the flat measurement domain. The minimal comparison transport is the flat Levi–Civita transport used above. The clock-body action attaches its positive response with $E_0 = \mu(J_0)$, and the controlled pulse preserves the complete clock-bearing boosted endpoint by Lemma 15.8. Normal traces, finite charges and exchange balance are verified by that construction and Proposition 15.6. The frame, duration, radar and endpoint laws therefore apply to actual joint records. The internal clock energy is part of the same body’s response.

Source-sector identity and omission. The metric projection answers the vacuum-baseline question, with $T_{\text{grav}} = 0$. The laboratory projection retains its finite sum of massive and null source measures, and the controlled body’s declared exchange. On a finite reference-clock segment, retain a labelled measure on the history records of each sector, including its event, tangent or null momentum, energy weight and original sector identity. The massive measure is the pushforward of proper length; the null measure is its momentum-normalized affine measure. The registered linear test readouts recover each sector’s stress pairing exactly. Adjacent half-open clock intervals add these measures, so restriction and concatenation preserve the ledger without double counting a shared endpoint. Vertex impulses and the external control current are separate readouts of the same protocol records. The source omission is therefore an explicit answer-preserving record, rather than a bare approximation label. In particular the laboratory equation residual on this background is

$$G[g_0] - \kappa T_{\text{lab}} = -\kappa T_{\text{lab}}.$$

It is generally nonzero and is never asserted to vanish. The quantum packet retains its own declared fixed-background scope; no unidentified quantum stress is silently equated to the clock-body tensor. This completes all the stated simultaneous constructions on $\mathcal{U}$. $\square$

Theorems 15.9–15.10 jointly inhabit the construction datum of Definition 15.1 and therefore the flat branch of Theorem 15.2, including its common-history conclusion. Its separately conditioned free quantum overlap retains its additional target. The witness uses a specified ideal action and finite scattering prescription, with explicit attachment, device, pulse and matching identities. These model choices demonstrate joint occupation after the reconstruction. The completed-role lemmas have their own proofs and are used for the derivational status of the selectors; occupation does not replace those implications. The vacuum recipient in this witness is a baseline sector, so this result supplies no estimate for neglecting the apparatus’s self-gravity. A controlled approximation to a nonvacuum realization requires an actual family and an error theorem.

15.3 Source identity and reception in the coupled Wheeler–DeWitt construction

Proposition 15.11 (Flat matched geometry and a common total source). *Let an Einstein recipient on a connected region U obey*

$$G[g] + \Lambda g = \kappa T_{\text{grav}}, \quad \kappa \neq 0,$$

with constant Λ . Suppose its actual metric identification agrees on U with the constant form of the special-relativistic response recipient. If $\Lambda = 0$ and the response source is the same

complete tensor, $T_{\text{resp}} = T_{\text{grav}}$ on U , then $T_{\text{resp}} = 0$ there. All of its well-defined surface charges supported in U vanish. Consequently this joined target has no positive-rest-energy occupant of Physical Condition 11.3.

For arbitrary Λ , the only possible common source is $T_{\text{resp}} = (\Lambda/\kappa)g$. In particular a nonzero localized worldline source cannot occupy this target. A homogeneous nonzero source also cannot supply the finite whole-space charges of Physical Condition 11.1 on the complete Minkowski carrier.

Proof. The identified constant metric is flat, so $G[g] = 0$. The displayed Einstein equation gives the claimed source identity, including when the source equation is interpreted distributionally. For $\Lambda = 0$ its pairing with every test section is zero, and every admitted trace or surface-charge pairing is zero. This contradicts $E_0 > 0$.

A positive worldline source has a strictly positive pairing with a nonnegative compact test section supported around a segment of the line. It is nonzero and is supported on that line. It cannot equal a nonzero smooth homogeneous tensor on an open neighborhood. On the complete flat carrier the constant-time energy integral of a nonzero homogeneous vacuum-type tensor diverges. Subtracting the very same homogeneous background leaves zero residual tensor and therefore cannot supply a positive residual rest charge. $\square$

Proposition 15.12 (Pulse source and closed gravitational total). *On a smooth Einstein recipient with nonzero constant coupling, the complete gravitational source is covariantly conserved. If an attached body source has nonzero exchange $\nabla_\nu T_{\text{body}}^{\mu\nu} = J^\mu$, it cannot during that exchange be identified with the entire closed Einstein source. It may be an explicitly registered contribution to it, provided the remaining contributions carry total exchange $-J$.*

Proof. Contracted Bianchi and metric compatibility give $\nabla_\nu T_{\text{grav}}^{\mu\nu} = 0$. Literal identification with the body source would force $J = 0$. For a decomposition $T_{\text{grav}} = T_{\text{body}} + T_{\text{remainder}}$, linearity instead gives $\nabla T_{\text{remainder}} = -J$. On the fixed flat response target, zero exchange and the stated vanishing side flux give zero impulse by Lemma 11.2; hence a nontrivial pulse on the positive-energy orbit cannot evade the distinction by relabelling the whole isolated source as the driven body. $\square$

The occupied flat target. The common source of Theorem 15.10 retains the vacuum Einstein projection, positive-energy laboratory response, clock-bearing body and declared source distinctions. Its maps satisfy the original queries on the fixed-background target. Propositions 15.11 and 15.12 specify the source identities needed to receive that laboratory in a coupled geometry. They do not establish an absence of coupled matter, physical clocks or source matching from the corpus. Those constructions already exist in the Wheeler–DeWitt arc.

The existing coupled receiving construction. WDW-4 first identifies the scalar action, attached covariant source and canonical source projections, then constructs an exact nonvacuum Einstein–Klein–Gordon occupant with a licensed development and a local material clock [18, Thms. W4.2, W4.3, W4.6]. Its Standard Model branch derives sector exchanges, total on-shell conservation and source projections from one action, and closes the joint Lorentz, gauge, spatial and normal constraint algebra on one reduced domain [18, Thms. W4B.4, W4B.7]. The nonzero gauge–Higgs occupant includes an exact correction of the gravitational and scalar data, with a source-attached coupled development. The integrated theorem retains that occupant, the pure scalar occupant, their actual lift and the continuing physical clock [18, Thms. W4B.9, W4I.1]. Thus gravitational response to the material source is already constructed. The

classical Standard Model witness has zero fermions within the displayed graded carrier; its existence does not by itself enlarge the later quantum target.

WDW-8 constructs an exact homogeneous, source-extended constrained family whose canonical and source projections agree in the constraint, metric- evolution and scalar-evolution columns. It proves a complete source and covariant response of order $O(\delta^2)$, a nonlinear remainder of order $O(\delta^4)$, and exact constraint, Ward and total-exchange identities on its core recovery subtype [29, Thms. 6.4, 6.5, 6.15]. These are actual coupled recovery results with estimates. Their compact hyperbolic, homogeneous scope is retained; Proposition 6.14 expressly prevents their promotion to a weak-field particle or Poincaré theorem. WDW-9 assembles compatible actual occupants of thirteen realization fibres, including the coupled history, physical quotient, clock, changing record and recovery construction [30, Thm. 0.1]. WDW-10 constructs plural clock presentations and their clock-neutral descent on the retained metric–real-scalar Branch-A lineage [20]. These theorems supply an occupied coupled receiving arc, with its quantum and clock constructions on their declared domains.

The specialization required for this laboratory. Receiving the present SR witness in that arc is a further, precisely identified construction. The bodies, records, signals and controller must be attached as actual subsystems of one admitted coupled history. Their source maps must identify each contribution to the gravitational total, with internal exchanges cancelling and with the pulse, clock and comparison queries preserved. The existence of a coupled field occupant does not alone identify this particular apparatus with its subsystems. Every such richer determinate history, source map and comparison itself necessarily incurs the kernel by Proposition 2.4; lawful refinement also requires the displayed source and query identities.

If quantum matching is included in that reception, the relevant task is to specialize an existing matching construction to these same attached subsystems. Module B already proves equality of total-source charges and translation generators, common active/Casimir mass and inertial Hamiltonian matching in its operator-presentable stable scalar envelope [24, Def. 5.1, Thms. 7.3, 26.1]. Its actual representation, common domain and source attachment must be matched to the receiving history; a shared numerical mass does not perform that identification.

Finally, a finite laboratory recovery of the present boost, charge and pulse laws requires a stated family or comparison map and estimates for those observables. The existing WDW-8 backreaction estimate concerns its printed homogeneous target; it is not already an estimate for this SR apparatus. A suitable local limit, curved conservation construction or asymptotic charge theorem would have to carry the specified finite experiments and their error bounds. Normal coordinates alone supply no such bound. These are specialization obligations for reception in the existing coupled arc. They are not additional completion conditions for Theorem 15.10 on its declared flat target.

15.4 Reception by the Wheeler–DeWitt, symmetry, and carrier constructions

The SR construction supplies results for the further physical constructions of the corpus. Its outputs are identified event and comparison domains, a calibrated causal metric, frame receipts, actual clock and endpoint laws, and attached source-response relations. Their reception retains the original physical subject and the hypotheses of the result being used. Each receiving construction, map and application necessarily incurs the kernel by Proposition 2.4; the additional interface proof establishes which physical answers it receives.

Order of dependence. The dependence is between specified results within the arcs. Pre-metric comparison, persistence, physical source identity and clock construction can support the SR proof before their later metric realizations receive its conclusions. In particular, the intrinsic route of Sections 5–6 constructs the Lorentzian form and frame law from its stated comparison inputs. The already metric tangent route of Section 12 is used for agreement with a dynamical recipient. It is not a premise of that intrinsic construction. The same distinction holds for clocks: the Wheeler–DeWitt source identifies an occupied material clock, while the SR results determine the duration and comparison laws of the calibrated ideal-clock role on the declared target. Physical calibration must identify those roles before either result is applied to the other clock.

Proposition 15.13 (Reception of the proved SR laws). *Let S be an occupied source of the SR construction, with its complete retained queries and operations, and let R be a witnessed source refinement of S in Definition 4.7, with source map π . Fix any of the frame, duration, endpoint, source-response or marked-history results proved above. Suppose its displayed hypotheses hold on the actual jointly occupied records being received, with every added selection, calibration and applicability test retained in the refinement profile. Then its conclusion holds for the received readouts on those records and descends to their minimal complete attained receiver. Original source-operation identities and original failures are retained; successful operations leaving an added selection remain recorded exits. Two successive such receptions compose on their common defined domain.*

If a receiving metric is claimed to be the constructed metric, that claim additionally requires the cone and calibration identification of Lemma 12.2, or the finite certificate of Lemma 12.3 in its stated quadratic class. The result does not extend the domain of a theorem beyond its displayed physical hypotheses.

Proof. Write each original readout on R as $a \circ \pi$. Evaluating a proved identity of readouts at $\pi(r)$ gives the identical relation on r wherever its hypotheses and original query domains hold. The complete refinement profile retains these answers, their applicability, joint incidences and failures. Theorem 4.8 therefore descends the relation to the attained receiver. For each admitted operation lift, the defining equation $\pi \circ f_R = f_S \circ \pi^{\times n}$ for an n -ary operation proves the corresponding commutative operation identity; its retained domain and exit queries preserve the stated distinctions. For successive maps $R_2 \xrightarrow{\pi_2} R_1 \xrightarrow{\pi_1} S$, substitution gives $(a \circ \pi_1) \circ \pi_2 = a \circ (\pi_1 \circ \pi_2)$ and the same composition for the operation equations. The metric conclusion is exactly the cited matching lemma after the original identification and conventions are fixed. No occupation or matching premise is supplied merely by composing maps. $\square$

Wheeler–DeWitt reception. The coupled Einstein–matter histories, physical material clocks and source closure already constructed in WDW-4, the constrained recovery and backreaction family of WDW-8, and the joint occupation and clock descent of WDW-9–10 provide established receiving structures [18, Thms. W4.3, W4.6, W4.I.1][29, Thms. 6.4–6.5, 6.15][30, Thm. 0.1][20, Thms. 5.4, 6.2, 6.4]. Theorem 12.6 supplies a Lorentz-path recipient for an actual marked history, and Lemma 12.2 fixes its agreement with the independently constructed comparison metric. The physical clock’s original readout must be related to geometric duration by an actual calibration; a scalar field used to label slices is not automatically the proper time of every body.

On a curved recipient these are local tangent, transport and duration statements. Corollary 12.8 gives exact affine SR continuation on its flat domain. Extended radar, finite-charge and pulse results are received only on their proved domains or through a separate controlled comparison. WDW-8’s semiclassical parameter and its backreaction estimates concern its

compact curved subtype; Proposition 6.14 of that source expressly withholds a weak-field particle or Poincaré recovery theorem. The specific laboratory embedding and any such limit therefore belong to the receiving coupled realization described in Section 15.3. Their construction is not an additional hypothesis needed to inhabit the flat SR theorem.

Symmetry reception. SYM-2 distinguishes IF-GEOMETRY, IF-PRESENTATION and IF-PASSIVE from the interfaces for active states, fields, generators and local Noether currents [28, Sec. 2.1, Table 2.1; Secs. 5.2, 6.1–6.3]. Theorems 5.2 and 5.7, Corollary 6.6, and the calibration results provide a constructed realization of the geometric and passive-frame portion on the identified domain. The convention $q = c^2 t^2 - |x|^2$ agrees with SYM-2's positive-timelike form; the gravitational convention used in the joint witness is $g_0 = -q$. This fixed sign translation changes no cone or calibrated proper duration.

Every supplied inertial receipt has the proved Lorentz form. The full proper orthochronous group requires the frame supply of Proposition 6.7; the full geometric Poincaré action additionally requires the stated translations. Global Minkowski realization uses Theorem 14.1, rather than extrapolation from one tangent space. Lemma 11.4 and Proposition 11.5 supply an active action on their specified invariant source prescription. SYM-2's state and field representations, spin covering data, invariant vacuum, spectral condition, generator domains and exponentiation retain their separately identified construction requirements [28, SYM2-T08, SYM2-T13]. A passive frame matrix alone does not supply those active objects.

Standard Model carrier reception. The carrier sequence fixes the primitive-standing and derivative-realization questions for its original carrier arena. SM-1 retains carrier identity together with representation, measurement, transport, route and projection records; SM-6 receives the sector closures and exhausts residual primitive standing on that same declared arena [31, Criterion 10.1, Thms. 13.1–13.3][32, Lemmas 9.1–9.2, 9.7, Thm. 10.1, Lemma 10.2, Thm. 20.1]. The persistence reconstruction used in Section 3 already preserves original carrier responses, their joint operations and failures [14, Thm. 3.5, Prop. 3.6]. The SR results supply a physical realization of the relevant spacetime comparisons and, where their source hypotheses hold, the energy–momentum and inertial responses of those identified carriers and composites.

Reception therefore fixes which events, clocks, transport changes and total sources a carrier's relativistic description concerns. For a stable rest branch with the proved finite-charge and impulse conditions, Theorem 11.9 attaches $E_0 = m_i c^2$ to that same source. The relation does not choose numerical particle masses, a carrier representation or a generation count; these remain the particular physical questions and constructions of the receiving sector. Module B already supplies source-charge and translation-generator matching on its specified scalar quantum overlap [24, Def. 5.1, Thms. 7.3, 26.1]; its use for a different carrier retains the exact source, representation and domain matching. Relativistic metric, coordinate and response descriptors acquire no independent primitive carrier standing merely from their occurrence in a formula. They receive their physical meaning through the retained carrier and the constructed comparison and response roles.

These three receptions give the SR manuscript a common function in the corpus: it proves the relativistic comparison and response laws available to further realizations, states their occupied domains, and supplies the exact preservation criterion for using them there. The receiving arcs supply their sector-specific coupled fields, representations and apparatus identifications. Theorems 15.9–15.10 already complete the jointly occupied flat target asserted here.

15.5 What the four effects explain

The necessary dependency is present in the subject of each effect. A time comparison requires the same clock and its ticks; a length comparison requires the same persistent endpoints; a simultaneity comparison requires identified distant events; an inertia comparison requires one retained source and its actual exchange history. The kernel makes these conditions explicit before the numerical laws can concern the claimed objects.

Effect	Retained physical content	What changes between the relevant comparisons
Time dilation	Two ticks of the same calibrated clock	Coordinate duration between those same events, relative to the clock's own duration
Length contraction	One persistent standard with two endpoints	Which endpoint events are simultaneous in the measuring frame
Relativity of simultaneity	Two identified distant events and a synchronization rule	The equal-time relation defined by a different inertial clock network
Mass–energy	One completed source, its rest profile, and controlled exchange	The total charge under active transport change and the impulse required to produce it

The first three follow from passive frame comparison together with their measurement certificates. The fourth additionally requires an active physical response law and complete exchange accounting. The distinction prevents a coordinate transformation from being mistaken for an applied force.

In particular, contraction is not inferred from a change in persistence identity, and time dilation does not imply a failure of standing for the moving clock. The original device and records remain fixed. The difference appears in the permitted joint readout of those records. Similarly, the relativity of distant simultaneity leaves coincidence, timelike causal order and the order of a clock's retained ticks intact.

15.6 Exact dependency boundary

Table 2 distinguishes unavoidable kernel work from the substantive physical facts selecting the SR realization. Every physical object, premise and calibration appearing in its third column already incurs that work by Corollary 2.2. The columns distinguish explanatory tasks within one dependent construction. Every construction in the third column necessarily incurs the kernel by its own determinate physical content. This includes the selection tests, their actual applications, the completed comparison roles and the laws received from them. The column records the substantive conditions and completion properties used at each stage of that same governed development. The role-completion lemmas establish the stated dependencies; the occupation theorem supplies an actual joint witness. Thus additional physical content remains inside the necessity argument at every level.

16 Relation to established physical constructions

16.1 The necessary primitive and its derived invariant boundary

The leading foundational comparison concerns what each construction receives before its geometric or dynamical equations are obtained. AASC reconstructs the necessary functions of

Table 2: Necessary foundation and physical specialization; every physical entry incurs K.

Passage	Established contribution	Physical specialization within that dependency
Every determinate construction	Necessary kernel at every stage, including richer selectors, maps and realized laws	Actual identified content and original occurrences at each construction stage
Physical source architecture	Forced characteristic quotient, first-class structure and exact attachment of actual SR records	Regular Hamiltonian observable evaluation; actual joint persistence and event records
Persistence to events	Constructed complete-context quotient and operation descent	Original coincidences and a supplied event/record domain
Comparison to affine events	Effective action quotient, vector torsor and affine receipt	Smooth abelian homogeneous comparisons; complete action profile, unique repetition, four-dimensional events and additive clock
Transport to cone	Completed-order separation gives a finite bound; isotropy constructs the quadratic cone	Pointed completed reach, weak clock, rest action with no fixed spatial vector, and full boundary isotropy
Cone to frame law	Unique normalized rest receipt and elimination of conformal scale	Ordered Archimedean interval role and unit, or certified calibration; rotations, frame composition and common-pair coverage
Frame law to all boosts	Full proper orthochronous Lorentz group	Transitive supply of timelike frame directions, with the admitted rotations
Frames to measurements	Unique duration, radar and endpoint comparison	Attached device readouts, ideal-clock response, stable endpoint standards and actual signals when radar is used
Metric comparison to continuation	Original transport identified with the Levi–Civita connection	Smooth first-order comparison-preserving transport and the minimal torsion-free branch
Charge to inertia	Invariant prescription supplies active response; Ward balance and impulse yield inertia	Complete source prescription and variation domains; finite charges, side-flux control, stable orbit and controlled pulses
Common source to SR	Finite original-query metric matching and Lorentz-path continuation	Original null and clock records, actual source identification and marked joint histories; flatness or controlled approximation
SR to free quantum limit	Exact dispersion estimate and strong unitary convergence	Same mass/action calibration, Hilbert translation realization and free positive-energy branch

determinate incidence and develops their AMetric boundary. The boundary is derived from the kernel, rather than supplied as an independent physical primitive. At its general level it invokes no primitive distance, order, probability or temporal process of authorization [2, Sec. 1.4, Thm. 7.5, Sec. 12.1]. The logical dependence is

$$\text{determinate incidence} \implies K \implies \text{AMetric boundary} \implies \text{licensed physical constructions.}$$

The final arrow receives the independently warranted physical content displayed throughout this paper. The boundary constrains admissible discrimination, equivariant selection, information-preserving interfaces and faithful transport; those constraints precede the choice of a metric or Hamiltonian. The exact group-action and quotient mathematics has established antecedents. The contribution claimed here is the necessity argument, its application to the identified source, and the physical construction descending from it.

A causal fermion system can start without a spacetime metric while already supplying Hilbert, operator, measure and action structure [44]. Entropic reconstruction supplies configurations, probabilities and inferential constraints, and its gravitational construction receives additional geometric and compatibility data [47]. Operational and causal reconstructions similarly receive particular events, comparisons or operational structures. Under Theorem 2.1, their determinate physical incidences already instantiate the necessary kernel functions. That dependency is prior to a comparison of how many familiar equations a framework recovers.

Functional minimality concerns indispensable work, rather than the number of primitive symbols: one sufficiently rich relation can implement all four functions [1, Thm. 3.6, Sec. 4]. The AMetric invariant concerns faithful preservation of bearer, question, warrant and comparison structure across eligible regimes; it does not identify different physical initial data or select one empirical law. Among the approaches examined, no matching necessity-and-boundary foundation followed by this physical construction has been identified. This is the bounded originality claim. Theorem 15.10 supplies a simultaneous occupant for the assembled theorem on the declared fixed-background branch; Section 15.3 proves the source restriction on identifying this laboratory with a complete gravitational source and locates its further reception in the existing coupled Wheeler–DeWitt construction. Section 15.4 states the results supplied to that arc, the symmetry arc and the Standard Model carrier sequence.

16.2 Kinematics, causal geometry and clocks

Einstein’s 1905 construction explicitly coordinates clocks and distances and derives the moving-system transformations and their measurement consequences [33]. The present formulas for dilation, contraction and simultaneity reproduce those results. The AASC task is to reconstruct the necessary conditions already incurred by their physical subjects and to prove how the same governed content passes into these comparisons. The dependency precedes the choice of inertial coordinates.

The Ignatowsky tradition studies how relativity, group structure and spatial symmetries restrict inertial transformations without beginning with a light postulate; a modern proof is given by Anker and Ziegler [35]. Our argument is a finite-propagation branch selection with explicit physical witness and calibration conditions. It does not prove that group composition alone excludes Galilean kinematics. Example 14.3 exhibits the missing finite-boundary content.

Alexandrov studies translated cone structures in affine space, while Zeeman reconstructs the causal automorphism group of Minkowski space without assuming continuity or affinity [37, 36]. Their geometric results are stronger than Lemma 5.10 in that respect. The present direct cone proof is deliberately elementary once its homogeneous displacement receipt is supplied. Its distinguishing feature is the connection to the AASC physical query, actual composition and a retained rest-clock scale; the cone-rigidity algebra is established mathematics.

The Ehlers–Pirani–Schild reconstruction develops spacetime geometry from light propagation and free fall [38]. It provides substantial prior art for physical inputs preceding a metric description and reaches curved-spacetime structure beyond the flat theorem here. Andr eka and collaborators formulate observer comparisons and relativistic effects in explicit logical axiom systems [39]. AASC therefore does not originate either operational reconstruction or rigorous axiomatization of SR.

Material and relational clocks also have a substantial history. Brown and Kuchař construct matter reference systems in canonical gravity [43]; Perlick separates radar constructions and physical clock parametrizations in Lorentzian spacetime [40]. The WDW arc’s contribution used here is its own occupied clock construction, quotient/readout certificate and exact continuation domain. A clock label, a spacetime foliation and a family of standard clock readings remain distinct mathematical objects.

16.3 Energy and inertia

Einstein’s mass–energy argument relates changes of a body’s energy to changes of inertia [34]. Laue’s complete-system analysis and its modern formulation by Giulini identify the conditions under which integrated sources transform as total four-momentum [41, 42]. The charge step of Section 11 belongs to that lineage.

The AASC-specific assembly is the attachment of that total source to the same retained persistence and response class, followed by a controlled impulse definition of measured inertia. Equation $E_0 = m_i c^2$ is not a new empirical prediction. The proved response receipt explains how the premetric inertia role acquires this numerical value in the declared Lorentz-covariant realization. It also identifies what must be retained for a composite: internal binding, fields, stress, source baseline and external exchange.

16.4 Common-source approaches to gravity and quantum dynamics

A broad claim that no earlier framework seeks quantum dynamics and Einstein gravity from a common source would be incorrect. Causal fermion systems provide a closely relevant comparison. Their abstract starting objects are a Hilbert space, a space of finite-rank operators, a measure and a causal action, rather than a prescribed spacetime metric; their textbook presents quantum mechanics and general relativity as limiting cases [44]. Their selected constructions nevertheless carry additional physical structure. Finster and Kroup’s Einstein-equation analysis uses smooth support and osculating vacua based on a Minkowski Dirac-sea reference [45, Sec. 3.2]. Finster, Kleiner and Paganini analyze a nonrelativistic effective Schr odinger equation and collapse corrections; their spinless reduction uses the nonrelativistic kinetic operator [46, Sec. 3.1]. These are genuine shared-framework precedents, although their primitive objects and receiving mechanisms differ from AASC.

Ipek and Caticha’s entropic geometrodynamics also combines quantum and gravitational dynamics in one scheme [47]. Its geometry and hypersurface-deformation requirements enter the construction, and its coupled dynamics has semiclassical features. It does not establish the same premetric persistence and standing route as the present source family. These differences justify a precise comparison; they do not justify erasing it from the historical record.

AASC’s foundational claim concerns the necessary conditions of physical objecthood, including the primitive objects used by other physical frameworks. A metric, Hilbert state, causal measure or material clock, when offered as a determinate physical bearer or relation, already incurs those conditions. A theory can implement the work implicitly; changing its terminology does not remove the dependency. This structural claim does not establish historical priority

by itself.

The corpus’s stronger constructive result supplies an occupied common source with compatible metric and quantum recipients. Universal kernel necessity alone would not identify distinct realized sources as one source: that identification comes from the capstone’s occupation, continuation and compatibility theorems. The present SR recipient is explicitly constructed on its certified marked coherent-history refinement. The completed capstone provides more than a shared vocabulary: it supplies occupied continuation and compatibility on its frozen target. The SR tangent and anti-development constructions then give lawful geometric and continuation recipients, while the independent finite-cone route explains which propagation and calibration facts select the same Lorentz form without first inserting a spacetime metric. The distinction between those two routes is essential: the tangent theorem inherits a Lorentz metric; the cone theorem constructs its form from the stated premetric transport receiving conditions.

16.5 The contribution established here

The derivation starts with the necessary functional conditions of determinate physical objecthood and follows their physical development. Section 3 proves the characteristic constraint architecture on its regular branch, the attachment of the actual SR records to that source, and the comparison and persistence constructions. The original comparison operations construct the affine receiving carrier; the completed transport body constructs its spatial and Lorentz forms. Metric licensing establishes the dependency of their physical use on the retained relata. The frame, device and response proofs then determine the quantitative laws.

Proposition 2.4 establishes direct kernel necessity for every richer stage, including each selector and law receipt. The source-refinement theorem connects these stages by exact maps preserving the original questions, operation domains, shared operands and failures. Theorem 10.1 and Proposition 10.3 add substantive downstream consequences: the minimal comparison transport is Levi–Civita, and a complete invariant on-shell response supplies the source conservation law used by the impulse theorem. The original source is therefore used in the geometry, measurement, energy and continuation constructions.

This is a kernel-rooted derivation through the stated physical source architecture. Its physical branch conditions are made explicit at the stage where they enter, as in Definition 15.1. The derivation does not require the premetric kernel to name a numerical spacetime metric. The completed, faithfully received source supplies the physical comparison and response content, and the proved construction determines the SR laws on that target. Uniqueness applies to the complete retained laws and questions; richer physical targets may possess additional structure.

The unification statement has the capstone’s exact occupation and eligibility scope. The common-history construction preserves that source and proves its relativistic recipient; the free quantum comparison proves its separate quantitative overlap. A universal interacting gravity–quantum law and absolute historical priority for the entire program are not conclusions of these results.

16.6 Why the explicit physical conditions matter

The theorem is strongest when its premise weight is visible. On the primary route, Theorem 5.2 constructs the affine four-dimensional receiver from its regular homogeneous comparison operations, and Theorem 5.7 constructs its spatial form. Dimension, homogeneous operation structure, completed order and clock separation, and actual rest isotropy remain explicit branch content. The strict clock condition may be received directly or obtained from the stated

rest-comparison completion. Physical condition 5.9 is the alternative entry for an already calibrated receiver. Lemma 5.10 proves linearity of homogeneous continuous frame receipt within that receiver; it does not derive spacetime dimension or the affine structure from the kernel. The finite bound and isotropy in Physical condition 5.11 then select the round finite cone.

The active source action in Physical condition 11.3 is also substantive. The mass theorem derives the rest response from closed-source Lorentz covariance and accessible impulse records. Its independence from a mass-shell assumption concerns that particular assumption; it is not independence from Lorentz covariance.

Similarly, the cone theorem defines a limiting boundary. Physical condition 7.3 supplies the working signal and device occurrences needed to populate the pulse-clock construction. A limiting path and an executed signal are distinct original queries. The proof explicitly joins geometry to these occupied physical records.

Theorem 5.7 constructs the spatial quadratic form from intrinsic isotropy. Proposition 5.6 derives its strict completed-clock premise from the stated completed-order and rest-comparison properties; Lemma 5.5 then proves the finite bound. The physical completion properties and their joint occupation remain explicit. Lemma 11.4 supplies solution and response transport, and Proposition 11.5 constructs active covariance; it does not follow from passive covariance alone.

Finally, Theorem 12.1 is the standard Lorentz-frame consequence of an already supplied tangent metric. It supplies the object-level recipient. Theorem 12.6 constructs the continuation action and its duration-preserving geometry, with a concrete coherent-history marking requirement. The separate finite-cone construction is the route that obtains the Lorentzian form before a spacetime metric is supplied. Appendix B states the exact source interfaces, distinguishing imported constructions from results proved within this paper.

17 Conclusion

Every non-degenerate determinate physical object or incidence already incurs Admissibility, Standing, Reference and Irreversibility. The kernel is necessary at the level of the physical subjects themselves: their identities, actual qualifications, combinations and original occurrences. Clocks, propagation laws, metric sources and calibration records bring this dependency with them into a derivation.

Special relativity is obtained here by constructing the intermediate physical machinery on its declared branches and following that dependency into a quantitative realization. The characteristic source quotient, the descent of the actual SR records, premetric comparison and persistence, affine displacement, compatible metric transport and attached Ward conservation are proved links in that construction. Premetric persistence and comparison receive an exact event and operation structure; homogeneous isotropic finite propagation fixes a round cone; composable frames and a common rest-clock scale force the Lorentz law. Attached clocks and persistent endpoints then give time dilation, length contraction and relativity of simultaneity. Closed Lorentz-covariant sources and controlled impulse balance determine the operational rest response $m_i = E_0/c^2$.

These effects concern the retained physical bearers, and the constructed comparison and response relations themselves necessarily incur the kernel in their own right. Preserving their identity, qualification and original occurrence is part of what makes the formulas physical claims. The explicit conditions selecting this realization add physical content whose own determinate realization is necessarily governed.

Theorem 15.10 realizes the regular source, energetic clocks, finite signals, controlled response and metric matching together. Its explicit continuation projects to the occupied vacuum-Einstein and full-ray Schrödinger packet, retaining the source and device records. The positive laboratory response remains distinct from the vacuum gravitational source; Proposition 15.11 excludes their identification on unchanged flat geometry. The foundational distinction is the necessary kernel and its derived AMetric boundary. The joint occupation inhabits Theorem 15.2 on its declared flat branch. The completed-role implications explain the stated selectors, and Proposition 2.4 applies to every richer construction and law in that chain. The coupled source and recovery results already established in the Wheeler–DeWitt arc are identified in Section 15.3. Proposition 15.13 gives the exact reception rule for the relativistic comparison and response results supplied to that arc, the symmetry arc and the Standard Model carrier sequence. Their further apparatus-specific coupling and recovery constructions retain the original source identities and lie beyond the flat target inhabited here.

A The kernel interface used in the derivation

This appendix states the foundational interface and its conservation consequence in the form used by the SR construction. Theorem 2.1 supplies the constitutive necessity argument. The clauses below expose the work already incurred by its physical subject matter; they introduce no further permission for kernel governance. The mathematical conservation argument is then stated separately, with its result-determinacy and independent-warrant hypotheses visible.

A.1 Original incidence and the four commitments

An original performance or incidence has identified relata, an occurrence or constructional position, and its original conditions. A question specifies the relevant bearer, use and conditions before an answer is assessed. A change of inscription need not change the occurrence; a change of original material or conditions must be represented as such. Non-degeneracy requires a nonempty realized fragment and at least one actual successful identity-bearing assessment or use. It does not require every proposed experiment or use to succeed. For example, identifying the occurrence of a failed experiment is compatible with negative standing for its attempted practical use [1, Defs. 2.1–2.3, Prop. 2.4].

Definition A.1 (Explicit kernel commitments). For a determinate physical target, with its evaluated questions and their complete conditions fixed independently of the answers, the commitments used here are:

- (A1) *Target determinacy.* Every evaluated incidence concerns an identified target; a presentation change claimed to retain that incidence retains the target under its original identification.
- (A2) *Assessment and use.* For an evaluated incidence and specified use under complete stated conditions, actual qualification distinguishes a warranted use from an unwarranted one. Positive qualification supports that use on those conditions; negative qualification cannot support it as though positive. Altering the use or conditions changes the question.
- (A3) *Original fixation.* In a retained history or construction, the original occurrence, conditions and assessment target remain identifiable. Correction may supply a further performance or a newly conditioned assessment; it cannot substitute new original data while claiming success of the unchanged original performance on its original data.

(A4) *Answer accountability.* Each answer offered with standing by a purported same-object presentation must have warrant at the original bearer and complete question. Warrant for another bearer, use or condition cannot be transferred by retaining a label.

These commitments have structural physical meanings. Qualification is whether the specified relation holds of the identified relata under its actual conditions, independently of endorsement or observation. Evaluation here need not be an algorithm, a measurement, or an observer's decision. An incompletely specified future experiment need not possess a selected outcome; a question about an outcome set, a distribution, or an actual realized record has its own result type. A question outside the evaluated fragment has not thereby acquired either a positive answer or a proved negative one [1, Sec. 2.3, Sec. 9.2].

Proposition A.2 (Reconstruction and functional minimality). *The commitments in Definition A.1 instantiate Reference, Standing, Irreversibility and Admissibility. A realization retaining the complete original determinate job cannot genuinely delete any of their functional work. A non-degenerate determinate physical incidence incurs these commitments by Theorem 2.1, including when the incidence is unobserved.*

Proof. (A1) supplies reference to the same target. The actual eligibility of an incidence for a specified use in (A2) supplies standing. (A3) retains the original occurrence and conditions, supplying irreversibility of attribution. The accountable entry into use required by (A2), with the target, original conditions and individual warrant retained by (A1), (A3) and (A4), supplies admissibility. One implementation can perform several or all of these functions.

If a function were genuinely deleted while the same complete job remained, applying this reconstruction to the retained job would recover that function. Either another component still performs it, so it was not functionally deleted, or the corresponding commitment is lost. The latter fails to retain the job. This is functional minimality; it does not assert logical independence of four primitive formal axioms [1, Thms. 3.1, 3.6, Prop. 3.7]. $\square$

The constitutive step from actual determinate physical incidence to (A1)–(A4) is argued in Theorem 2.1 and the source's Proposition 2.4. It is not inferred from the existence of an uninterpreted set element, nor proved merely by introducing an adequacy predicate. The reconstruction above identifies the indispensable functions of those commitments. The following theorem is the distinct mathematical implication for independently grounded descriptions [1, Sec. 7.1, App. C].

A.2 Independent warrant and representation conservation

For each bearer b let $\mathcal{Q}_b$ be its independently specified complete questions, and let $V_{b,q}$ be the result type for question q . Fix a relation

$$J(b, q, v), \quad q \in \mathcal{Q}_b, \quad v \in V_{b,q},$$

meaning that v is incident as an answer to that original question. Its physical or mathematical specification is fixed before comparing proposed readers. In particular, neither the agreement of those readers nor a proposed quotient defines J .

Theorem A.3 (Same-bearer, same-query conservation). *Fix $b, q \in \mathcal{Q}_b$, and independently specified result incidence J . Suppose:*

(i) *the question is determinate at this level of result:*

$$J(b, q, v) \wedge J(b, q, w) \implies v = w;$$

- (ii) the two original presentation uses independently retain b and the complete q ;
- (iii) each proposed output has its own warrant: $J(b, q, v)$ and $J(b, q, w)$.

Then $v = w$. If these conditions hold for every question in the complete original query family, the two presentations have equal complete answer profiles. No enumeration or finite syntax for that family is required.

Proof. Independent retention of the bearer and complete question puts the two warranted results in the same incidence fibre. Apply (i). Repeating the argument for each original query gives pointwise profile equality. No equality between outputs was used to establish either answer's warrant [1, Thms. 6.2, 6.3, 7.1]. $\square$

The theorem does not require a result for every proposed question. If a complete answer distinguishes success, undefinedness and several failures, its result type includes those distinctions, and each such answer requires warrant for the distinction asserted. Lack of an answer is not by itself proof of an undefined or negative answer. Similarly, changing a frame changes a frame-indexed coordinate question; the theorem then applies to the complete covariant comparison, rather than demanding equal coordinate numbers for different questions.

Corollary A.4 (Same-query exclusion of a discordant answer). *Under Theorem A.3, an answer different from the established answer cannot also have standing at the unchanged bearer and complete question. A proposed alternative either changes the bearer or question, agrees on the original answer, or lacks the claimed answer-standing there.*

Proof. If the bearer and query are unchanged and both unequal answers have warrant, the theorem yields equality, a contradiction. This is the same-object exhaustion used in the source's Theorem 7.1 [1]. $\square$

A.3 Governance and the fidelity of a receiver

A representing inscription, mistaken calculation or failed apparatus can itself be a non-degenerate determinate object and thereby incur the kernel. Its existence does not confer positive standing as an answer about another bearer. That further use must meet (A4) and the independent-warrant conditions above. Thus universal governance neither certifies every proposed map nor disappears when a map fails.

For the physical receivers in this paper, Lemma 4.2 proves the exact readout criterion: an original readout descends precisely when it is constant on each encoded-input fibre. Lemma 4.3 adds saturation of the operation domain and constancy of the retained output, thereby preserving and reflecting applicability. Proposition 4.4 supplies finite compositional preservation. These proofs implement the inherited obligation for the specified event, clock, endpoint and exchange target. They do not assume a representation is faithful merely because its fields carry kernel terminology.

Consequently the kernel's universal necessity and the receiving theorem have different quantifiers. Every non-degenerate determinate physical source already incurs the kernel. A particular receiver is faithful only on the original queries and operation domains whose required preservation is established. Each substantive physical specialization used to select the Lorentzian regime is itself determinate physical content and inherits the first statement; its faithful metric realization must also satisfy the second.

B Physical source interfaces and the exact common-source claim

This appendix states the physical interfaces used by the manuscript without requiring the reader to reconstruct them from the whole corpus. Every carrier and condition below already incurs the kernel through determinate objecthood. The issue recorded here is which additional physical information a source theorem receives and which conclusion it supplies. The imported results are used at the interfaces stated here. Section 3 reproduces the source forcing and construction steps needed by the primary derivation; Section 10 proves the compatible-transport and source-closure bridges. This appendix retains the broader imported hypotheses and conclusions, including constructions not independently reproduced in full.

The cone, frame, duration, radar, endpoint, impulse and free-limit proofs in this paper can be checked from their displayed hypotheses. The common-source identification additionally uses the imported realization results. Module B and the full WDW atlas provide compatibility and construction context; neither is a hidden premise of the direct Lorentz or impulse proofs.

B.1 Structural gravity, comparison, transport, and metric realization

Gravity as structural necessity. The regular AER interface consists of a local symplectic carrier (S, ω) , a smooth standing surface $C \subseteq S$, a smooth quotient submersion $\pi : C \rightarrow S_{\text{phys}}$, and physical observables constant on its fibres. Their Poisson brackets must descend independently of ambient extensions; their differentials span the annihilator of $D = \ker d\pi$ and separate quotient classes; maximal same-scope observability excludes an undetected direction outside $\ker \omega_C$. These hypotheses force

$$(TC)^{\perp \omega} \subseteq D = \ker \omega_C \subseteq TC.$$

Thus C is coisotropic, admissible redescription is its characteristic foliation, and local constraint generators are first class. If presentation-parametrized Hamiltonians induce the same reduced physical continuation, modulo dynamically inert reduced content (AER-P), they have reduced-plus-constraint-multiplier form [6, Thms. 6.3, 7.8, 7.10, 8.2]. This is the necessary gravity role in that physical comparison class. Theorem 3.2 and Corollary 3.3 prove this architecture in the present paper, and Lemma 3.4 applies it to the actual SR record and operation source. It supplies neither spacetime dimension nor a Lorentzian line element; the state-space symplectic carrier is not the later spacetime receiver.

Comparison before numerical metric. R4 receives induced continuation-neighborhood domains, actual stable overlap/comparison witnesses, path-coherent standing transport and quotient-stable redescription. Quotienting those witnesses by their same-role comparisons produces a comparison carrier preserving the original identity, distinguishability and overlap/transport obligations. Its representation is a partial map where a unique witness class is fixed, and otherwise retains the admitted comparison relation [8, Thm. 11.1]. The carrier need not be numerical, additive, symmetric or totally ordered [8, Def. 3.9, Rem. 3.11]. This supplies the comparison role that the present event and displacement readouts must faithfully realize. An affine $3 + 1$ torsor, real additive duration and Euclidean spatial norm remain the expressly declared strengthening of that role, not conclusions of carrier induction.

Transport before finite metric speed. R5 receives bounded standing localization, induced adjacency and continuation neighborhoods, and the preceding quotient comparison carrier. Its transport witnesses preserve source and target burden, boundary traces, identity, restoration

and comparison class; compatible witnesses compose on matching intermediate boundaries. Standing-relevant propagation must be mediated by this admitted transport organization rather than an unwitnessed pathless transfer [9, Secs. 2, 7–10, 15]. The source establishes its T2–T4 layers; it explicitly reserves finite physical speed T5 and relativistic causal structure T6. Here it supplies the physical witness and composition objects on which the cone theorem operates. Uniform metric-speed bounds, rescaling/limiting realization and full spatial isotropy are additional receiver conditions.

Licensed metricity and symmetry. The licensing theorem receives a non-degenerate determinate background with the AMetric exclusion result, and an admissible same-background metric use becoming physically interpretable at a local projection interface. Such use must articulate already standing-positive relata while preserving redescription and continuation; it cannot authorize its own target or act as a boundary selector, repair or channel [15, Thm. 6.2, Cors. 6.3–6.4]. This fixes the dependence of every interval and calibration used here without generating its numerical geometry. SYM-2 supplies a fixed Minkowski/Poincaré target and classifies faithful passive presentation separately from nontrivial active state/field action [28, Secs. 5–6, 10]. It supports that distinction in the mass theorem; passive Lorentz receipt alone supplies no active body-boost orbit.

Null structure and occupied signals. The null-propagation result receives a relativistic theory with an admissible metric representation in which vacuum light is null in the classical/geometric-optics regime. Under physical-content-preserving redescriptions, null structure and its propagation role remain invariant while numerical speed depends on calibration [16, Thm. 1]. Here it identifies the invariant content of a certified limiting signal. It does not populate the cone boundary: the clock and radar constructions separately receive actual signal, endpoint and readout witnesses.

B.2 Persistence and physical clocks

Distributed persistence. CSP–R starts with participating local continuation domains and actual persistence profiles

$$\Phi_D(U) = (B_U, S_U, I_U, R_U, P_U, Q_U, \partial U, T_U), \quad U \in \text{dom } \Phi_D,$$

retaining bounded support, standing, identity memory, restoration, perturbation burden, comparison, boundary and transport, together with actual local configurations, partial operations, overlaps and comparison witnesses. These are structural physical data before a numerical spacetime metric or species assignment [14, Prop. 1.2]. Premetric therefore retains local comparison; it does not mean featureless.

For the independently fixed structural or occupied physical target T , let U_T be its native presentation domain, e_T its evaluation into the existing receiving structure, and $R_{\text{nat}}^T, R_{\text{ARC}}^T$ the complete original response records, including licensed contexts. The reconstruction theorem states

$$\ker R_{\text{nat}}^T = \ker(R_{\text{ARC}}^T \circ e_T).$$

The induced attained quotients are isomorphic as many-sorted partial-operation structures, preserving and reflecting domains, failures and shared-operand incidences [14, Thm. 3.5]. Reconstruction of the finer retained source class additionally retains the original source comparisons and joint diagrams; it does not invert all raw construction history [14, Prop. 3.6]. A new SR query absent from that target requires an explicit enlargement or a proved factorization. The present event construction supplies that requirement for its stated original queries.

Occupied local scalar clock. The WDW-4 construction supplies an actual local scalar-clock chart on its specified nonvacuum Einstein–Klein–Gordon source. On the initial compact slice its scalar field has spatial gradient zero and positive normal derivative ε , so, in signature $(-+++)$,

$$\chi = g^{-1}(\mathrm{d}\phi, \mathrm{d}\phi) = -\varepsilon^2 < 0.$$

Continuity gives a neighborhood with $\chi < -\varepsilon^2/2$. The future vector

$$V_\phi = \frac{\nabla\phi}{\chi}, \quad \mathrm{d}\phi(V_\phi) = 1$$

then gives a local oriented clock flow. The complete clock object retains the source anchor, interval, quotient, readout, continuation and first-failure boundary. Transversality and the guarded proper-refoliation comparison give local relational descent [18, Secs. 40–43, Prop. 43.1]. An unattached relational quantity is not promoted to a measured tick record [18, Rem. 43.2].

WDW-7’s uniqueness applies to the inherited fixed clock-use lineage, including its source, readout and failures [19, Def. 7.1, Thm. 8.2]. WDW-10 constructs a populated plural clock atlas on its enlarged fixed source target and glues its complete records on actual overlaps; it does not select one global numerical time [20, Thms. 3.8, 5.4, 6.2, 6.4]. These results support the source and readout discipline used here. They supply neither a vacuum Minkowski matter clock nor an executed null pulse at every point of a limiting cone. The pulse-clock theorem in this paper has its own occupied signal and endpoint conditions, and the scalar-flow formula is separately reconstructed on its declared flat background.

B.3 Module A: the total source actually imported

Let M carry the admitted smooth metric g and let

$$\mathcal{V}_g = \Gamma_c(S^2T^*M).$$

A source-bearing original response has an attached complete operational metric witness $W : \mathcal{V}_g \rightarrow \mathbb{R}$ that is continuous and linear in the compact test perturbation, with its source identity, lawful comparison, boundary and exchange data retained. Module A’s representation theorem gives a unique symmetric contravariant tensor-density distribution T_W with

$$W(h) = \frac{1}{2} \langle \mathsf{T}_W, h \rangle \quad (h \in \mathcal{V}_g).$$

This follows from the continuous dual of compactly supported smooth test sections; equal complete witnesses have equal distributional sources [23, Thm. 5.1]. A non-source-bearing response is not assigned a stress tensor merely by giving it a source label.

The additional Ward identity

$$W(\mathcal{L}_X g) = -\langle J, X \rangle \quad \text{for every } X \in \Gamma_c(TM)$$

is equivalent to $\nabla_\mu T^{\mu\nu} = J^\nu$ in distributions, after the density and index conventions are fixed [23, Thm. 8.2]. Indeed symmetry gives $\frac{1}{2} \langle T, \mathcal{L}_X g \rangle = \langle T, \nabla X \rangle = -\langle \nabla \cdot T, X \rangle$. Source representation alone does not imply closure. Proposition 10.3 constructs the $J = 0$ identity from complete invariant response and actual source equations; equation (26) constructs it from a complete finite operational ledger when the external sum vanishes.

For a finite completed composite, the source certificate lists every contribution exactly once, retains its witness and exchange current, and balances internal signed incidences. Finite summation then yields the unique total witness and tensor, cancellation of internal exchanges

and the declared external divergence [23, Thm. 12.1]. This is the source interface of the mass theorem.

It does not supply finite hypersurface charges, their normal traces, vanishing side flux, a stable active Lorentz orbit or a lossless pulse protocol. Those are the explicit conditions of the present total-charge and impulse construction. Once received, Gauss–Green balance and the active covariance calculation in Section 11 prove the operational inertia result. The mass-shell relation is a conclusion of that calculation.

B.4 The imported Einstein and Schrödinger normal forms

Einstein recipient. The imported Einstein theorem receives the regular AER gravity role, a faithful four-dimensional Lorentzian metric projection, a fixed regularity/locality/boundary regime with minimal compatible connection, and a closed same-scope source. Its geometric candidate is a local symmetric rank-two tensor built from that metric alone, linear in the first curvature response, with no curvature products or derivatives, extra fields, torsion, nonmetricity, nonlocal terms, boundary selectors or hidden source terms. Its general curvature-response branch is nonzero. These are the explicit SameScopeMetric₁ conditions [21, Defs. 14.2, 20.3, Rem. 14.3].

The result is

$$E_{\mu\nu} = aR_{\mu\nu} + bRg_{\mu\nu} + cg_{\mu\nu}, \quad b = -a/2, \quad a \neq 0,$$

and hence, after coupling and vacuum normalization,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

Curvature exhaustion and closed-source divergence compatibility supply the indicated restrictions [21, Thm. 30.1]. This theorem fixes the equation form inside those anchors; it does not derive Lorentzian signature, four dimensions, source existence or Cauchy well-posedness.

Schrödinger recipient. The quantum theorem first imports its exact standing-quotient construction. It then receives a faithful closed scalar nonrelativistic projection into $\mathbb{P}L^2(\mathbb{R}^d)$ that preserves transition weights, continuation and declared independent composition. Reversible transition-preserving ray evolution, a continuous phase lift and strong continuity give a unitary group and self-adjoint Stone generator. The fixed spatial target supplies Euclidean comparison, measure, dense domain/core, a certified scalar potential, and the absence of undeclared gauge, spin, relativistic, nonlinear, open, update or boundary terms [22, Def. 16.1, Thm. 36.1].

The SSQ₁ conditions further require a local linear scalar differential response of at most second spatial order, uniform metric-proportional principal symbol and the declared self-adjoint domain. Their normal form is $H = a\Delta + M_V$. The numerical coefficient requires the admitted action scale $\hbar$, mass m and inertial dispersion $E(p) = |p|^2/(2m)$, giving

$$H = -\frac{\hbar^2}{2m}\Delta + M_V, \quad i\hbar\partial_t\psi = H\psi$$

[22, Def. 24.3, Thms. 25.10, 27.3, 36.1]. The present free-limit theorem independently relates that kinetic term to the positive-energy relativistic branch on a common quantum translation realization; it does not infer Hilbert structure from classical SR.

B.5 Module C: common histories and their dynamical eligibility

Module C fixes a regular local Banach chart U_D of the lawful RPI quotient, an embedded standing locus S_D , a registered failure locus, a certified local clock τ , the complete original RPI roles, and a locally Lipschitz vector field V_D tangent to S_D . The field obeys

$$d\tau(V_D) = 1$$

and the stated failure, exchange and profile-closure conditions. The seed is supplied physical tensor content; the kernel and the clock do not uniquely choose it [25, Defs. 1.1, 4.1–6.4, Rem. 6.2].

Local ODE existence and uniqueness construct its flow $\alpha_{t,s}$. Admitted Moore segments retain exact durations and endpoint-compatible concatenation. Quotienting only by lawful fixed-clock history comparisons yields the nonempty history category used in the capstone; positive-duration segments exist in the exhibited packets [25, Thms. 7.1, 11.1]. Thus the upstream history is formed before downstream solutions are evaluated.

For each role profile Π_X , descent needs a precise additional property. On a smooth metric profile, the seed satisfies a projectability identity

$$d\Pi_g(V_D) = F_g \circ \Pi_g.$$

On a general quotient profile it satisfies, on common defined intervals,

$$\Pi_X(x) = \Pi_X(y) \implies \Pi_X(\alpha_{t,s}x) = \Pi_X(\alpha_{t,s}y).$$

The descended process is then uniquely $\beta_{t,s}^X(\Pi_X x) = \Pi_X(\alpha_{t,s}x)$. These are the mathematical conditions making the process well defined, rather than independent generation of a selected dynamical law [25, Defs. 14.1, 14.3, Thms. 14.2–15.2].

The two physical classifications require further eligibility:

- (i) *Metric Cauchy eligibility.* Fix the four-dimensional Lorentzian and source conditions above, a smooth hyperbolic Einstein–matter reduction, declared gauge and boundary data, constraint-satisfying initial data, the attached matter source, a local gauge slice and local Cauchy uniqueness. Crucially, the projected upstream seed must satisfy

$$dP_g^\sigma(V_D) = F_{\text{Einstein}} \circ P_g^\sigma$$

for that independently defined reduced Cauchy vector field. The projected curve then equals its unique Cauchy development [25, Def. 16.2, Thm. 17.3]. The infinitesimal compatibility identity is an explicit dynamical certificate, not inferred from the word common.

- (ii) *Quantum eligibility.* The quantum profile is surjective onto the full ray space of a declared Hilbert carrier. Its exact quotient process preserves all transition probabilities and has a strongly continuous all-real-parameter group extension. The generator then has the spatial/domain eligibility stated above. Wigner, phase lifting, Stone and the imported normal-form theorem classify the descended process as Schrödinger evolution [25, Defs. 18.1–18.4, Thms. 19.1–19.4].

Record and compatibility projections carry their exact original quotient closure. Declared exchanges use a common incidence monoid and retain separate closed-prediction and exchange-extended maps; their directed cells satisfy the source-target-correct cocycle [25, Thms. 20.3, 21.3, 22.6, 23.2–23.3].

What the occupied example establishes. The complete closed packet is explicit:

$$U_\star = J \times Q_\star \times \mathbb{R}, \quad V_\star = \partial_\theta + \partial_r, \quad \alpha_t(\theta, q, r) = (\theta + t, q, r + t).$$

Its metric projection is the Minkowski vacuum datum with certified zero source. Its quantum projection is

$$\Pi_H(\theta, q, r) = R_\theta q, \quad R_t[c_0, c_1, \dots] = [e^{-iE_0 t/\hbar} c_0, e^{-iE_1 t/\hbar} c_1, \dots],$$

where the specified full-ray group has a harmonic-oscillator realization. Therefore $\Pi_H \alpha_t = R_t \Pi_H$, while the upstream q coordinate itself is fixed [25, Defs. 24.1, 24.3, 24.5, Thm. 24.7]. This exhibits a nonempty common history and compatible recipients. The nontrivial quantum dynamical information is carried by the clock-conditioned projection; it is not obtained from the clock-translation seed alone. The example establishes neither a new mutual gravity–quantum interaction nor a unique empirical choice of the seed, spectrum or potential. In particular, it is a Module C history/projection witness, with a certified-zero metric source and a fixed-background oscillator recipient. It does not identify the oscillator energy with that zero gravitational source, and it does not by itself verify Module B’s nonzero source/mass matching or all Modules A–D hypotheses at once. The source explicitly distinguishes its strict continuation certificate from a nonzero interacting model [25, after Thm. 24.7; Sec. 25.1].

The capstone imports these conclusions on exactly these eligibility classes and also records a complete declared-open packet [27, Thms. 16.1, 33.4, 35.1–35.5, 79.1].

B.6 Modules B and D: matching and completed occupation

Module B is a separate overlap result. Its matching object includes an attached normalized-ray source family with polarization and common-domain control; a strongly continuous positive-energy Poincaré representation with irreducible reference algebra and invariant baseline; integrated total-source charges implementing the same translations; a stable positive-mass spin-zero Wigner sector; regular total-stress form factors; a regulated stationary Breit source; a weak-field Einstein recipient; an explicit scalar Wigner contraction; and common calibration [24, Def. 5.1]. The conclusions include equality of the integrated source charges with the Stone generators, zero-transfer normalization $A(0) = 1$, common active/Casimir mass, Bargmann central mass and the Schrödinger kinetic coefficient [24, Thms. 7.3, 26.1]. Because Poincaré symmetry and a massive Wigner sector enter this matching envelope, Module B is not used to derive those structures in the present cone or operational-inertia proof. It supports their compatibility with an independently specified quantum recipient.

Module D completes one fixed target. Its input is the completed source-locked Modules A–C parameter diagram: the actual source system, state, seed, potential, initial data, matching parameters, generators, overlaps and accepted probes. The source-locked tensor interfaces have explicit covariant completions and a colimit; the joint probe fixes the maximal lawful kernel-pair equivalence. Over each of the eight fixed role parameters, the relevant presentation groupoid is nonempty and connected. The candidate and compatibility analysis then gives the exact live residue, a compatible bijection from the required roles, and one lawful assignment orbit [26, Thms. 30.1–30.3, Prop. 31.1]. The capstone imports the result as completed occupation of the frozen target [27, Thms. 17.1, 80.1].

The conclusion is substantive about coverage, gluing and occupation at that fixed diagram. It is not a uniqueness theorem across different physical seeds, source systems, spectra or initial data, and it supplies no new dynamical equation. The theorem’s restriction is in its parameter base, not an optional restriction of kernel necessity.

Reception by the three physical arcs. The coupled Wheeler–DeWitt source and clock constructions include the integrated classical branches of WDW-4 and the constrained recovery family of WDW-8; WDW-9 assembles its thirteen compatible predecessor occupants and WDW-10 constructs clock-neutral descent [18, Thm. W4.I.1][29, Thms. 6.4–6.5, 6.15][30, Thm. 0.1][20, Thms. 5.4, 6.2, 6.4]. Section 15.3 distinguishes those established results from matching the particular SR apparatus. Section 15.4 identifies the geometric and passive-frame receiving interfaces in SYM-2 and the transport, measurement and response fields of the carrier sequence. Proposition 15.13 preserves the actual source and operation domains of each imported SR result; it does not assume that different occupied targets have already been physically identified.

B.7 Exactly what the SR construction adds

Theorem 4.8 makes these imports an explicit source-to-condition interface. Its actual-record and independent-warrant hypotheses are required for physical occupation; its quotient proof constructs exact readout and operation reception. It supplies neither a witness for an empty refinement nor truth for an arbitrary added predicate.

There are two cone routes. Proposition 5.3 supplies an effective nonperiodic comparison group from complete action records and unique repetition on its stated branch. The primary route constructs affine displacement from original comparison composition in Theorem 5.2; Theorem 5.7 then constructs the spatial comparison and Lorentz form. The calibrated alternative starts with its already supplied Euclidean comparison. For the primary route, an additive clock strictly positive on the completed witness cone and an intrinsically isotropic rest-comparison action reconstruct the spatial quadratic form itself. Proposition 5.6 derives strict completed-cone positivity from completed antisymmetry, weak clock monotonicity and the actual rest action with no fixed spatial vector. Lemma 5.5 then gives the finite bound. These completion properties have their own source warrants; they are not inferred from R5 path mediation alone. Lemma 6.2 supplies common rest calibration on the ordered interval-comparison branch. Dimension and the isotropy of actual comparisons remain visible physical content.

The tangent route starts with an occupied Lorentzian metric image and therefore inherits signature. Lemma 12.2 proves equality with the independently transport-constructed form when their actual source identification preserves the same cone and one timelike calibration. Lemma 12.3 supplies a finite certificate from nine original null-direction answers and one clock standard. A common source name or flatness alone is insufficient for this matching. Theorem 12.6 then does additional work: Levi–Civita transport explicitly computes a Lorentz-path recipient of each actual marked causal history, with proved identity, concatenation, causal-norm, geometric-duration and naturality laws. Corollary 12.8 identifies its output with affine SR displacements on the flat branch. The continuation action is constructed, on a certified nonempty marked refinement over the joint Einstein–Schrödinger eligible domain. Theorem 15.10 constructs such a refinement on the declared fixed-background branch, including its energetic clocks, signal records, controlled response and exact matching; its vacuum metric source and positive laboratory response retain distinct query identities.

The original source-object identifier and complete endpoint records make source composability available in the recipient; the retained record factor preserves the original applicability and failures. Arbitrary geometric curves are not thereby physically occupied. Device readings still require their existing calibration, and anti-development does not preserve arbitrary ambient finite-pair intervals or entire radar networks in curved spacetime.

Corollary 12.7 postcomposes the existing source projection with this constructed action.

The final functor-composition inference remains elementary; the construction supplies the explicit action and its properties. Module C and the capstone supply their original occupied histories and eligibility classes. A mere match of notation with another clock construction cannot prove that their marked physical subtypes intersect.

Lemma 11.4 differentiates the complete invariant source prescription to establish stationary-solution and response-graph transport. Proposition 11.5 then derives tensor and charge covariance on the stated charge domain. The source representation alone does not choose that prescription. Proposition 11.14 verifies a particular ideal response model, including hyperplane traces and a smooth controlled history; it does not replace an empirical apparatus certificate.

These constructions support the joined theorem: necessary kernel dependence, exact witnessed refinement, and uniqueness of the derived SR laws on the unchanged source target. Proposition 2.4 applies directly to every richer determinate construction in the chain, including the selectors, maps, physical conditions and received laws. The source-refinement and construction proofs determine how those inputs yield their numerical realization without supplying any omitted physical witness by terminology.

References

- [1] A. J. Maley, *Non-Degenerate Construction and the Kernel of Admissibility: An Exhaustion Theorem for Standing, Reference, and Irreversibility*. Corpus reference edition selected in the September 2026 source contracts.
- [2] A. J. Maley, *The AMetric Boundary as a Unique Regime-Invariant Constraint on Admissibility-Bearing Construction*. 20 September 2026.
- [3] A. J. Maley, *A Fixed-Domain Exhaustion Theorem: Standing, Consequence, and Compositional Closure*. Publication reference edition, 21 September 2026.
- [4] A. J. Maley, *Anchor, Tensor, and Skin: A Fixed-Domain Decomposition Theorem for Admissible Description*. Publication reference edition, 21 September 2026.
- [5] A. J. Maley, *Closure by Exhaustion for Same-Scope Operators under Admissibility*. Publication reference edition, 21 September 2026.
- [6] A. J. Maley, *Gravity as Structural Necessity*. Corpus manuscript, 2026.
- [7] A. J. Maley, *The Realized Physical Interior: Locality, Metric Role, Fields, Particles, Gauge Structure, Mass, Vacuum, Time, Causality, and the Standard Physics Stack under AASC*. June 2026.
- [8] A. J. Maley, *The Metric Role from Continuation Comparison*. Realized Physics Arc, R4, May 2026.
- [9] A. J. Maley, *Bounded Perturbation Transport and the Origin of Finite-Propagation Constraint: Continuation-Comparison Constraints on Standing-Relevant Transport*. Realized Physics Arc, R5, May 2026.
- [10] A. J. Maley, *Fields as Persistence Carriers: The Minimal Carrier Role of Physical Field Structure under Bounded Perturbation Transport*. Realized Physics Arc, R6, May 2026.
- [11] A. J. Maley, *Particles as Localized Persistence Classes: The Minimal Object Role of Particle Identity under Field-Carrier Mediation*. Realized Physics Arc, R7, May 2026.

- [12] A. J. Maley, *Mass, Inertia, and Persistence Compression*. Realized Physics Arc, R9, May 2026.
- [13] A. J. Maley, *Time and Causality from Irreversible Continuation Ordering*. Realized Physics Arc, R11, May 2026.
- [14] A. J. Maley, *The Persistence Bearer and the Reconstruction of Standard-Model Roles*. CSP–R, primitive-standing reference edition, 21 September 2026.
- [15] A. J. Maley, *Metricity as Licensed Description: The L-metric Boundary and the Realized Physical Interior*. May 2026.
- [16] A. J. Maley, *Null Propagation as the Causal-Limit Invariant: Why the Speed of Light Is a Metric Representative*. Corpus manuscript, inspected September 2026.
- [17] A. J. Maley, *The Physical Clock as a Quotient*. Archive edition with audit appendices, June 2026. [doi:10.5281/zenodo.20833048](https://doi.org/10.5281/zenodo.20833048).
- [18] A. J. Maley, *Classical Matter Occupancy and Joint-Constraint Closure under AASC*. Wheeler–DeWitt Arc, WDW-04, publication edition, July 2026.
- [19] A. J. Maley, *Profile-Clock Exhaustion and Constraint-Induced Cauchy-Germ Evolution*. Wheeler–DeWitt Arc, WDW-07, publication edition, 4 August 2026.
- [20] A. J. Maley, *Global Clock-Neutral Relational Descent and Universality Closure*. Wheeler–DeWitt Arc, WDW-10, publication edition, 6 August 2026.
- [21] A. J. Maley, *Einstein Dynamics as the Unique Minimal First-Response Metric Projection of AASC Gravity*. 2026, 76-page edition; Definition 14.2, Remark 14.3, Theorem 30.1. Inspected PDF generated in July 2026.
- [22] A. J. Maley, *Schrödinger Dynamics as the Unique Minimal First-Response Hamiltonian Projection of AASC Quantum Standing*. 2026, 67-page edition; Theorems 25.10, 27.3, 36.1. Inspected PDF generated in August 2026.
- [23] A. J. Maley, *Operational Metric-Source Representation and Composite Ledger Closure under AASC*. Module A, AASC-GRQM-MA-SRC-1.0, v1.1.0-attachment-corrected, 2026.
- [24] A. J. Maley, *Common Mass-Charge Matching and Universal Inertial–Active Proportionality under AASC*. Module B, AASC-GRQM-MB-MASS-1.0, v1.3.0-normalized-ray-source-lift-corrected, 2026.
- [25] A. J. Maley, *Kernel-Native Upstream Continuation and Projection Intertwining under AASC*. Module C, AASC-GRQM-MC-CONT-1.0, standalone v1.2.0-cycle2-corrected, 2026. The capstone imports the mathematically unchanged v1.2.1 editorial clarification.
- [26] A. J. Maley, *Exact Certificate-Fibre Exhaustion and Unique Role-Occupancy Closure under AASC*. Module D, AASC-GRQM-MD-OCC-1.0, v1.4.0-cosieve-support-corrected, 2026.
- [27] A. J. Maley, *Complete Certificate Closure for a Constraint-Native Gravity–Quantum Realization under AASC*. Modules A–D closure edition, v1.6.0-moduleABCD-integrated-fep1.0, 2026, 92 numbered pages plus title.
- [28] A. J. Maley, *Spacetime Symmetry as Domain Anchor*. Symmetry Arc, SYM-2, full reference edition, 2026.

- [29] A. J. Maley, *Covariant–Canonical Reconciliation and Controlled Semiclassical Recovery for the Wheeler–DeWitt Constraint*. Wheeler–DeWitt Arc, WDW-08, August 2026; Theorems 6.4–6.5, 6.15 and Proposition 6.14.
- [30] A. J. Maley, *Complete Joint Constraint–Kernel Realization Closure for the Wheeler–DeWitt System*. Wheeler–DeWitt Arc, WDW-09, 5 August 2026; Theorem 0.1.
- [31] A. J. Maley, *Primitive Carriers, Derivative Skins, and Classifier-Level Prediction Closure in the Standard Model*. Standard Model carrier sequence, SM-1, July 2026.
- [32] A. J. Maley, *The Standard Model Carrier Exhaustion Closure*. Standard Model carrier sequence, SM-6, primitive-standing reference edition, 21 September 2026.
- [33] A. Einstein, Zur Elektrodynamik bewegter Körper, *Annalen der Physik* **17** (1905), 891–921. [doi:10.1002/andp.19053221004](https://doi.org/10.1002/andp.19053221004).
- [34] A. Einstein, Ist die Trägheit eines Körpers von seinem Energieinhalt abhängig?, *Annalen der Physik* **18** (1905), 639–641. [doi:10.1002/andp.19053231314](https://doi.org/10.1002/andp.19053231314).
- [35] J.-P. Anker and F. Ziegler, Relativity without light: A new proof of Ignatowski’s theorem, *Journal of Geometry and Physics* **158** (2020), 103871. [arXiv:2007.09301](https://arxiv.org/abs/2007.09301).
- [36] E. C. Zeeman, Causality implies the Lorentz group, *Journal of Mathematical Physics* **5** (1964), 490–493. [doi:10.1063/1.1704140](https://doi.org/10.1063/1.1704140).
- [37] A. D. Alexandrov, A contribution to chronogeometry, *Canadian Journal of Mathematics* **19** (1967), 1119–1128. [doi:10.4153/CJM-1967-102-6](https://doi.org/10.4153/CJM-1967-102-6).
- [38] J. Ehlers, F. A. E. Pirani, and A. Schild, The geometry of free fall and light propagation, in *General Relativity: Papers in Honour of J. L. Synge*, Oxford University Press, 1972, pp. 63–84. Reprinted in *General Relativity and Gravitation* **44** (2012), 1587–1609.
- [39] H. Andr  ka, J. X. Madar  sz, I. N  meti, and G. Sz  kely, A logic road from special relativity to general relativity, 2010. [arXiv:1005.0960](https://arxiv.org/abs/1005.0960).
- [40] V. Perlick, On the radar method in general-relativistic spacetimes, 2007. [arXiv:0708.0170](https://arxiv.org/abs/0708.0170).
- [41] M. von Laue, Zur Dynamik der Relativit  tstheorie, *Annalen der Physik* **35** (1911), 524–542. [doi:10.1002/andp.19113400808](https://doi.org/10.1002/andp.19113400808).
- [42] D. Giulini, Laue’s theorem revisited: Energy-momentum tensors, symmetries, and the habitat of globally conserved quantities, *International Journal of Geometric Methods in Modern Physics* **15** (2018), 1850182. [arXiv:1808.09320](https://arxiv.org/abs/1808.09320).
- [43] J. D. Brown and K. V. Kuchař, Dust as a standard of space and time in canonical quantum gravity, *Physical Review D* **51** (1995), 5600–5629. [arXiv:gr-qc/9409001](https://arxiv.org/abs/gr-qc/9409001).
- [44] F. Finster, S. Kindermann, and J.-H. Treude, *Causal Fermion Systems: An Introduction to Fundamental Structures, Methods and Applications*. Cambridge University Press, 2025. [arXiv:2411.06450](https://arxiv.org/abs/2411.06450).
- [45] F. Finster and C. K  rpoun, A geometric derivation of the Einstein equations from the causal action principle, 2026. [arXiv:2607.13871](https://arxiv.org/abs/2607.13871).
- [46] F. Finster, J. Kleiner, and C. F. Paganini, Causal fermion systems as an effective collapse theory, *Journal of Physics A* **57** (2024), 395303. [arXiv:2405.19254](https://arxiv.org/abs/2405.19254).
- [47] S. Ipek and A. Caticha, An entropic dynamics approach to geometrodynamics, 2019. [arXiv:1910.01188](https://arxiv.org/abs/1910.01188).